GEK-36544

INSTRUCTIONS

ELECTROHYDRAULIC CONTROL

FOR

BOILER FEEDPUMP TURBINE

3S7513TC300A18

GENERAL ELECTRIC

TABLE OF CONTENTS

	PAGE
INTRODUCTION	3
RECEIVING, HANDLING AND STORAGE	3
DESCRIPTION	3
Speed Control	3
Speed Indication	4
Speed Reference or Set Point	5
High Pressure/Low Pressure Stop Valve Indication	5
Low Speed Switch	6
Power Supplies	6
Series Regulator	6
MONITORING	7
Power Supply Alarm	7
Ground Alarm (24 VDC)	7
HSS/LSS Alarm	7
Cable Check Alarm	7
Lamp Test	7
Test Switch	8
INSTALLATION AND ADJUSTMENT	8
OPERATION	8
MAINTENANCE	8
TROUBLESHOOTING	8
RENEWAL PARTS	9
METER RELAY LAMP REPLACEMENT	9
Pt. No. 4868K32P001 Lamp	9
Pt. No. 4872K12G700 Lamp & Holder	9
Reassembly	10
DRAWING LIST	11

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company

INTRODUCTION

The electronic portion of an electrohydraulic control system for a boiler feedpump turbine drive is described in these instructions.

Any adjustment or maintenance of this equipment should be performed either by qualified turbine personnel or supervised by such personnel.

This equipment is an electronic speed governor whose speed set point can be varied by an automatic boiler controller or by an operator from a remote control unit. Electronic redundancy and mechanical overspeed trip are used to provide a high degree of reliability and safety.

The elementary diagram and alignment diagram for this unit are included in these instructions. Since both the number and arrangement of auxiliary components vary with particular application, it is necessary to refer to the drawings furnished with each equipment to determine the correct connections and auxiliary components furnished in each instance.

RECEIVING, HANDLING AND STORAGE

RECEIVING AND HANDLING

Immediately upon receipt, the equipment should be carefully unpacked to avoid damaging the apparatus. Particular care should be exercised to prevent small parts being mislaid or thrown away in the packing material.

As soon as the equipment is unpacked it should be examined for any damage that might have been sustained in transit. If injury or rough handling is evident, a damage claim should be filed immediately with the transportation company and the nearest General Electric Sales Office should be notified promptly.

STORAGE

If the equipment is not to be used as soon as it is unpacked, it should be stored in a clean, dry place and protected from accidental damage. Particular care should be exercised to avoid storing the equipment in locations where construction work is in progress. If the equipment is to be placed in long term storage, the interior cabinet temperature should be 5°C above ambient.

DESCRIPTION

SPEED CONTROL

The speed control loop will be described first. Starting at zone B6 and B7 of the alignment drawing, the coil of the electrohydraulic servo valve receives a low power signal from the electronic unit and controls the flow of oil to a double acting hydraulic cylinder which positions the turbine steam control valve. Turbine speed is sensed by three magnetic sensors mounted near the periphery of a shaft-driven gear. Pulses whose frequency is directly proportional to turbine

DESCRIPTION (continued)

SPEED CONTROL (cont'd)

speed are generated as each gear tooth passes a sensor. Primary and secondary speed channels in the electronic control produce a DC voltage proportional to the frequency of the sensor outputs. Outputs of two speed channels are fed into high value gate number one, which rejects the lower of the two speed signals and passes on the higher valued speed signal. This redundancy permits one channel to fail without significantly affecting turbine speed. The secondary channel is adjusted for an output 1% less than the primary channel. It is therefore uncontrolling, except in case of failure of the primary channel or if a backup amplifier test switch in the cabinet is actuated which increases the output of the secondary channel to 1% greater than the primary channel. A test meter in the cabinet can be used to determine the controlling channel.

The speed signal from high value gate one is summed with the speed set point to obtain a speed error. (See zone B3 of the alignment diagram.) The speed error is passed through a stability compensation and error amplifier, which includes a regulation adjustment to obtain a valve lift demand signal. This is summed with a valve position signal to obtain a valve position error which is passed through the servo amplifier to the servo valve. Increasing the speed set point increases valve lift demand, valve position error, and valve opening. Speed will increase until the rise matches the set point increases.

The valve position signal is obtained from a Linear Variable Differential Transformer (LVDT) whose core is positioned by the control valve. Three kilohertz excitation for the LVDT is generated in the electronic cabinet. A demodulator in the cabinet produces a DC signal proportional to the AC output of the LVDT, which varies with core position.

A control valve closing bias is applied directly to the servo coil of the control valve in the tripped condition. Before the EHC can reset the feedpump turbine, a valve closing signal from the servo amplifier (2SA) must be present to operate the Close Signal Check (CSC) relay and remove the bias signal that had been previously applied directly to the servo valve coil. Relay circuitry also requires that the electrical connectors must be plugged into the servo valve, the LVDT, and the speed sensors before starting. The LVDT signal is biased to close the control valve if the LVDT connector is removed with the turbine running. The servo valve contains a 5% mechanical bias which will close the control valve in the absence of an electrical signal.

SPEED INDICATION

At zone B9 of the alignment diagram, a third magnetic sensor and circuitry identical to the previously described speed signal are used for speed indication. An analog signal is obtained for speed indication on the EHC remote control panel, local panel, and customer use.

DESCRIPTION (continued)

SPEED INDICATION (cont'd)

A clamp signal from the 10% speed voltage comparator holds the turbine to less than 10% speed if both the primary and secondary speed channels show less than 10% speed.

At 105% speed, a voltage comparator operates a relay that unlocks the lock out valve in the emergency trip exerciser scheme. At 110% speed another voltage comparator operates the 110SX relay to indicate that 110% speed has been reached. A contact of this relay is provided for the customer.

SPEED REFERENCE OR SET POINT

Identical motor driven LVDT's are used to generate the manual and automatic speed references. The core position is varied by a two-phase stepping motor which moves in discrete steps and locks in position when not running. Raise and Lower pushbuttons on the operator's manual control panel cause the manual reference to run from the LSS stop to the HSS stop in 80 to 90 seconds. Raise Fast and Lower Fast pushbuttons will run the manual reference from stop to stop in approximately 10 seconds. The automatic speed reference circuits contain interface circuitry for compatibility with a specific boiler controller. The auto speed reference is adjusted to run from the LSS to the HSS in 10 seconds. The automatic and manual speed references are fed into a low value gate which rejects the higher reference and passes on the lower reference to control turbine speed. A relay operated from this gate indicates which of the two references is controlling.

An overspeed test can be performed after first running both manual and automatic references to the high speed stop. The Overspeed Test pushbutton is then actuated, causing an additional speed reference to be generated which increases linearly with time at approximately 12% speed per minute. This is added to both manual and automatic references. If turbine speed rises to 120% of rated, a speed signal will override the ramp generator and prevent further speed increase. Upon release of the Test pushbutton, the reference ramps back to normal at the same rate as it increased.

HIGH PRESSURE/LOW PRESSURE STOP VALVE INDICATION

The EHC cabinet contains circuitry for driving HPSV and LPSV position indicators located on the control panel supplied with the EHC cabinet. The signals for these position indicators are obtained from LVDT's mounted on their respective stop valves. This function is for position indication only and is not part of the control circuitry for the feedpump. (Refer to zone C7, C8 of the alignment drawing.)

DESCRIPTION (continued)

LOW SPEED SWITCH

An electronic Low Speed Switch (LSS) is provided in the EHC cabinet to automatically, if selected, engage the turning gear on decreasing turbine speed less than 2 rpm.

If at turbine speeds greater than 200 rpm, pulses are lost from the LSPU shown at zone A9 of the alignment drawing, the LSS will prevent the LSSX relay from energizing and prevent automatic engagement of the turning gear by the EHC.

The LSS provides an alarm under the following conditions:

1. If the turbine is at standstill, 0 rpm, and then is increased in speed and pulses are not detected from the LSPU, the LSS 2K contacts will energize the LSS alarm X relay when 200 rpm is reached.

2. If the pick-up pulses are detected from the LSPU but lost before 200 rpm is reached, the LSS 2K contacts will energize the LSS alarm X relay when 200 rpm is reached.

3. If pick-up pulses are lost from the LSPU at turbine speeds greater than 200 rpm, the LSS 2K contacts immediately energize the LSS alarm X relay.

4. Once the LSS alarm X relay is energized, the automatic engagement of the turning gear is prevented until pulses from the LSPU are obtained.

POWER SUPPLIES

Two redundant power supplies, referred to as line A and line B, are provided, They are the full wave rectifier, capacitor filtered variety. Each receives its 120 VAC from the customer's power.

The 120 VAC is fed to a transformer that has three secondary windings. The output of each of the rectified, filtered supplies is connected to three identical series regulators that are adjusted for 30 VDC, 24 VDC and -22 VDC.

The outputs of the series regulators on the line A supply are adjusted slightly higher so they are carrying load and the line B supply is the backup power supply.

SERIES REGULATOR

Operation of the series regulators is described as follows. Output of the voltage sensing bridge consisting of 15R, 16R, 17R, 1P and 2ZD is fed to 7Q and 8Q. An increase in output voltage raises the voltage on 8Q, increasing its conduction and lowering conduction of 7Q. Decreased current in 7Q decreases current in 4Q, 5Q, and 6Q, dropping the output voltage until balance is restored. Excessive current output through 10R turns 3Q on, 2Q off, and 1Q on, grounding the base of 7Q. With 7Q off, series regulators 4Q, 5Q, and 6Q are off and the output goes to zero. When 1C discharges, 3Q is turned off and output is restored. The circuit will oscillate between off and on until the cause of the high current is removed.

MONITORING

Located in the front of the cabinet are two meter panels that contain several alarms, meters, indicating lights and test meter that will allow checking or monitoring of the power supplies and most amplifiers in the cabinet.

POWER SUPPLY ALARM

For example, an alarm will occur if the 30 volt line A power supply should exceed the 95% or 105% set point of the meter relay in the cabinet. There will be three lights that will be illuminated. First, the power lamp on the control panel supplied with the cabinet will change from a steady glow to a flashing indication, the alarm lamp in the cabinet will be illuminated and the illuminated pushbutton above the line A 30 volt meter will indicate either a high or low voltage.

The three alarm lamps can be cleared after the malfunction is corrected by depressing the lighted pushbutton above the line A 30 volt meter. Identical functions are provided for both the line A and line B 30 V, 24 V, and -22 V power supplies and 3 KHz 6 V RMS power supply in the EHC cabinet. If any of these power supplies are not at their proper voltage level when the 120 V 60 Hz is applied, startup of the feedpump will be prevented.

GROUND ALARM (24 VDC)

If a ground were to occur on the positive or negative side of either the line A or line B 24 volt power supply, the power lamp on the control panel would flash, the +24 V GND/-24 V GND lamp in the cabinet would be illuminated, and the alarm lamp in the cabinet would be illuminated. These lamps will be cleared when the ground is removed. Depressing the +24, -24 V lamp in the cabinet is not required to clear the lamp.

HSS/LSS ALARM

HSS and LSS indication is provided both in the cabinet and on the control panel for both the manual and auto speed reference travel limits. These lamps will be cleared when the respective speed reference is driven off the limit.

CABLE CHECK ALARM

The cable check will prevent startup of the feedpump if the cables are not properly connected. The power lamp on the control panel will flash, the cable check lamp in the cabinet will be illuminated, and the alarm lamp in the cabinet will be illuminated until the cable is properly reconnected.

LAMP TEST

A lamp test can be performed on the indicating lamps in the cabinet by depressing the Alarm pushbutton. There aren't any lamps in the Backup Speed Amplifier Test pushbutton.

MONITORING (continued)

LAMP TEST (cont'd)

Depressing the Power pushbutton on the control panel will test the lamps on the control panel. There aren't any lamps in the Lower Fast, Lower, Raise, Raise Fast pushbuttons.

TEST SWITCH

The amount of current being supplied by each series regulator can be observed by the test switch and meter in the cabinet. When measuring series regulator current, the meter is calibrated to read approximately 5 amps full scale.

The voltage output of amplifiers 1 thru 12 can be observed by using the test switch and meter. When measuring amplifier outputs, the meter is calibrated to read 10 volts full scale. The amplifier numbers and function are shown on the alignment drawing and elementary diagram.

INSTALLATION AND ADJUSTMENT

The electronic control cabinet should preferably be mounted in a clean, dry, air-conditioned environment. Ambient temperature should not exceed 104°F unless otherwise noted on the outline drawing for a specific unit.

Equipment adjustment and the required test equipment are covered in detail in an alignment diagram and alignment instructions furnished with each unit.

The location of subassemblies within the cabinet is shown on the elementary diagram furnished with each control. Printed circuit board locations are shown on sheets 1 and 2 of the elementary diagram.

OPERATION

Refer to the instruction book for the turbine.

MAINTENANCE

No periodic maintenance is required. Outage time due to component failure is best minimized by maintaining a complete set of spare parts. Elementary diagrams of the printed circuit boards are included in this instruction book. The assembly drawings showing component location on the printed circuit boards are furnished with each order except where component identifications are marked on the boards.

TROUBLESHOOTING

Most troubleshooting problems can be quickly isolated to a particular section of the control by use of the built-in test meter, the meter relays, and alarm lights. These, used in conjunction with the alignment diagram, indicate the operating condition of each section. An oscilloscope is recommended for pinpointing a defective component or components. Each printed circuit board can be removed from the rack for servicing without disconnection due to the use of terminal boards.

RENEWAL PARTS

When ordering renewal parts, address the nearest Apparatus Sales Office of General Electric Company and supply the following information.

a. Catalog number stamped on the part and a complete description of the part, including its use and location.

b. Complete nameplate data appearing on the assembly of which the part is a component.

c. If possible, data on original order on which equipment was first supplied, including all numerical references.

Since operation of a steam turbine-generator is dependent upon operation of the feedpump control, it is suggested that an adequate stock of spare parts be carried on hand to minimize outage time due to component failure.

METER RELAY LAMP REPLACEMENT

Type 195 Meter Relays have been made with two versions of lamps; one without leads attached to the lamp (GE Pt. No. 4868K32P001) and the other with leads attached to the lamp and lamp holder assembly (GE Pt. No. 4872K12G700). These replacement parts can be ordered directly from: the nearest General Electric Sales Office or Utility & Process Automation Products Department, General Electric Company, 40 Federal Street, West Lynn, Mass. 01910.

In the event of lamp failure, the applicable lamp or lamp assembly can be replaced when the control unit is detached from the rear of the indicator set point unit. Then, the lamp-access plate (see figure) must be removed (screws are beneath adhesive labels on some models) to expose the lamp holder which can be pulled out.

PT.NO. 4868K32P001 LAMP

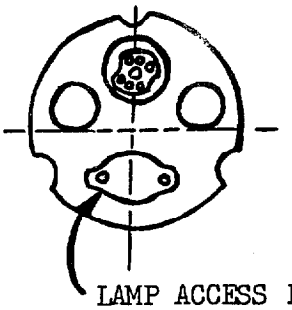
When installing the bayonet-type Pt. No. 4868K32P001 (without leads), twist the lamp in the holder assembly until the lamp filament will be in the horizontal position when the lamp holder is replaced in the indicator set point unit.

PT. NO. 4872K12G700 LAMP & HOLDER

The Pt. No. 4872K12G700 lamp has been fixed in the lamp holder at the factory so the filament position will be in the correct position when the keyed assembly is inserted in the indicator set point unit. Make sure the lamp-lead clips are engaged by the screws when the lamp access cover is replaced.

NOTE: If relay operation is not within $\pm$ 2% of set point index, the lamp holder position must be rotated 180°.

METER RELAY LAMP REPLACEMENT (continued)



REASSEMBLY

After the new lamp or lamp assembly has been installed, make sure the lamp access plate is replaced (and screws are tight) before control unit is reassembled on rear of indicator set point unit.

DRAWING LIST

DESCRIPTION	DRAWING NUMBER
Diode Board, 1DB	44A300243
Reference Low Value Gate, 2LVG	44A300281
Compensation and Regulation, 1CR	44A301400
Overspeed Integrator, 1OSI	44A302692
Signal Low Value Gate, 1LVG	44A304763
Alignment Instructions	44A308804
Series Regulator Board #1, 1REC	44A306392
Renewal Parts	44A376102
Cable Wiring Diagram, Signal & Low Voltage, 1CO	44B300423
Control Panel Outline	44B309476
Low Speed Switch Power Supply, LPS	44B306144
Cable Wiring Diagram Power, 2CO & 3CO	44B306148
Series Regulator Board #2, 1SR	44B306197
Detector Indicator, 1DI	44C300609
Operational Amplifier, 10A	44C300610
Analog Output, 1AO2	44C301216
Analog Output, 3AO2	44C301622
Frequency to Voltage Converter, 2FVC	44C301682
Clock and Stepping Motor Drive, CI&SMD	44C303567
Servo Amplifier, 2SA	44C304444
3 KHz Oscillator, 60CS	44C321478
Signal Converter, 3SC	44C304898
Voltage Comparator, 1VC	44C305259
Elementary Drawing, Master	44C321492
Interconnection Drawing	44C321493
Low Speed Switch, LSS	44C306444
Boiler Control Interface, 7BCI	44C306492
Line A & Line B Power Supply	44C306526
Cabinet Outline	44D300043
Alignment Drawing	44D309956
Control Panel Connection Diagram	44D309986
Relay Panel Connection Diagram	44D308137
Summary	44X321562

GENERAL ELECTRIC

44A300243
CONT ON SHEET SH NO.

REV NO. 0

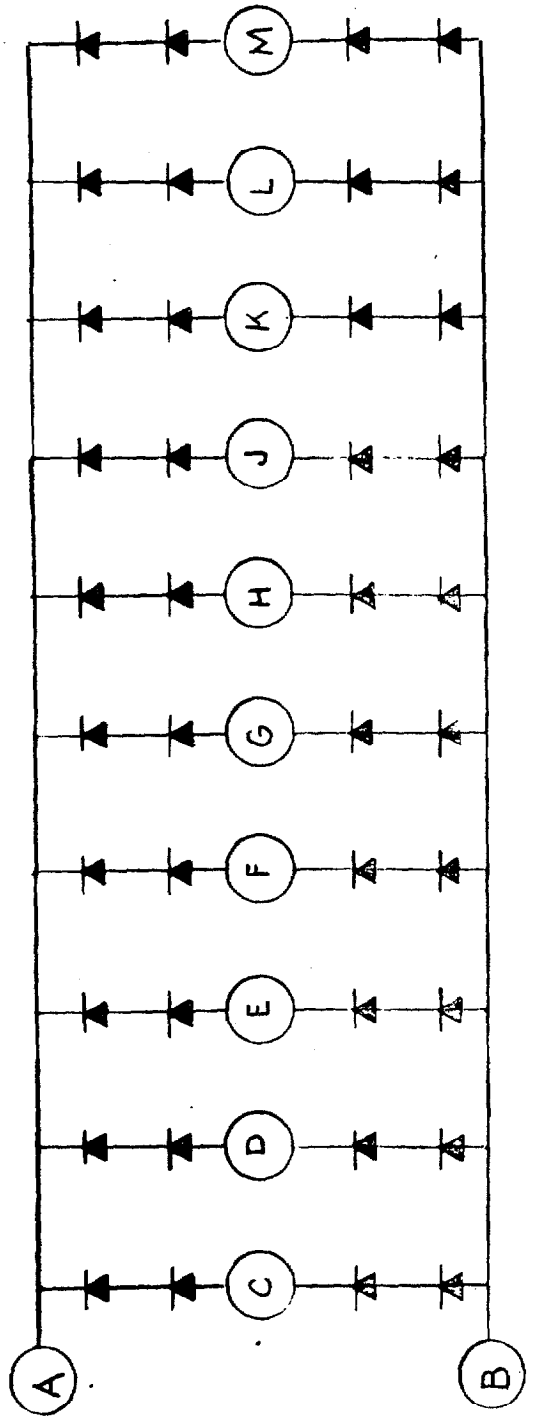
CONT ON SHEET SH NO.

TITLE

DIODE BOARD IDB

FIRST MADE FOR 44D236370-G01

REVISIONS



ALL DIODES 200 MA; 200 V.

11

DFB 12P

PRINTS TO

MADE BY R. Chambers 1/15/68

ISSUED MAR 12 1968

APPROVALS MBC

SPECIALTY CONTROL DIV OR DEPT.
WAYNESBORO, VA. LOCATION

44A300243
CONT ON SHEET SH NO.

CODE IDENT NO. 2

FF-803-WF (2-67)
PRINTED IN U.S.A.

GENERAL ELECTRIC

44A300281

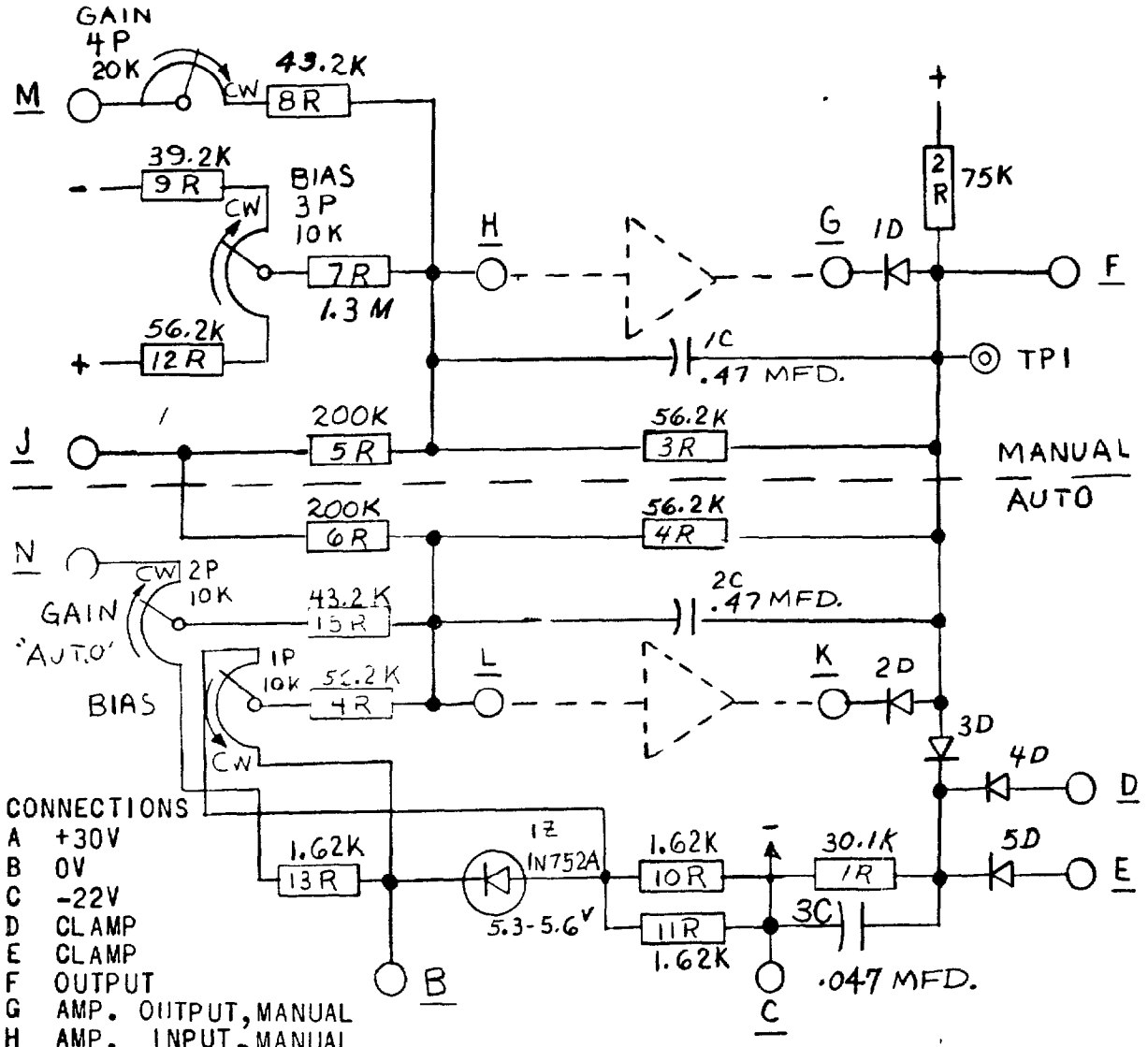
REV NO. (1) C KENNEDY

CONT ON SHEET SH NO.

TITLE

REFERENCE LOW VALUE GATE (2LVG)

FIRST MADE FOR 44D236378-G01



CONNECTIONS

A +30V
B 0V
C -22V
D CLAMP
E CLAMP
F OUTPUT
G AMP. OUTPUT, MANUAL
H AMP. INPUT, MANUAL
J OVERSPEED TEST INPUT
K AMP. OUTPUT, AUTO
L AMP. INPUT, AUTO
M MANUAL INPUT
N AUTO INPUT (SIGNAL CONVERTER)

CLAMP NOTE: ZERO VOLTS ON D OR E WILL HOLD OUTPUT (F) AT ZERO.

ADJUSTMENT:

1: WITH AUTO INPUT AT HSS AND MANUAL AT LSS, ADJUST BIAS FOR ZERO OUTPUT AT F.

2: WITH BOTH GAINS MAX. (CW) AND INPUTS AT HSS, ADJUST MANUAL GAIN FOR 5 VOLTS OUTPUT.

3: DECREASE AUTO GAIN UNTIL OUTPUT OF THE ASSOCIATED AMPLIFIER SWITCHES NEGATIVE TO POSITIVE.

4: AUTO BIAS IS SET CCW INITIALLY, FINAL SETTING PER APPLICATION REQUIREMENT.

REVISIONS

1	CHANGED SR, GR BY V.A.L. 4-5-1969 BUKOSKEY	MAY 5 1969
2	E.A. Penn	OCT 19 1970
3	Yadin	Nov 17, 1971
11		
12P DFB		
FG-W		

PRINTS TO

MADE BY H Chambers 1968

ISSUED APR 17 1968

APPROVALS RW

SPECIALTY CONTROL DIV OR DEPT

WAYNESBORO, VA LOCATION

44A300281

CONT ON SHEET SH NO.

FF-803-WF (9-66) PRINTED IN U.S.A.

CODE IDENT NO

GENERAL ELECTRIC

44A301400
CONT ON SHEET SH NO.

REV NO.

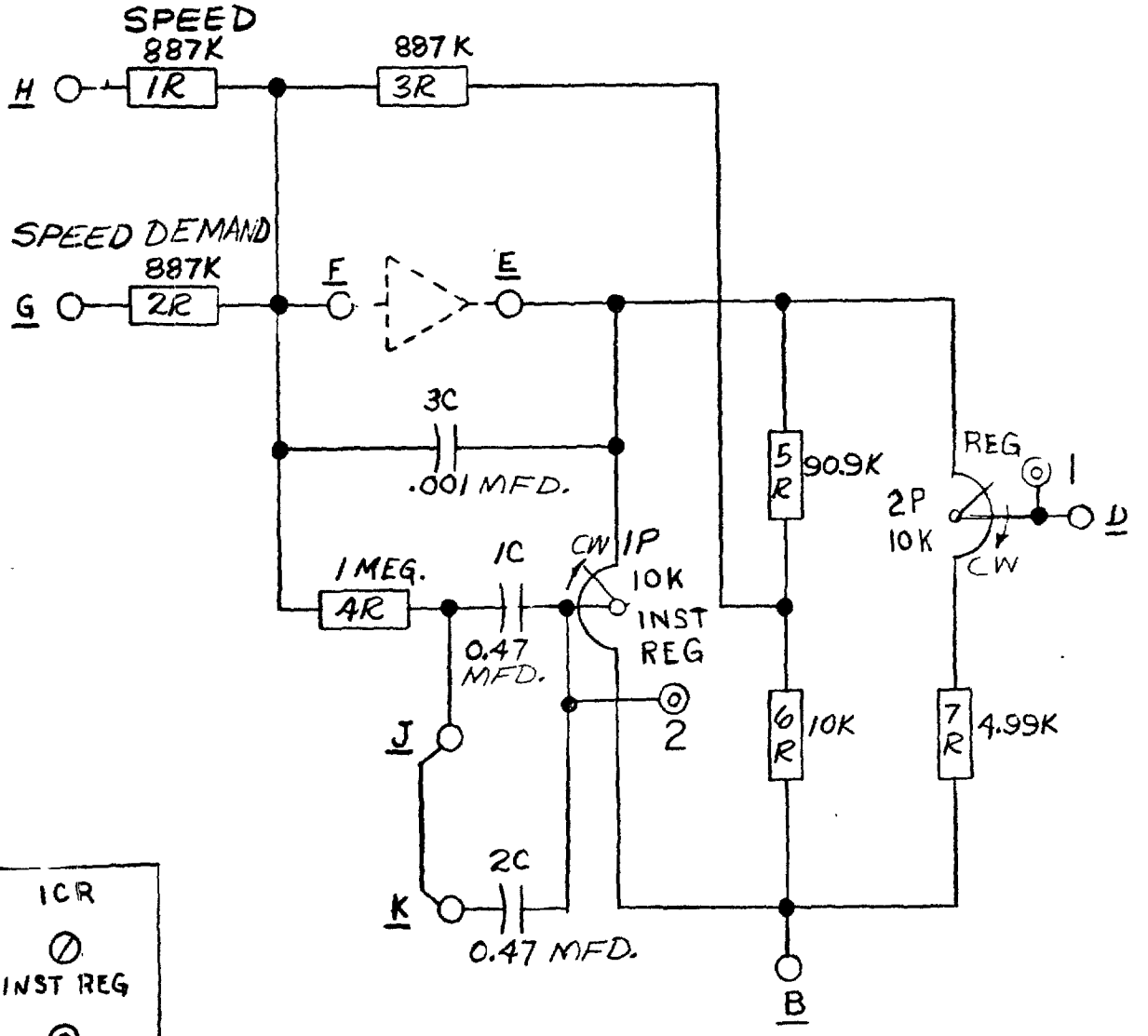
44A301400
CONT ON SHEET SH NO.

TITLE
COMPENSATION & REGULATION (ICR)

FIRST MADE FOR 44D236388-G01

REVISIONS

1	R M. BUKOSKEY	APR 1 1969
2	E.A. Ellis	Sept. 24, 1971
3	R. Ellis	Nov. 17, 1971
4	H. Brooks	Aug. 28, 1972



ICR
INST REG
REG

11

PRINTS TO

MADE BY R.M. Bukoskey
ISSUED FEB 18 1969

APPROVALS R.W. Wimmer

SPECIALTY CONTROL DIV OR DEPT.
WAYNESBORO, VA. LOCATION

44A301400
CONT ON SHEET SH NO.

CODE IDENT NO.

FF-803-W1P (9-67)
PRINTED IN U.S.A.

GENERAL ELECTRIC

44A302692'

CONT ON SHEET SH NO.

REV NO. (1) C. KENNEDY

44A302692

CONT ON SHEET SH NO.

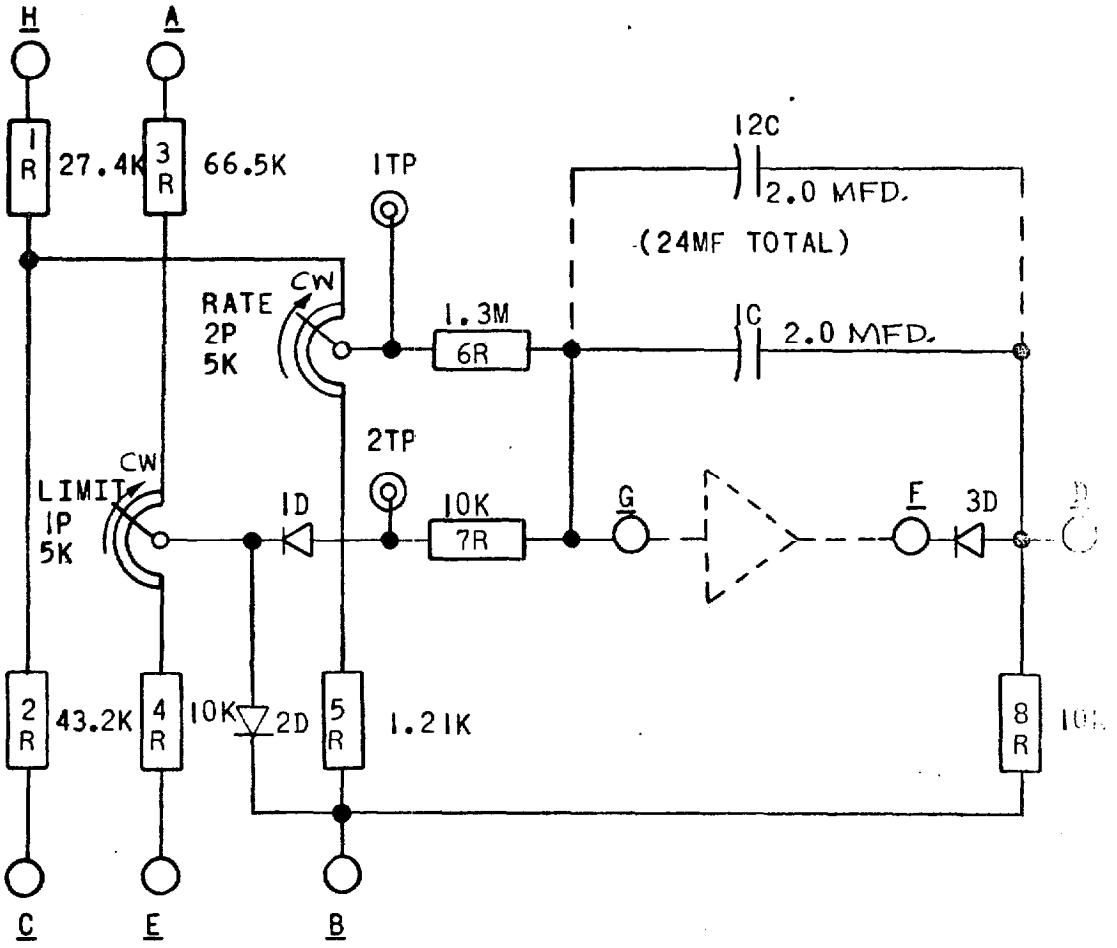
TITLE

OVERSPEED INTEGRATOR

IØSI

FIRST MADE FOR 44D236373-G02

REVISIONS



CONNECTIONS

A + 30V
B 0V
C -22V
D OUTPUT
E SPEED SIGNAL
F AMPLIFIER OUTPUT
G AMPLIFIER INPUT
H POS. SUPPLY THROUGH RELAY
J N.C.

MADE BY H. Templeton 3-16-70

ISSUED APR 8 1970

APPROVALS R. Kuhner

Semiconductor Devices DIV OR DEPT.

Waynesboro, Va. LOCATION

44A30

CONT ON SHEET

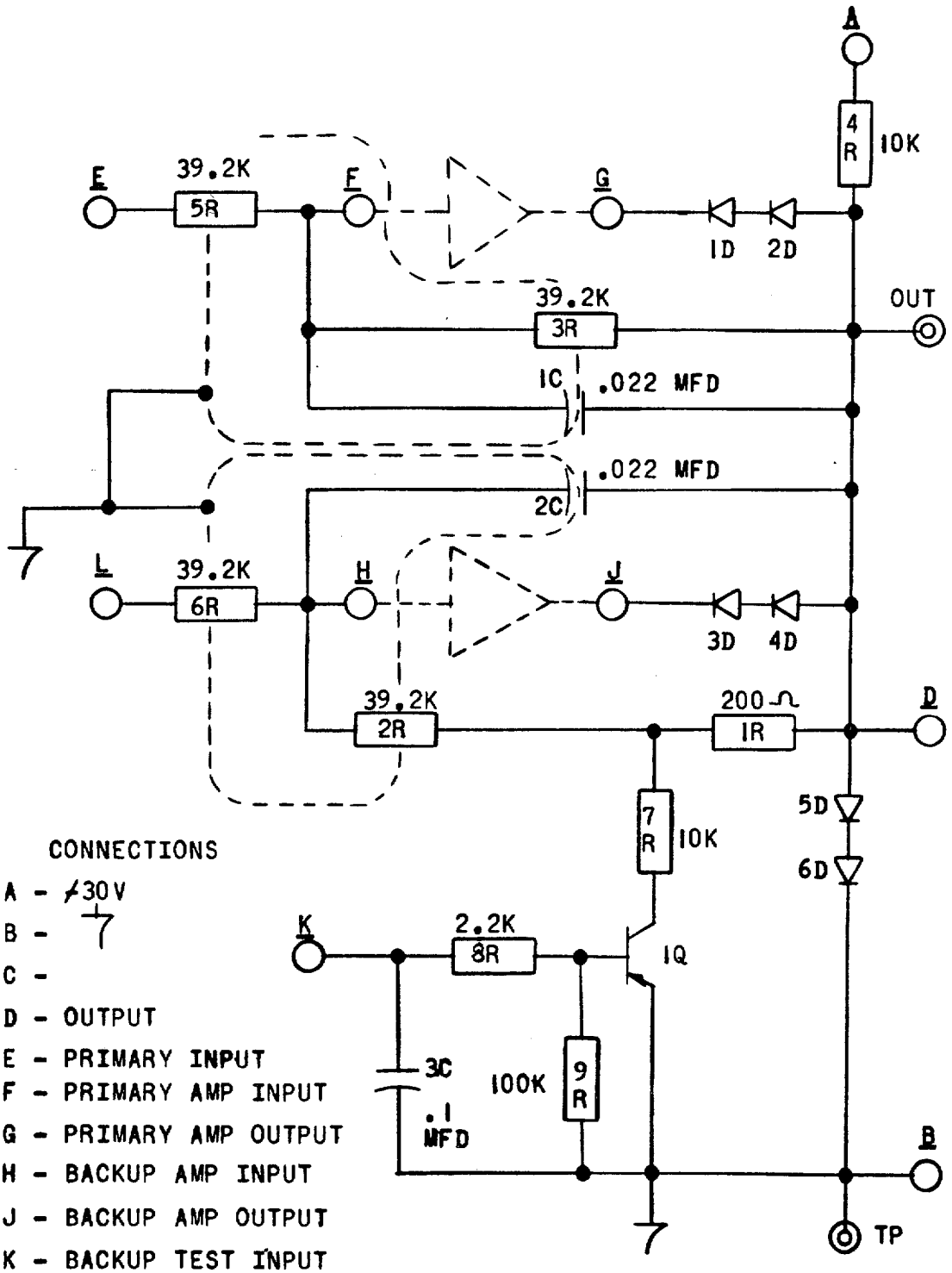
FF-803-WF (10-67)
PRINTED IN U.S.A.

2-1

GENERAL ELECTRIC

REV NO.	TITLE	44A304763 CONT ON SHEET SH NO.
44A304763 CONT ON SHEET SH NO.	SIGNAL LOW VALUE GATE	ILVG
	FIRST MADE FOR 44D236372-G02	

REVISIONS



CONNECTIONS

A - +30V
B - 7
C -
D - OUTPUT
E - PRIMARY INPUT
F - PRIMARY AMP INPUT
G - PRIMARY AMP OUTPUT
H - BACKUP AMP INPUT
J - BACKUP AMP OUTPUT
K - BACKUP TEST INPUT
L - BACKUP INPUT

11
12P DFB
FG-W

MADE BY	APPROVALS		PRINTS TO
H. Eagleton 8-13-71	R. Culver	COMM. & CONT. DEVICES DIV OR DEPT	44A304763
ISSUED Aug. 16, 1971	D. L. Mayfield	WAYNESBORO, VA LOCATION	CONT ON SHEET SH NO.

FF-803-WF (2-62)
PRINTED IN U.S.A.

GENERAL ELECTRIC

44A308804
CONT ON SHEET 2 SH NO. 1

REV NO.

CONT ON SHEET SH NO.

TITLE LOUISVILLE GAS & ELECTRIC
ALIGNMENT INSTRUCTIONS
ELECTRONIC FEED PUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A15 TURBINE 122948

REVISIONS

REFERENCE ALIGNMENT DIAGRAM 44D308104

TEST EQUIPMENT

1. AUDIO OSCILLATOR: HEWLETT PACKARD 201C, 200CD; GENERAL RADIO 1210C, OR EQUIVALENT. OUTPUT MUST BE 10 VOLTS RMS MINIMUM INTO 600 OHMS.

2. AC/DC DIGITAL VOLTMETER: 0.1 MV TO 100 V MAX.

3. DIGITAL FREQUENCY COUNTER: 6 VOLT RMS SENSITIVITY. 30 TO 5000 HZ.

4. TEN TURN POTENTIOMETER; NOT MORE THAN 5000 OHMS RESISTANCE.

5. TWO INPUT OSCILLOSCOPE, CALIBRATED SWEEP: X1 AND X10 PROBES.

6. OPTIONAL: BRUSH RECORDER SPECIAL TESTS ARE DESIRED.

7. VOLT - OHM - MILLIAMMETER

8. TWO 100K RESISTORS, TWO 1 MFD CAPACITORS (TO ADJUST OSCILLATOR OFFSET)

I. PRELIMINARY

1. CHECK INTERCONNECTION OF CONTROL CABINET, OPERATOR'S CONTROL DEVICES, TURBINE DEVICES, ETC.

2. MECHANICALLY ZERO TEST METER IN CABINET AND OPERATOR'S TACHOMETER AND OTHER INSTRUMENTS.

II. LINE "A" AND LINE "B" POWER SUPPLIES LOC. 12 AND 13

1. SET UP CONNECT 120V, 60 HZ TO LINE "A" POWER SUPPLY TERMINALS T24 AND T25. REMOVE POWER FROM LINE "B" TERMINAL T22 & 23.

2. DC SERIES REGULATORS, ADJUST POWER SUPPLIES AND METER CALIBRATIONS PER CHART. IF THE VOLTAGES LISTED CANNOT BE OBTAINED IT MAY BE NECESSARY TO ADJUST THE TAPS ON TRANSFORMER 1T, LOC. 12 & 13.

LINE "A" SUPPLY	TEST POINTS	ADJUST
+30V	BUS BARS	0308
-22V	BUS BARS	0316
24V	+F LOC. 0322; -G, LOC. 0324	0324

11
12P-DFB 2
FG-W
12P-FS 2

PRINTS TO

DRAWN BY C. Weaver 4-8-75
APR 15 1975

APPROVALS
R Cubes

DRIVE SYSTEMS PROD. DEPT. DEPT.
WAYNESBORO, VA. LOCATION

44A308804
CONT ON SHEET 2 SH NO. 1

CODE IDENT NO. 1

FF-803 WF (1-72)
PRINTED IN U.S.A.

REVISIONS

3. METER RELAY ALARMS VARY ALARM SET POINT OF LINE "A" POWER SUPPLY METER RELAYS TO CHECK ALARM OPERATION. SET ALARMS FOR 95% AND 105%.

4. CONNECT POWER TO LINE "B" POWER SUPPLY BY JUMPERING TERMINAL T24 TO T22 AND T25 TO T23, SO THAT POWER IS APPLIED TO BOTH SUPPLIES.

5. ADJUST THE INDIVIDUAL LINE "B" SERIES REGULATOR BOARDS SO THAT THEY ARE CARRYING ONLY 0.1 AMP AND THE LINE "A" SUPPLIES THE REMAINING LOAD. WHILE NOTING THIS CONDITION IS MAINTAINED MAKE FINAL ADJUSTMENTS IF NECESSARY PER CHART.

SUPPLY	LINE B	LINE A	TESTPOINTS
+30	0310	0308	BUS BAR
-22	0318	0316	BUS BAR
+24	0326	0324	LINE A, + F, 0322; -G, 0324 LINE B, + F, 0328; -G, 0326

6. TEST AND SET LINE "B" METER RELAY ALARMS AT 95% AND 105%

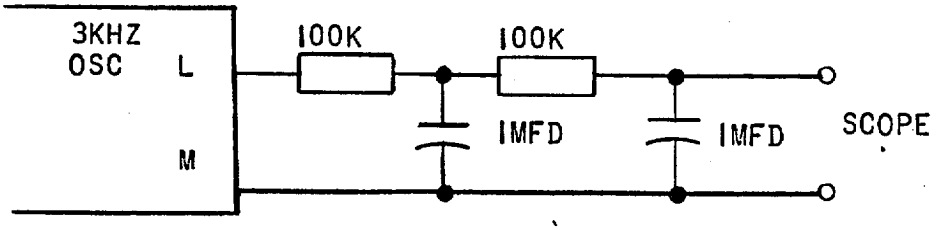
III. 3KHZ OSCILLATOR, -0217

1. VERIFY OUTPUT FREQUENCY AT 0217 IS 3000HZ. ± 300 HZ. USING COUNTER AT TERMINALS L&M.

2. ADJUST VOLTAGE LEVEL FOR 6.000 ± .006V. SET METER FOR 100%. CHECK METER RELAY ALARMS AND SET AT 95 AND 105%.

3. OFFSET ADJUSTMENT, CONNECT A FILTER CIRCUIT AS SHOWN:

LOC. 0217



3.1 REMOVE 3KHZ OSC BD FROM RACK AND ADJUST OFFSET FOR ZERO D.C. VOLTS AS OBSERVED ON OSCILLOSCOPE.

IV. SSPU SPEED INDICATION CIRCUITS

1. SET UP, REMOVE LEAD FROM T28 AND T29 AND CONNECT AUDIO OSCILLATOR. SET OSCILLATOR TO SSPU RATED HZ.

11

12P-DFB 2

FG-W

12P-FS 2

PRINTS TO

2. FREQUENCY TO VOLTAGE CONVERTER

2FVC LOC. 0224. ADJUST COARSE AND FINE POTS TO OBTAIN ⌿5.00V AT TERMINAL J.

3. VOLTAGE COMPARATORS IVC LOC., 0218, 0219, 0220, 0221. SET EACH VOLTAGE COMPARATOR (IVC) PER CHART ON ALIGNMENT DRAWING. RED LIGHT ON IVC SHOULD BE ON WHEN RELAY ENERGIZED. CHECK EACH IVC SETTING BY SLOWLY VARYING FREQUENCY ABOUT THE OPERATING POINT.

4. TACHOMETER 3AØ2 LOC. 0108 WITH OSCILLATOR AT ZERO FREQUENCY ADJUST 2A BIAS FOR ZERO INDICATION ON REMOTE CONTROL PANEL. SET OSCILLATOR TO RATED SSPU FREQUENCY AND ADJUST GAIN AND I OUT AS REQUIRED FOR CORRECT INDICATION OF TURBINE RPM.

4.1 REPEAT ZERO FREQUENCY AND RATED SPEED SETTINGS AND MAKE FINAL ADJUSTMENTS IF REQUIRED.

5. SECOND TACHOMETER 3AØ2 LOC. 0110 WHERE REQUIRED ADJUST 1A & 2A USING PROCEDURE IN STEPS 4 AND 4.1 ABOVE.

6. REMOVE OSCILLATOR FROM T28 AND T29 AND RECONNECT INCOMING LEADS.

V. SSPU SPEED MEASURING CIRCUITS

1. SET-UP JUMPER T4 TO T7 AND T5 TO T8. CONNECT OSCILLATOR AND COUNTER TO T4 AND T5.

2. FREQUENCY TO VOLTAGE CONVERTERS 2FVC, LOC. 0203 AND 0206 HIGH SPEED GATE ILVG, LOC. 0209.

2.1 PRIMARY AT RATED SHAFT SPEED PICKUP FREQUENCY, ADJUST 2FVC LOC. 0203 FOR -5.000 VOLTS AT OUTPUT TESTPOINT OF ILVG, 0209. (OBSERVE WITH TEST METER IN POSITIONS 6 AND 7 THAT AMPLIFIER 7 IS MORE POSITIVE THAN 6; IF NECESSARY ADJUST 2FVC, 0206, TO OBTAIN THIS CONDITION).

2.2 BACKUP DEPRESS BACKUP SPEED AMPLIFIER TEST SWITCH AND ADJUST 2FVC, 0206, FOR -5.05 VOLTS AT OUTPUT TP, 0209.

3. REMOVE TERMINAL POINT JUMPERS AND OSCILLATOR AND COUNTER LEADS.

VI. CONTROL VALVE RAM SERVO LOOP

1. CHECK LVDT CORE PER LEVER DIAGRAM.

REVISIONS

2. DETECTOR INDICATOR 1DI, LOC. 0216. WITH CONTROL VALVE FULL OPEN ADJUST NULL OF 1DI FOR 0.00 ± 0.01 VOLTS AT OUT TP. NOTE VOLTAGE READING ON ALIGNMENT DIAGRAM. WITH VALVE IN CLOSED OVERTRAVEL POSITION RECORD OUT TP VOLTAGE ON ALIGN. DIAG. ADJUST ZERO FOR 0 INDICATION OF CONTROL VALVE POSITION ON OPERATOR'S CONTROL CONSOLE. RETURN VALVE TO FULL OPEN POSITION AND ADJUST CALIBRATE FOR 100% POSITION INDICATION.

3. DETERMINE GAIN OF DETECTOR CIRCUIT IN VOLTS PER INCH BY DIVIDING THE DIFFERENCE IN OUT TP VOLTS AT EXTREME VALVE POSITIONS BY THE TOTAL VALVE TRAVEL IN INCHES, AND RECORD THIS FIGURE ON THE ALIGNMENT DIAGRAM.

3.1 ANALOG CONTROL VALVE POSITION SIGNAL

1AØ2 LOC. 0106. WITH VALVE CLOSED ADJUST 1A BIAS FOR ZERO POSITION ON OPERATORS METER.

3.2 WITH VALVE OPEN ADJUST 1A GAIN AND I OUT FOR FULL OPEN INDICATION.

3.3 REPEAT 3.1 AND 3.3 AS NECESSARY.

4. MAKE ELECTRICAL CONNECTIONS TO SERVO VALVE.

5. SERVO AMPLIFIER, 2SA, LOC. 0214.

5.1 PS10 TRIP CHECK FOR APPROX. -0.7 VOLTS AT "I" TP UNDER TRIP CONDITION.

5.2 CLOSE SIGNAL CHECK WITH TURBINE TRIPPED, DISCONNECT DEMAND TERMINAL E AND LIFT TERMINAL D OF 2SA. CONNECT TEST POT WITH NEGATIVE OUTPUT TO TERMINAL D AND SET POT FOR ABOUT -3 VOLTS AT LIFT TP. THE OUTPUT OF AMPLIFIER 9 ON TEST METER POSITION 9 SHOULD BE ABOUT -10 VOLTS. TURN BIAS ADJUSTMENT CCW UNTIL AMP. 9 GOES TO ≠10 VOLTS. THERE SHOULD BE LITTLE OR NO CONTROL OF 9 AT INTERMEDIATE VOLTAGES. LEAVE 9 AT ≠10 VOLTS AND RESET TURBINE. SLOWLY REDUCE TOWARDS 0 THE TEST POT SETTING AND NOTE THAT 9 SWINGS FROM ≠10 TO A MINIMUM VALUE OF APPROX. -6.2 VOLTS. NOW APPLY MORE NEGATIVE VOLTAGE FROM THE TEST POT AND NOTE THAT 9 HAS A MAXIMUM VALUE OF APPROX. ≠6.2 VOLTS.

5.3 GAIN REFER TO ALIGNMENT DRAWING TO OBTAIN LIFT TEST POINT VOLTAGE REQUIRED TO PRODUCE 1 VOLT AT I TEST POINT, WHEN GAIN IS PROPERLY ADJUSTED.

11
12P-DFB 2
FG-W
12P-FS 2

PRINTS TO

REVISIONS

5.31 VARY THE TEST POT SETTING UNTIL I TEST POINT IS 0.000 VOLTS AND NOTE LIFT TEST POINT VOLTAGE. CHANGE THIS VOLTAGE WITH THE TEST POT BY PLUS LIFT TEST POINT VOLTAGE FROM STEP 5.3, ADJUST GAIN UNTIL I TEST POINT READS MINUS 1.000 VOLTS.

5.32 RECHECK ZERO POINT BY RETURNING TEST POT SO THAT ZERO VOLTS IS ONCE AGAIN AT I TEST POINT AND TEST POT VOLTAGE IS THE SAME AS NOTED IN 5.31.

5.33 ADJUST TEST POT FOR MINUS LIFT TEST POINT VOLTAGE AND NOTE THAT I TEST POINT IS PLUS 1.000 VOLT.

5.4 BIAS DISCONNECT TEST POT AND RECONNECT THE LEAD REMOVED FROM D. WITH THE VALVE RAM AT ITS INTERNAL STOP AND ZERO VOLTS A DEMAND TEST POINT SET BIAS FOR -0.35 ± .01V AT I TEST POINT. RECONNECT TERMINAL E.

6. AFTER REMOVING THE ELECTRICAL CONNECTOR, MOUNT THE SERVO VALVE AND VERIFY THAT THE MECHANICAL BIAS IS IN THE VALVE CLOSING DIRECTION.

VII. REGULATION AND COMPENSATION ICR, LOC. 0212

1. REGULATION CONNECT THE OUTPUT OF THE TEST POT TO SUPPLY A NEGATIVE VOLTAGE TO TERMINAL E. ADJUST REGULATION TO OBTAIN VALVE TRAVEL FROM NO LOAD LIFT TO FULL LOAD LIFT FOR 3.00 VOLTS CHANGE IN THE OUTPUT OF THE TEST POT.

2. COMPENSATION VERIFY THAT JUMPER J-K IS NOT CONNECTED.

3. INSTANTANEOUS REGULATION SET TEST POT FOR -1 VOLT OUTPUT AND ADJUST INST. REG. FOR VOLTAGE EQUAL TO .242 VOLTS AT INST REG. T.P.

11

12P-DFB2

FG-W

12P-FS2

PRINTS TO

VIII. MANUAL SPEED REFERENCE (STAND-STILL CHECKS)

1. THIS REFERENCE IS OBTAINED FROM THE UPPER OF THE TWO MOTOR DRIVEN LVDT'S. WHEN POWER IS APPLIED IT SHOULD RUN TO LSS AND LIGHT THE LSS INDICATING LIGHT.

2. SLOW REFERENCE SPEED, CL & SMD, LOC. 0118. SHUT OFF HIGH PRESSURE OIL SUPPLY OR DISCONNECT SERVO VALVE BEFORE RAISING MANUAL REFERENCE. ACTUATE RAISE PUSHBUTTON AND TIME TRAVEL TO HSS LIGHT INDICATION. TIME SHOULD BE 80-90 SECONDS.

3. FAST REFERENCE SPEED, CL & SMD, LOC. 0118. ACTUATE LOWER FAST PUSHBUTTON AND TIME MOTOR TRAVEL FROM HSS LIGHT TO LSS LIGHT. IT SHOULD BE 8 TO 10 SECONDS.

4. BIAS AND GAIN, MANUAL REFERENCE, 2LVG, LOC. 0116. DISCONNECT K OF 2LVG. RAISE MANUAL REFERENCE OFF LSS AND BACK INTO IT AT SLOW RATE WITH LOWER PUSHBUTTON. ADJUST MANUAL BIAS FOR 0.000 VOLTS AT OUTPUT TP, 0116. RAISE TO HSS AT SLOW RATE AND ADJUST MANUAL GAIN FOR OUTPUT VOLTAGE AT TP 0116 AS SPECIFIED BY ALIGNMENT DIAGRAM.

11

12P-DFB2

FG-W

12P-FS2

PRINTS TO

REVISIONS

IX. AUTO SPEED REFERENCE, (STAND-STILL CHECK)

1. BOILER CONTROLLER INTERFACE, 73CI, LOC. 0122, CONFIRM THAT THE INTERCONNECTING CABLE BETWEEN THE FEEDWATER CONTROLLER AND EHC CABINET IS A SHIELDED CABLE, AND THE SHIELD OF THE CABLE IS CONNECTED TO COMMON ON THE FEEDWATER CONTROLLER AND TO T34 ON THE EHC CABINET.

   THE ADJUSTMENT OF IP (CMR) IS TO ELIMINATE ANY NOISE THAT MAY BE PICKED UP BETWEEN THE FEEDWATER CONTROLLER AND THE EHC CABINET.

2. USING A SCOPE, DETERMINE IF NOISE IS PRESENT BETWEEN TERMINALS B AND D OF 77CI, LOC. 0122. IF A NOISE SIGNAL (PROBABLY 60 HZ OR 120 HZ) EXISTS, SET FEEDWATER CONTROLLER SIGNAL FOR LSS AND ADJUST IP FOR MINIMUM NOISE AT 1A OUT.

3. IF THERE ISN'T ANY NOISE SIGNAL BETWEEN TERMINALS D AND B OF 73CI LOC. 0122, THEN DISCONNECT CUSTOMER'S INCOMING SHIELDED CABLE AT EHC TERMINALS T32, T33, AND T34 AND TEST EHC AS FOLLOWS. PLACE A JUMPER BETWEEN T32 AND T33 AND CONNECT OSCILLATOR BETWEEN T32 AND T34. SET OSCILLATOR AT 6 V RMS, 60 HZ AND ADJUST IP FOR MINIMUM AT 1A OUT. REMOVE OSCILLATOR AND JUMPER AND RECONNECT CUSTOMER LEADS TO T32, T33 AND T34.

4. WITH FEEDWATER CONTROLLER AT LSS, ADJUST 2P BIAS FOR OVDC AT 1A OUT.

5. SIGNAL CONVERTER 3SC, LOC. 0125

   WITH FEEDWATER CONTROLLER AT LSS. CHECK THAT 2LS LIMIT SWITCH CONTACTS BETWEEN 1TB(D)& 1TB(E) ON AUTO SPEED REF. LOC. 0005 ARE CLOSED. IF NECESSARY, REDUCE INPUT FROM FEEDWATER CONTROLLER UNTIL 2LS LIMIT SWITCH CLOSES. JUMPER SIGNAL FAILURE HOLD CONTACTS T77 AND T78 WITH 2LS LIMIT SWITCH CLOSED AND FEEDWATER CONTROLLER SIGNAL AT LSS. CONNECT VOLTMETER TO 1TP OF 3SC LOC. 0125 AND CHECK FOR ZERO VOLTS. IF NECESSARY, OBTAIN ZERO VOLTS BY WITHDRAWING 3SC FROM RACK AND ADJUSTING 5P.

6. REMOVE T77-T78 JUMPER AND SET FEEDWATER CONTROLLER SIGNAL TO HSS. CHECK THAT 1LS LIMIT SWITCH CONTACTS BETWEEN 1TB(D) AND 1TB(C) ON AUTO SPEED REF. LOC. 0005 ARE CLOSED. IF NECESSARY, ADJUST IP CW ON 3SC LOC. 0125 AND/OR 2P GAIN UNTIL 1LS LIMIT SWITCH CLOSES. WITH FEEDWATER CONTROLLER AT HSS AND 1LS LIMIT SWITCH CLOSED. JUMPER SIGNAL FAILURE HOLD CONTACTS T77-T78 AND ADJUST 2P GAIN FOR MINUS 5.00 V ±.01 VDC AT 1TP.

11
12P-DFB 2
FG-W
12P-FS 2

PRINTS TO

7. WITH T77-T78 STILL JUMPERED, ADJUST IP B.C. REF. FOR ZERO VOLTS AT 2TP. REMOVE JUMPER.

8. WITH FEEDWATER CONTROLLER AT LSS CHECK FOR ZERO VOLTS AT ITP AND WITH FEEDWATER CONTROLLER AT HSS CHECK FOR ZERO VOLTS AT 2TP.

9. ADJUST 3P DEADBAND FOR MINUS 2VDC (0.1% DEADBAND) AT 3TP. DISCONNECT WIRE AT OUTPUT TERMINAL F OF 3SC LOC. 0125.

10. ADJUST "BOILER CONTROLLER" INPUT TO LSS (-9±1VDC AT 2TP). CONNECT AN OSCILLOSCOPE TO TERMINAL F AND CHECK FOR A PULSE WIDTH OF 10 MSEC. IF NECESSARY, WITHDRAW BOARD AND ADJUST FREQUENCY CONTROL POTENTIOMETER 4P. RECONNECT WIRE TO TERMINAL F. AUTO SPEED REFERENCE SHOULD DRIVE FROM LSS TO HSS IN 10 SECONDS.

11. AUTO SPEED REFERENCE STOPS, LSS, 2LVG, LOC. 0116 DISCONNECT G, 0116 AND WITH AUTO REFERENCE AT LSS ADJUST SC BIAS, 0116 FOR OUTPUT TEST POINT VOLTS PER ALIGNMENT DIAGRAM.

12. HSS, 2LVG, LOC. 0116. WITH AUTO REFERENCE AT HSS ADJUST SC GAIN, 0113 FOR OUTPUT TEST POINT VOLTS PER ALIGNMENT DIAGRAM. RECONNECT G.

13. LOCAL/REMOTE CONTROL LIGHT, IVC, LOC. 0117. ADJUST REFERENCE, 0117 FOR 6.5 VOLTS AT L TP. CHECK THAT LOCAL CONTROL INDICATION IS OBTAINED WHEN AUTO REFERENCE IS AT LSS AND MANUAL REFERENCE IS LOWERED SO THAT 2LVG OUTPUT IS LESS THAN APPROXIMATELY 1 VOLT.

14. AUTO SPEED REFERENCE INDICATION, IDI LOC. 0128. ADJUST ZERO AT LSS AND CAL AT HSS TO OBTAIN 0 AND 100 INDICATIONS ON OPERATORS POSITION INDICATOR.

X. OVERSPEED TEST INTEGRATOR, 10SI, LOC. 0112

1. RATE CONNECT VOLTMETER TO OUTPUT TEST POINT 2LVG, 0116. WITH MANUAL AND AUTO REFERENCE AT HSS JUMPER T54 TO T58. VOLTMETER READING SHOULD INCREASE 0.6 ± 0.1 VOLTS PER MINUTE FOR A (12% ± 2%) SPEED INCREASE PER MINUTE. ADJUST RATE, 0109 AS REQUIRED.

2. LIMIT CONNECT OSCILLATOR AND COUNTER TO T4 AND T5. SET FREQUENCY FOR 120% SSPU FREQUENCY. OBSERVE OUTPUT OF AMPLIFIER 5 WITH TEST METER. JUMPER T54 TO T58 AND ADJUST LIMIT, 0112 FOR POINT AT WHICH AMPLIFIER SWITCHES FROM POSITIVE SATURATION TO ZERO VOLTS. CHECK BY REDUCING OSCILLATOR FREQUENCY AND SLOWLY INCREASING IT TO THE SWITCHING POINT FREQUENCY.

11.
12P-DFB2
FG-W
12P-FS2

PRINTS TO

REVISIONS

XI. LP STOP VALVE POSITION INDICATOR 1DI, LOC. 0102. WITH STOP VALVE FULLY CLOSED ADJUST DIFF. TRANS. NULL FOR -3 VOLTS AT OUT TP AND METER ZERO FOR 0 POSITION. ADJUST CAL. FOR 100% WITH VALVE FULL OPEN.

XII. HP STOP VALVE POSITION INDICATOR 1DI, LOC. 0104. SAME AS XI.

XIII. LOW SPEED SWITCH POWER SUPPLY, LPS, LOC 0227

1. CHECK FOR 5.6 VDC AT TERMINAL D.

2. CHECK FOR 12VDC AT TERMINAL E.

XIV. LOW SPEED SWITCH LSS LOC. 0225

1. STANDSTILL CHECKS - CHECK THAT TERMINALS F&G ARE +5.6VDC WITH RESPECT TO B. THIS WILL CONFIRM 1TDX & 200RPM RELAYS ARE ENERGIZED.
2. RELAY CONTACTS AT TERMINAL D AND E SHOULD BE CLOSED.
3. 1TP SHOULD BE 5.6 VDC.
4. REMOVE WIRE FROM F AND JUMPER F TO B. RELAY CONTACTS AT H & K CLOSE. REMOVE JUMPER AND CONNECT WIRE TO F.
5. OPERATE TURNING GEAR TO OPEN CONTACTS AT H & K.

OPERATIONAL CHECKS

1. CHECK FOR 0-5 VOLT PULSE AT TERMINAL N WHEN TURNING GEAR OPERATED.
2. POSITIVE PULSES SHOULD BE PRESENT AT 2TP WHEN TURNING GEAR OPERATED.
3. 1TP SHOULD BE ZERO VOLT WHEN TURNING GEAR OPERATED.
4. RELAY CONTACTS AT D & E SHOULD BE OPEN WHEN TURNING GEAR OPERATED.
5. WITH TURNING GEAR OFF AND TURBINE AT STANDSTILL RELAY CONTACTS AT D & E SHOULD BE CLOSED.

XV. SEE LEVER DIAGRAM FOR RUNNING CHECKS OF SPEED REGULATION.

11
12P-DFB2
FG-W
12P-FS2

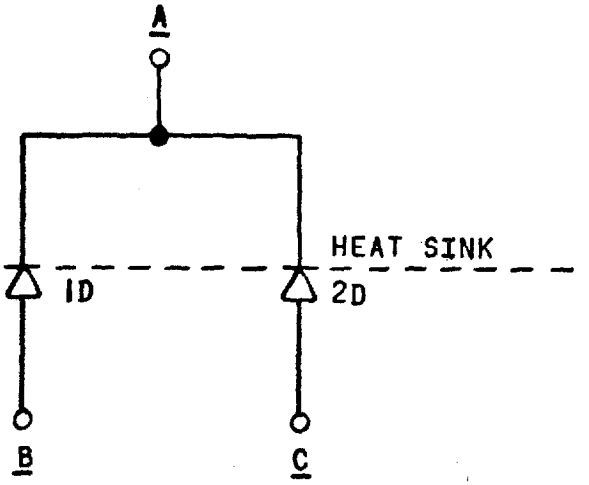
PRINTS TO

GENERAL ELECTRIC

44A306392
CONT ON SHEET SH NO.

REV NO.	TITLE
44A306392	RECTIFIER 1 REC
CONT ON SHEET SH NO.	FIRST MADE FOR 44B317827-G01

REVISIONS



11
FG-W
12F-DFR2

PRINTS TO

MADE BY	APPROVALS		
Eleanor Ellison 4/24/73	R.G. Sadler	DRIVE SYSTEMS PROD. DEPT OR DEPT.	44A306392
ISSUED D. Moyer April 26, 1973	R.J. Culver	WAYNESBORO, VA LOCATION	CONT ON SHEET SH NO.

FF-803-WF (8-71)
PRINTED IN U.S.A.

CODE IDENT NO. 3

3S7513TC300A18 EHC BOILER FD PUMP

QTY	PART NUMBER	DESCRIPTION
2	3S7513KF300A5	LINE POWER PANEL
1	3S7513KF306A9	METER PANEL
1	3S7513KF306A11	METER PANEL ASM
1	3S7513KF307A16	RELAY PANEL
5	44B3317761-G01	PWB VOLT COMPARATR
1	44D236389-G02	PWB SERVO AMPLI
5	44D381585-G01	PWB DETECTOR IND
9	44D381589-G01	PWB OPER AMPLIFIER
1	44B371594-G01	PWB 3K HZ 7-OSC
1	44D236379-G01	PWB ANALOG OUTPUT
2	44D236379-G03	PWB ANALOG OUTPUT
2	44D331918-G01	PWB CL MOTOR DRIVE
3	44D236365-G02	PWB 2FREQ VOLT CNV
1	44B331743-G01	PWB SIG CONV 3SC
1	44B331746-G03	PWB BOILR CON INTF
3	44D236370-G01	PWB DIODE BOARD
6	44B317827-G01	PWB SERIES REG 1
6	44D236371-G01	PWB SERIES REG 2
1	44D236373-G02	PWB OVERSPD INTEG
1	44D236372-G02	PWB SIG LO VAL
1	44D236388-G01	PWB COMP*REG ICR
2	44D315178-G01	PWB DIFF TRANS DR
1	44D236378-G01	PWB REF LOW VALUE
1	44B337328-G01	PWB LPF PWR SUPPLY
1	44B337327-G01	PWB LOW SPD SW LSS
3	44C319112-G01	RACK ASM PWB
2	K9774719L50	CAP 500UF 50VDC

3S7513KF300A5 LINE POWER PANEL

QTY	PART NUMBER	DESCRIPTION
2	44B334726-G01	SUPPLY TRANSF
18	44A316724-M19	CAP 1900UF 75VDC
6	44A333453-D36	RES 360HM 100W
6	44B214518-G01	REACTOR

3S7513KF306A9 METER PANEL

QTY	PART NUMBER	DESCRIPTION
5	44A317759-018	SWITCH
20	GE387	LAMP

REVISIONS

LTR	DATE
	7/11
	7/26

3S7513KF306A11 METER PANEL ASM

QTY	PART NUMBER	DESCRIPTION
1	128A4057H200	RES .200 MEGO .5W
2	44A317759-018	SWITCH
2	44A317759-019	SWITCH
1	44A317759-012	SWITCH
3	44A317759-020	SWITCH
3	128A4057G301	RES 30.1K OHMS .5W
20	GE387	LAMP

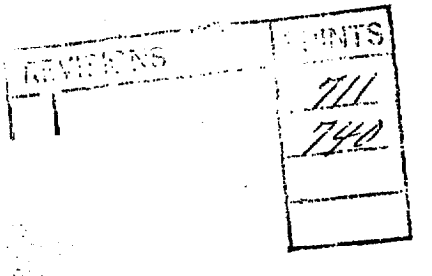
2

3S7513KF307A16 RELAY PANEL

QTY	PART NUMBER	DESCRIPTION
20	3S2791G138D4	MINIATURE RELAY
2	3SAA1343A2	RELAY
1	12HMA11B3	RELAY
1	44B312431-005	TIME DELAY RELAY
1	44A230536-001	SIGNAL FLASHER
34	44B232019-004	RECT 1A 400V
2	44A317994-G01	COMP BD ASM
4	3S2791G138D3	MINIATURE RELAY
2	K9774723K40	CAP 40UF 450VDC
2	44B314000-G01	D C REACTOR
2	K5908303G1	T B ASM
26	933B545P2	RELAY SOCKET
10	44B232011-001	RECT 200MA 200VDC

3S7513KL300A16 CONTROL PANEL ASM

QTY	PART NUMBER	DESCRIPTION
3	CR2940U207	CONTACT BLOCK
2	CR2940UA202C	PUSH BUTTON
1	44A317759-007	PUSHBUTTON SWITCH
4	44A317759-001	PUSHBUTTON SWITCH
3	44A317759-008	PUSHBUTTON SWITCH
1	440392721-G01	DIODE BOARD
12	GE387	LAMP
3	CR2940UA200C	OPERATOR
1	44A317759-058	ILLUMINATED PB SW



GENERAL ELECTRIC

44B300423 CONT ON SHEET SH NO

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS ±	DECIMALS ±	ANGLES ±
	✓			

REV NO 44B300423 CONT ON SHEET SH NO

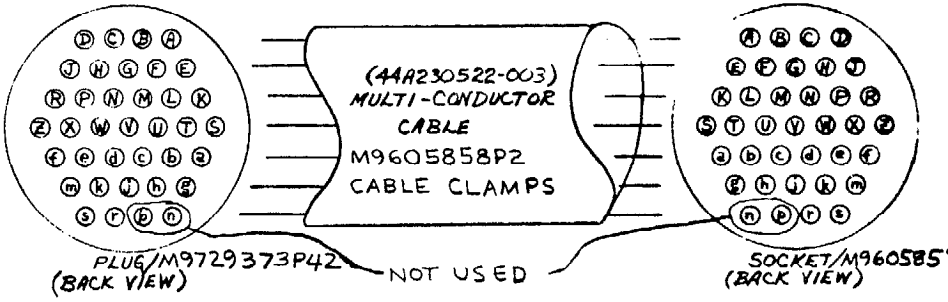
TITLE

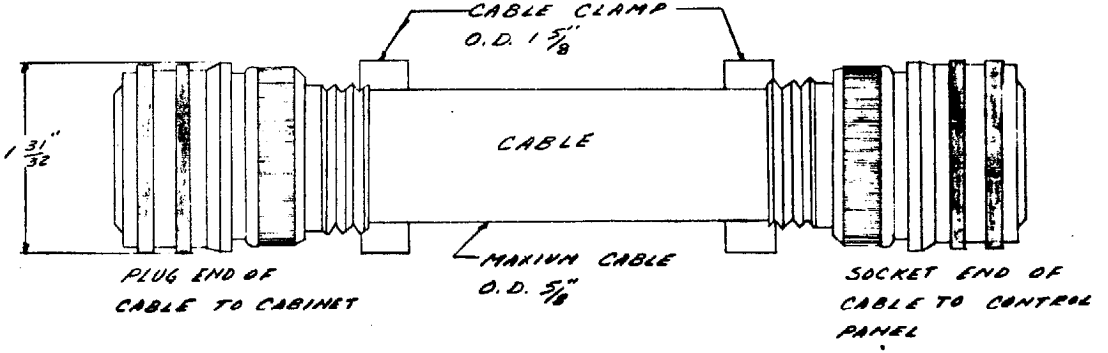
CABLE WIRING

FIRST MADE FOR 3S7513TC300A2

SIGNAL & LOW VOLTAGE

	PLUG TERMINAL	WIRE	SOCKET TERMINAL
SHIELDED CABLE	V	SHIELD	V
	M	WHITE/BLACK/VIOLET	M
	N	BLACK	N
	W	BROWN	W
SHIELDED CABLE	d	WHITE/BLACK/GRAY	d
	c	SHIELD	c
	U	RED	U
SHIELDED CABLE	T	WHITE/BROWN/RED	T
	L	SHIELD	L
	F	ORANGE	F
SHIELDED CABLE	G	WHITE/BROWN/ORANGE	G
	H	SHIELD	H
	P	WHITE	P
SHIELDED CABLE	X	WHITE/BLACK	X
	e	SHIELD	e
	k	WHITE/BROWN/YELLOW	k
SHIELDED CABLE	j	WHITE/BROWN/GREEN	j
	h	SHIELD	h
	b	WHITE/BROWN	b
SHIELDED CABLE	g	WHITE/RED	g
	a	SHIELD	a
	S	WHITE/ORANGE	S
	K	WHITE/YELLOW	K
	E	WHITE/GREEN	E
	A	WHITE/BLUE	A
	B	WHITE/VIOLET	B
	C	WHITE/GRAY	C
	D	WHITE/BLACK/BROWN	D
	J	WHITE/BLACK/RED	J
	R	WHITE/BLACK/ORANGE	R
	Z	WHITE/BLACK/YELLOW	Z
	f	WHITE/BLACK/GREEN	f
	m	WHITE/BLACK/BLUE	m
	s	WHITE/BROWN/BLUE	s
	r	WHITE/BROWN/VIOLET	r





NOTES:

1. USE PLASTIC SLEEVING OVER WIRE AND TERMINALS TO INSULATE CONNECTIONS FROM EACH OTHER.

REVISIONS		PRINTS TO	
1	Mod F AUG 19 1969		11
2	H. Tamilia	12P-0	EB
3	ADDED NOTE: 3Bx.13 D.K. McVee 1722	FG-W	

MADE BY R.M. Bukoshey ISSUED MAY 22 1968

APPROVALS R.M.B

SPECIALTY CONTROL DIV OR DEPT WAYNESBORO, VA. LOCATION

44B300423 CONT ON SHEET SH NO

FORM 000-F (10-66) PRINTED IN U.S.A.

44D307416

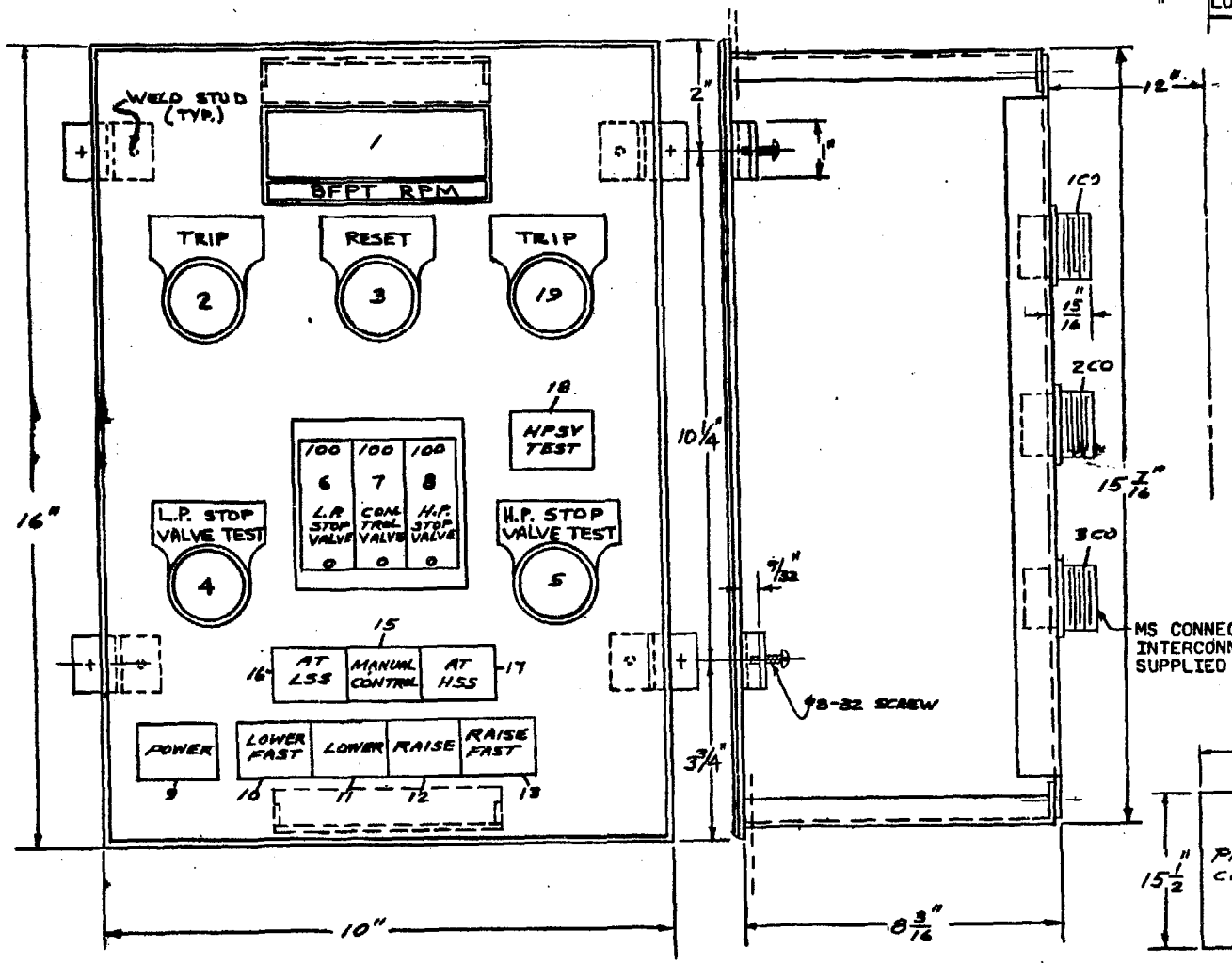
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

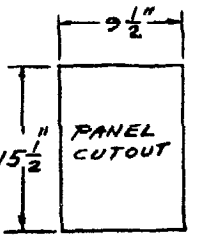
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

REV NO. | CONT ON SHEET | SH NO.

TITLE (OUTLINE)
FEEDPUMP TURBINE CONTROL PANEL

FIRST MADE FOR 3S7513KL300A16 TURBINE 153308
LOUISVILLE GAS & ELECTRIC, MILLCREEK #4
REQN. 091-41975





PARTS LIST

A. 1 DIGITAL TACHOMETER
B. 2,319 GE CR2940UA203C PUSHBUTTON WITH GUARD.
C. 4,5 GE CR2940UA202C PUSHBUTTON WITH GUARD.
D. 6, 7, 8 INSTRUMENTS, GE TYPE 185.
E. 9 ILLUMINATED PUSHBUTTON RED, POWER ON, CONDITIONS NORMAL, YELLOW (FLASHING) POWER ON, CONDITIONS ABNORMAL. DEPRESS TO TEST ALL LAMPS.
F. 10-13, PUSHBUTTONS. (WHITE)
G. 15, 16, 17 INDICATING LIGHTS. (WHITE)
I. 9-17, MASTER SPECIALTIES SERIES 12, VERTICAL BARRIER.
J. 18 - GREEN LIGHT "TEST PERMISSIBLE"
K. FINISH PURCHASER TO SUPPLY DARK. GREEN PORTER PAINT #23694
L. PURCHASER WILL MOUNT & SUPPLY ALL NAMEPLATES.

REVISIONS			PRINTS TO
1	T.L. Lee IRO	OCT 1 1976	711
2	R. GREEN DARK WAS LIGHT #23694 WAS 23693 W2 Wilson RO	FEB 3 1977	FG-1A1 12P-FS2

MADE BY W.L. Shipley ISSUED May 19 1976 MAY 20 1976
APPROVALS E.K. R.O. Luhr
DRIVE SYSTEMS PROD. DEPT. DIV. OR DEPT. WAYNESBORO, VA. LOCATION
44B309476 CONT ON SHEET SH NO.
3+1PL

FF-606-F M-7N PRINTED IN U.S.A.

GENERAL ELECTRIC

44B306144

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING: —

APPLIED PRACTICES	SURFACES	TOLERANCES IN MACHINE DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

REV NO.

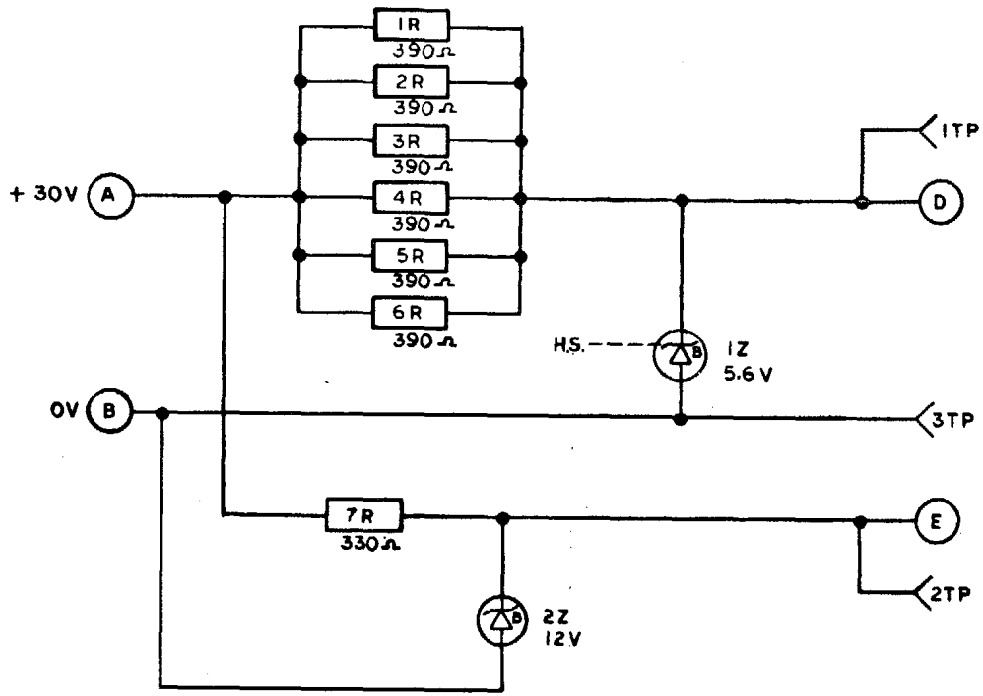
CONT ON SHEET SH NO.

TITLE

LSS PWR SUPPLY

FIRST MADE FOR

ELEM DIAGRAM



REVISIONS

PRINTS TO

11

12P-3PB

FG-W

MADE BY Guy Rouzie 9-11-72

ISSUED D.K. Mayer Dept. 12, 1972

APPROVED

DRIVE SYSTEMS PROD. DEPT. OF WAYNESBORO, VA. LOCATION

44B306144 CONT ON SHEET SH NO.

3

PP-502-F 10-70
PRINTED IN U.S.A.

GENERAL ELECTRIC

44B306148

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:— | APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

REV NO

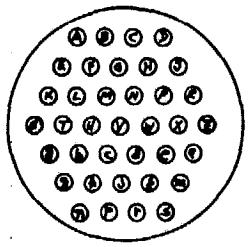
CONT ON SHEET SH NO

TITLE
POWER CABLE WIRING, ELECTRONIC BOILER FEED PUMP TRUBINE CONTROL
FIRST MADE FOR 3S7513TC 300A10
2CO, 3CO POWER CABLE

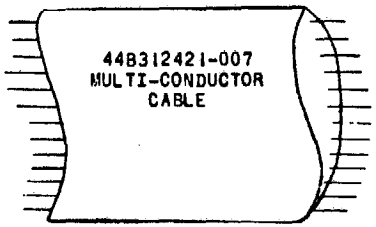
ASSEMBLY NOTES

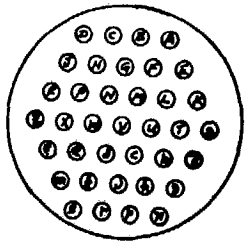
THE SPECIFIC COLOR OF WIRE CONNECTED TO A PIN IS NOT IMPORTANT, BUT THE SAME WIRE SHOULD BE CONNECTED TO THE SAME PIN AT THE OTHER CABLE END AND CONNECTOR.
EXAMPLE: IF A RED WIRE IS CONNECTED TO PIN "A" ON THE CONNECTOR PLUG, THE OTHER END OF THAT RED WIRE MUST BE CONNECTED TO PIN "A" OF THE SOCKET CONNECTOR AT THE OTHER END.

CONNECT SAME COLOR WIRE TO THESE PINS AT BOTH CABLE ENDS	DO NOT CONNECT ANY WIRES TO THESE PINS
A	B
C	D
T	G
E	L
F	M
	N
H X	P
J Z	R
K	S
	U
	V
	W

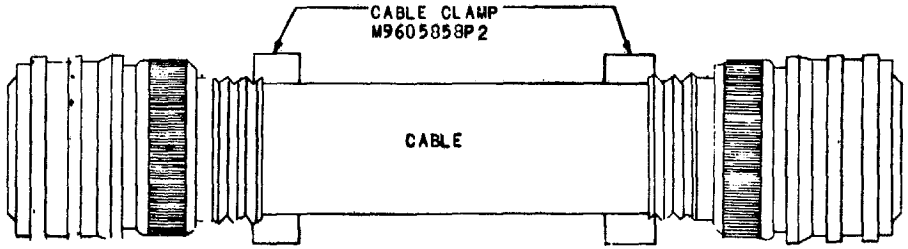


SOCKET
M9605859P18





PLUG
M9605859P42



SOCKET END OF
CABLE TO EHC REMOTE CONTROL PANEL.

PLUG END OF
CABLE TO EHC CABINET

REVISIONS

PRINTS TO
11
EG-W

MADE BY A. Templeton 10-18-72
BY S. Morgan Oct. 20, 1972
APPROVED J. Pulliver
COMM. & CONT. DEVICES DIV OR DEPT
WAYNESBORO, VA. LOCATION

44B306148
CONT ON SHEET SH NO

3

FP 809 F 12 69
PRINTED IN U.S.A.

GENERAL ELECTRIC

44B306197 CONT ON SHEET SH NO

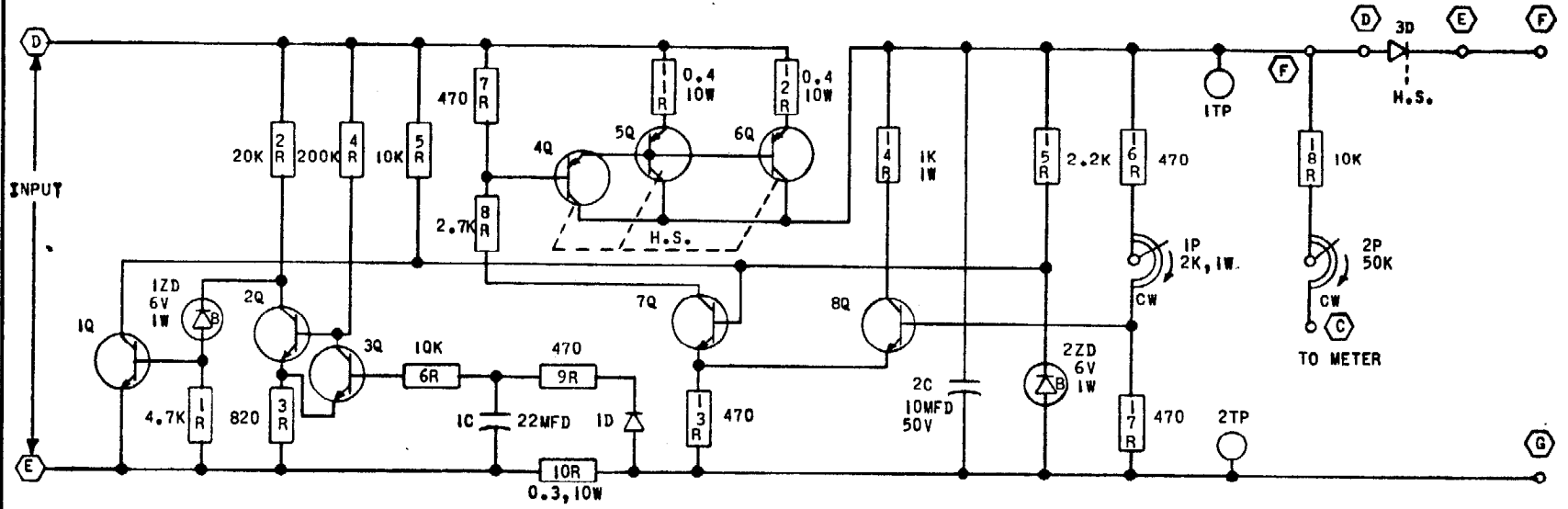
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

REV NO 44B306197 CONT ON SHEET SH NO

TITLE SERIES REGULATOR 1SR

FIRST MADE FOR 44B236371-G01



REVISIONS

PRINTS TO

11
FG-W
12C-D4B2

MADE BY Elmer Ellison 9/29/73

ISSUED BY Moyer April 26, 1973

APPROVALS R G Shipman, R J Allen

DRIVE SYSTEMS PROD. DEPT. DIV OR DEPT
WAYNESBORO, VA. LOCATION

44B306197 CONT ON SHEET SH NO

PF-808-F (9-72)
PRINTED IN U.S.A.

GENERAL ELECTRIC

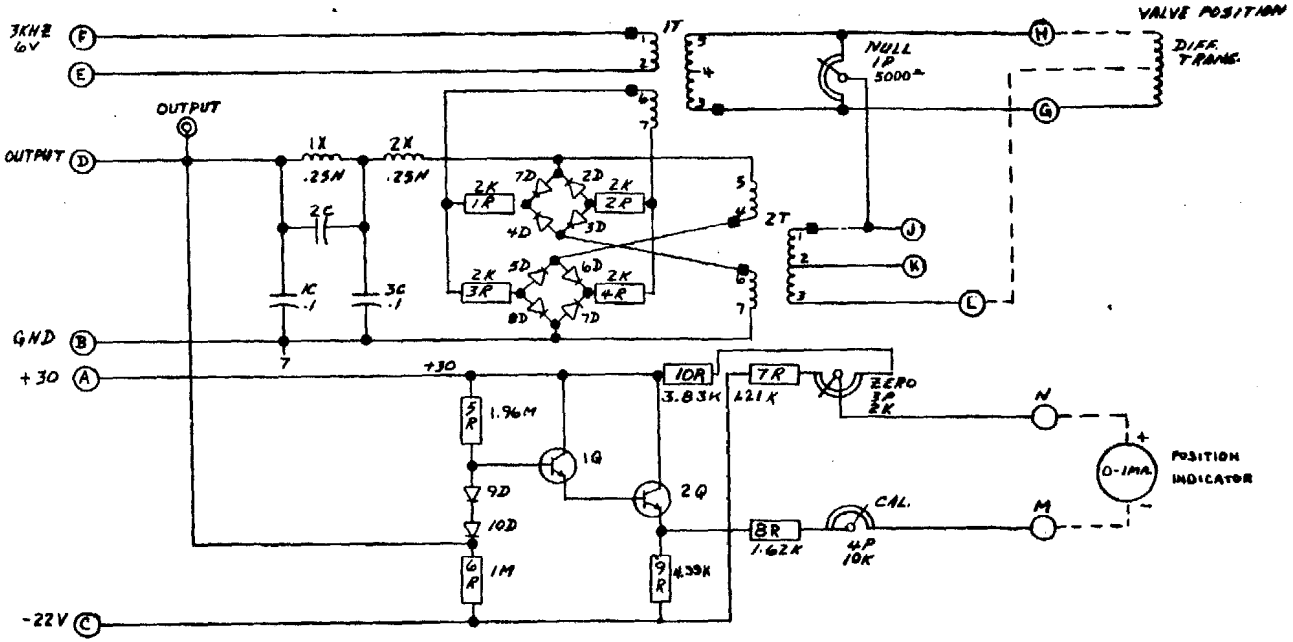
44C300609

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE: DETECTOR INDICATOR 1DI

FIRST MADE FOR 44D381585-GD1



NOTES:

1. CONNECTIONS

A	+30V	G, H, J, K, L, M	DIFFERENTIAL TRANSFORMER CONNECTIONS
B	GROUND		
C	-22V		
D	OUTPUT		
E, F	6V, 3 KHZ		
N			OUTPUT METER 0-1MA

44C300609 REV NO. 0

REVISIONS		PRINTS TO
1	R.M. Carr NOV 7 1968	H
2	R.M. Carr MAY 5 1969	FG-M
3	R Marr OCT 19 1969	12P-DFB
4	J.J. DEC 18 1969	
5	J.K. MAY 18 1970	
6	[illegible] OCT 19 1970	

DRAWN	E. M. Templeton JAN 26 1968	APPROVALS	SPECIALTY CONTROL DIV. OR DEPT.	44C300609 SHT. NO.
CHECKED	APR 17 1968	Abe	WAYNESBORO, VA LOCATION	CONT. ON SHEET

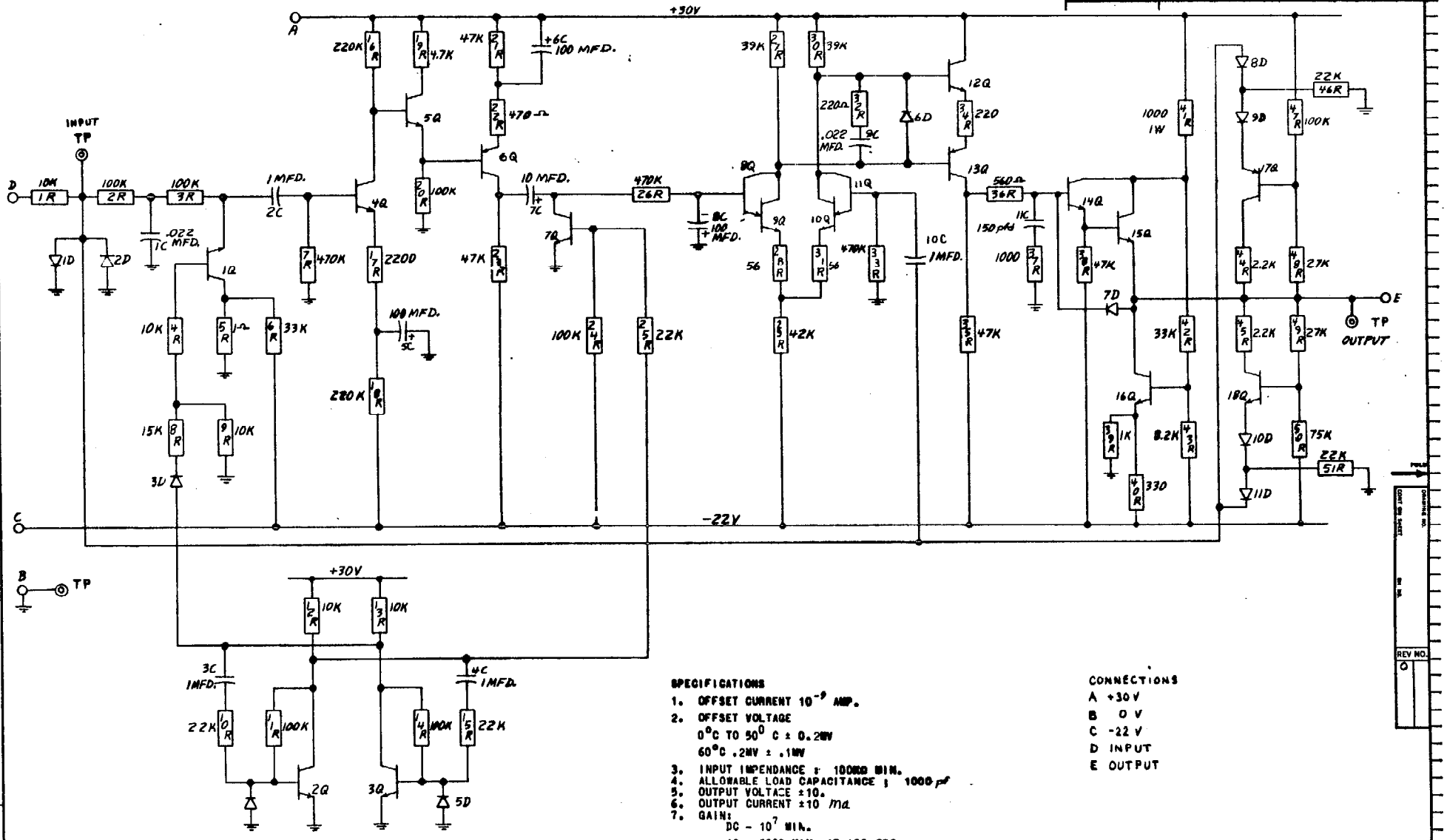
GENERAL ELECTRIC

44C300610 1ØA

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS		
	✓	FRACTIONS ±	DECIMALS ±	ANGLES ±

TITLE
CHOPPER STABILIZED
OPERATIONAL AMPLIFIER
FIRST MADE FOR 44D381589-G01



SPECIFICATIONS

1. OFFSET CURRENT 10^{-9} AMP.
2. OFFSET VOLTAGE
   0°C TO 50° C ± 0.2MV
   60°C .2MV ± .1MV
3. INPUT IMPENDANCE : 100000 MIN.
4. ALLOWABLE LOAD CAPACITANCE : 1000 pf
5. OUTPUT VOLTAGE ±10.
6. OUTPUT CURRENT ±10 ma
7. GAIN:
   DC - 10^7 MIN.
   AC - 5000 MIN. AT 100 CPS
   CHARACTERISTIC: BREAK FREQUENCY - 250 CPS
   20 db PER DECADE TO CROSSOVER.
8. CHOPPER STABILIZER RECOVERY TIME: 1 MINUTE
9. POWER REQUIREMENTS: 30 VOLTS - 25 ma
   -22 VOLTS - 20 ma
10. AMPLIFIER PROTECTION:
    A. OUTPUT CURRENT LIMITED TO APPROX 10 ma
    B. OUTPUT LIMITED TO 10 VOLTS
    C. ALLOWABLE INPUT VOLTAGE BEFORE DAMAGE - 70 VOLTS.

CONNECTIONS

A +30 V
B 0 V
C -22 V
D INPUT
E OUTPUT

REVISIONS		PRINTS TO
1	M. Nov 17, 1967	11
2	E.A. Allen Aug 9, 1968	12P-DFB

MADE BY	DRAWING		
J. C. Glenn JAN 22 1968	18C	SPECIALTY CONTROL DEPT.	44C300610
ISSUED MAR 12 1968		WAYNESBORO, VA. LOCATION	SHEET NO.

REV NO. 0

GENERAL ELECTRIC

44C301216

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

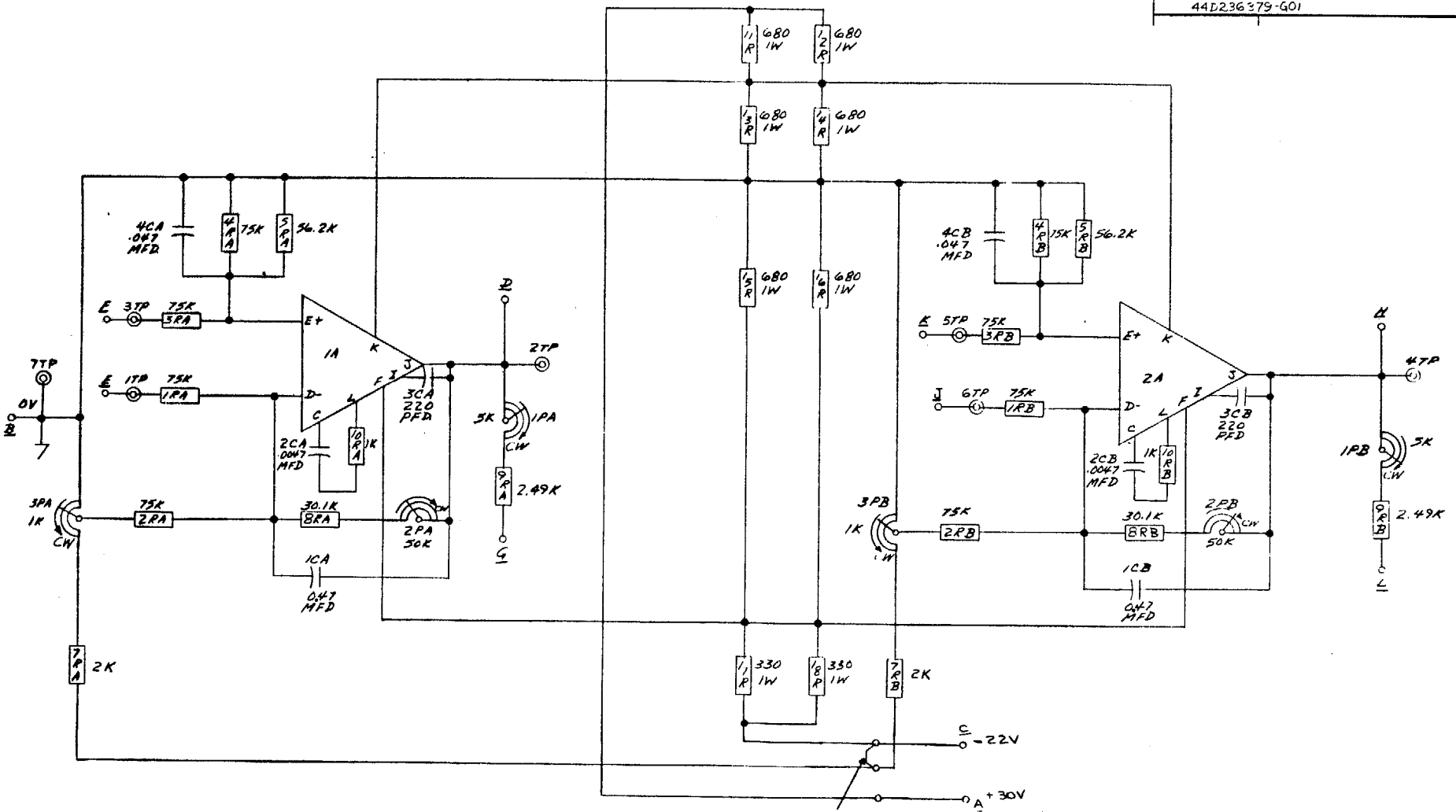
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE

ANALOG OUTPUT 1AØ2

FIRST MADE FOR

44D236379-G01



JUMPER POINTS PROVIDED ON THE PRINTED CIRCUIT BOARD SO RESISTORS 7RA & 7RB MAY BE CONNECTED TO EITHER +30V OR -22V.

CONNECTIONS

A = +30V
B = 0V
C = -22V
D = 1A VOLTAGE OUTPUT
E = 1A INPUT
G = 1A CURRENT OUTPUT
H = 2A VOLTAGE OUTPUT
J = 2A INPUT
L = 2A CURRENT OUTPUT

44C301216 SH NO

REV NO

REVISIONS		PRINTS TO
1	SEP 30 1968	
2	OCT 9 1969	
3	FEB 3 1970	
4	Nov 17, 1971	

ISSUED	APPROVALS		
R. W. Templeton SEP 15 1968			44C301216
SEP 24 1968	E182	WAYNE	

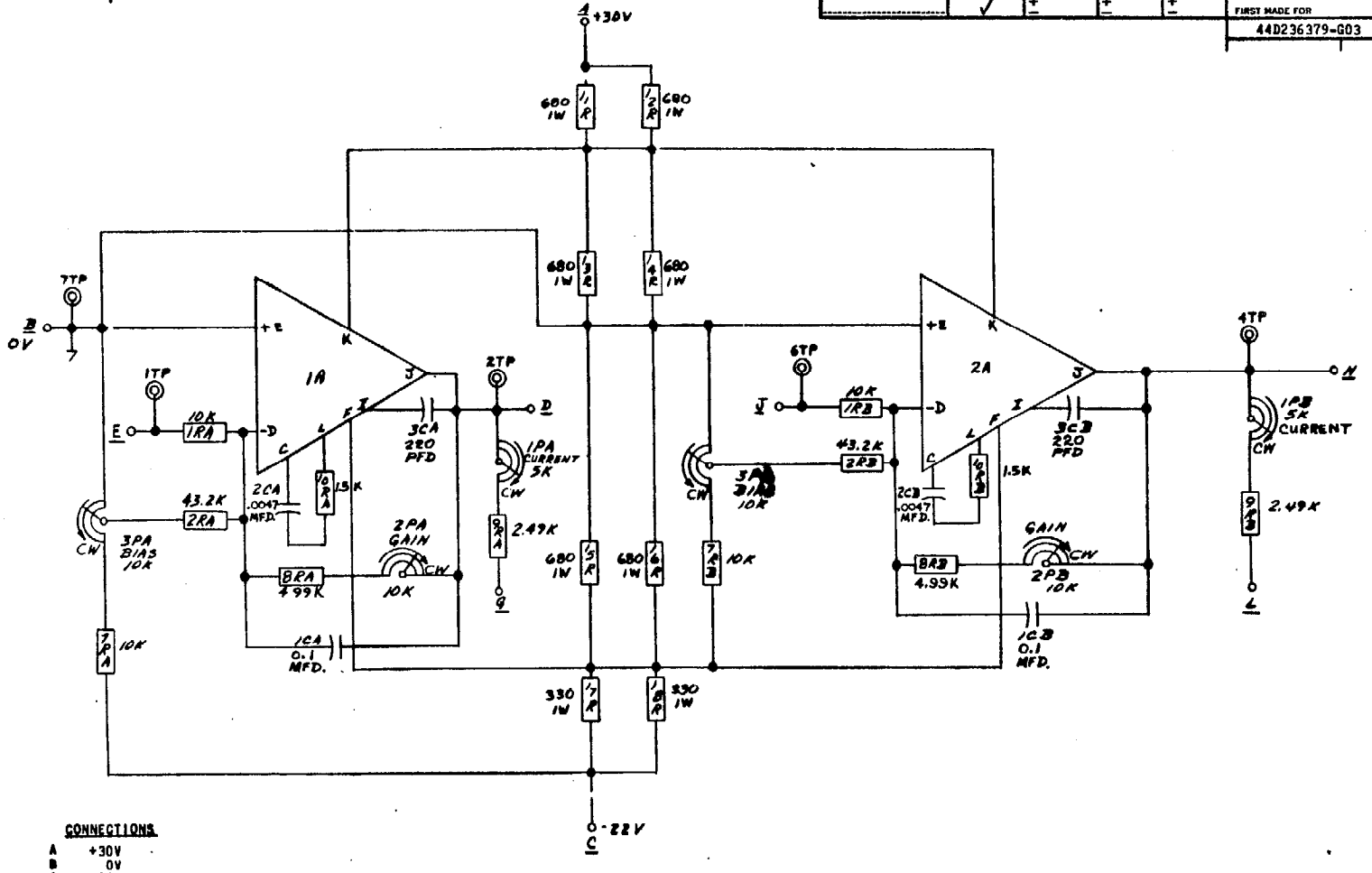
GENERAL ELECTRIC

44C301622

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ANALOG OUTPUT 3A#2

FIRST MADE FOR
44D236379-G03



CONNECTIONS

A	+30V
B	0V
C	-22V
D	1A VOLTAGE OUTPUT
E	1A INPUT
G	1A CURRENT OUTPUT
H	2A VOLTAGE OUTPUT
J	2A INPUT
L	2A CURRENT OUTPUT

44C301622
REV NO.

REVISIONS

1	[illegible]	
2	[illegible]	
3	[illegible]	

PRINTS TO: 11

NEW Templeton SEP 17 1968
ISSUED SEP 19 1968
APPROVALS MBC
SPECIALTY CONTROL DEPT WAYNESBORO, VA LOCATION
44C301622 CONT ON SHEET SH NO

GENERAL ELECTRIC

44C301682

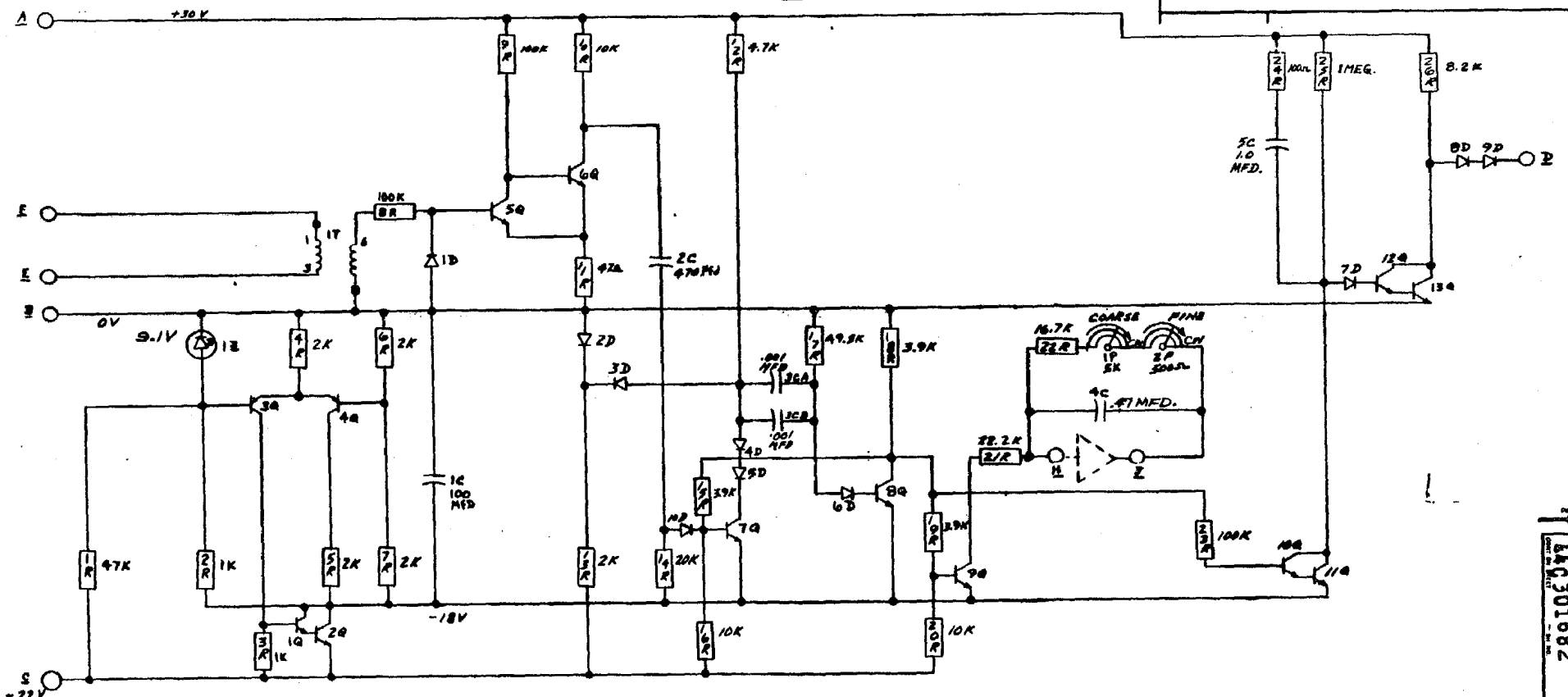
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE

FREQUENCY-VOLTAGE CONVERTER (2FVC)

FIRST MADE FOR 44D236365-G02



REVISIONS			PRINTS TO
1		Nov 17 1961	
2	C.H. Card	Dec 6 1961	12P-DFB
3	C.O. Clemens	Dec 7 1961	FG-W

DRAWN	R. W. Templeton NOV 15 1961	APPROVED	SPECIALTY CONTROL DIV OR DEPT	44C301682
ENGR	OCT 9 1961		WAYNESBORO, PA LOCATION	CONT ON SHEET

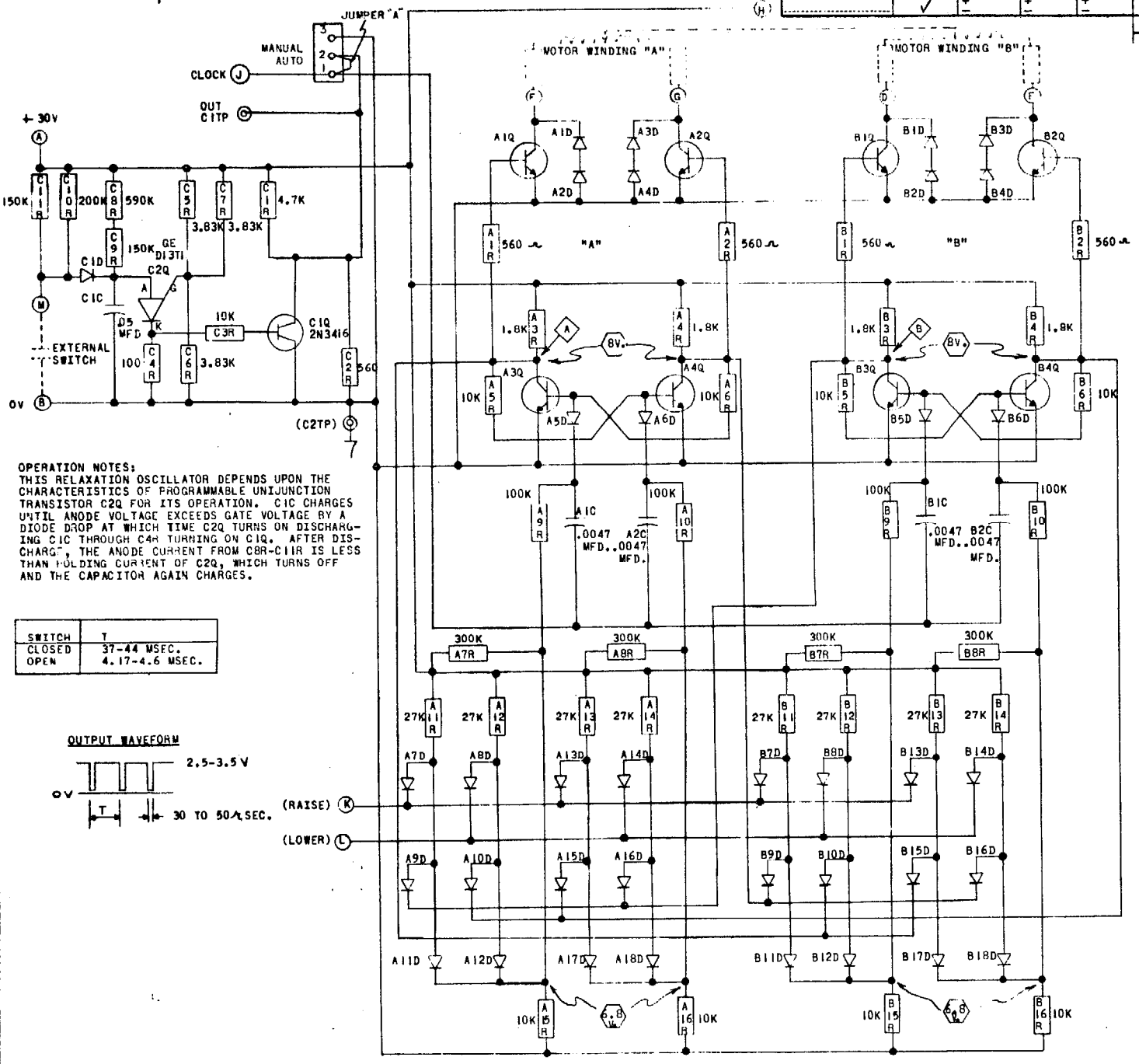
44C301682 REV NO.

GENERAL ELECTRIC 44C303567

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING — APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE: CLOCK & STEPPING MOTOR DRIVE

FIRST MADE FOR

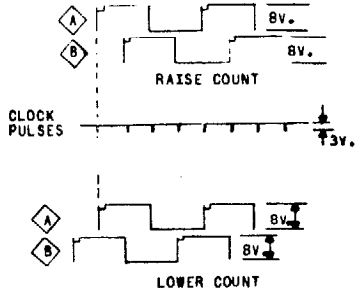


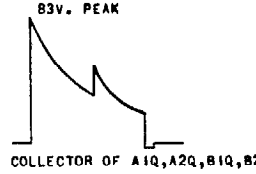
1. THIS BOARD CONSISTS OF A RELAXATION OSCILLATOR CLOCK IDENTIFIED BY "C" COMPONENTS AND ALSO TWO FLIP-FLOPS "A" AND "B" WITH DIODE LOGIC TO FORM A REVERSIBLE COUNTER. OUTPUT OF THE FLIP-FLOPS FEEDS POWER AMPLIFIERS A1Q, A2Q, B1Q & B2Q. THE STEPPING MOTOR WINDINGS ARE CONNECTED DIRECTLY TO THE COLLECTORS OF THESE FOUR TRANSISTORS.
2. THE CLOCK PULSES ARE THREE VOLTS AMPLITUDE WITH FAST FALL TIME. NORMAL CLOCK FREQUENCY IS 240 PULSES PER SECOND.
3. RAISE COUNTING RESULTS FROM 0 TO 1 VOLT ON TERMINAL (K) & +6 VOLTS OR MORE ON TERMINAL (L). REVERSING THE VOLTAGE ON TERMINALS (K) & (L) RESULTS IN LOWER COUNTING. A POSITIVE VOLTAGE ON TERMINAL (K) & (L) WILL INHIBIT COUNTING. 0 TO 1 VOLT ON BOTH (K) & (L) IS NOT PERMISSIBLE.
4. CONNECT JUMPER "A" TO 1 & 2 FOR MANUAL SPEED REFERENCE; CONNECT JUMPER "A" TO 2 & 3 FOR AUTO SPEED REFERENCE.

OPERATION NOTES:
THIS RELAXATION OSCILLATOR DEPENDS UPON THE CHARACTERISTICS OF PROGRAMMABLE UNIJUNCTION TRANSISTOR C2Q FOR ITS OPERATION. C1C CHARGES UNTIL ANODE VOLTAGE EXCEEDS GATE VOLTAGE BY A DIODE DROP AT WHICH TIME C2Q TURNS ON DISCHARGING C1C THROUGH C4R TURNING ON C1Q. AFTER DISCHARGE, THE ANODE CURRENT FROM C8R-C1IR IS LESS THAN HOLDING CURRENT OF C2Q, WHICH TURNS OFF AND THE CAPACITOR AGAIN CHARGES.

SWITCH	T
CLOSED	37-44 MSEC.
OPEN	4.17-4.6 MSEC.

OUTPUT WAVEFORM





REVISIONS			PRINTS TO
1	SR	MAR. 9, 1970	11
2		July 24, 1971	12P-DF B²
3		Nov. 17, 1971	FG-W

MADE BY ISSUED MAR 3 1970 — APPROVALS MBC — COMM. & CONT. DEVICES, WAYNESBORO, VA. — 44C303567

REV NO.

GENERAL ELECTRIC

44C304444

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
SERVO AMPLIFIER (2SA)
FIRST MADE FOR 44D236389-G02
357513TC300

+30V
A
34.8K
6R
2.4K 1R
2.4K 2R
750Ω 1W 26R
9R 2K
13R 470Ω 1W
470Ω 1W 14R
3Z 12V
6C .022 MFD
IN752A 6V 12Z 22Z
BIAS 1P 10K CW
34.8K 3R
15K 36R
1D 2D
D- K E+ J C L F I 1A
10K 8R
1Q 2N3416
3D
3Q 2N3054
4D 5D 6D
20Ω 35R
DEMAND E 4R 43.2K
LIFT D 5R 100K
2.2M 28R .001MFD 1C
4Z 12V 10R 10K
7D 20Ω 36R
2Q 2N3416
4Q 2N3741
330Ω 1W 35R 10K 1R 2K 2R
270Ω 1W 7R 270Ω 1W 8R
GAIN V
L
F
H
I
J
K
24R 806Ω
2P GAIN 1K CW
20Ω 21R
806Ω
22R 100Ω
23R
1K 1W 9R
1K 1W 20R
B 0V
C -22V
G +30V
180Ω 21R
1K (STX)
180Ω 27R
2K (CSC)
M
N
8D 9D
(A) +30V
15K 31R
2M 32R
56K 25R
10D 11D
D- K E+ J C L F I 2A
33R 3.3K
5Q 2N3416
12D

SIMPLIFIED SCHEMATIC
DEMAND E
LIFT D
BIAS
M N CSC V
G (STX)
L T F
I H
GAIN J
K

2SA
BIAS ⊘
GAIN ⊘
LIFT ◎
DEMAND ◎
V ◎
I ◎

NOTE
1. NOMINAL OUTPUT F-H ± 10 VOLTS; ±70 MA.
2. CLOSE SIGNAL CHECK RELAY 2K PICKS UP WHEN "V" VOLTAGE IS LESS THAN -5 VOLTS

REVISIONS
1 ea Cleaver JUN 17 1971

PRINTS TO
12P DP8
FG-W

MADE BY ea Cleaver 2/1/71
ISSUED MAR 26 1971
APPROVALS
COMM. & CONT. DEVICES DIV. OR DEPT. WAYNESBORO, VA. LOCATION
44C304444 CONT ON SHEET

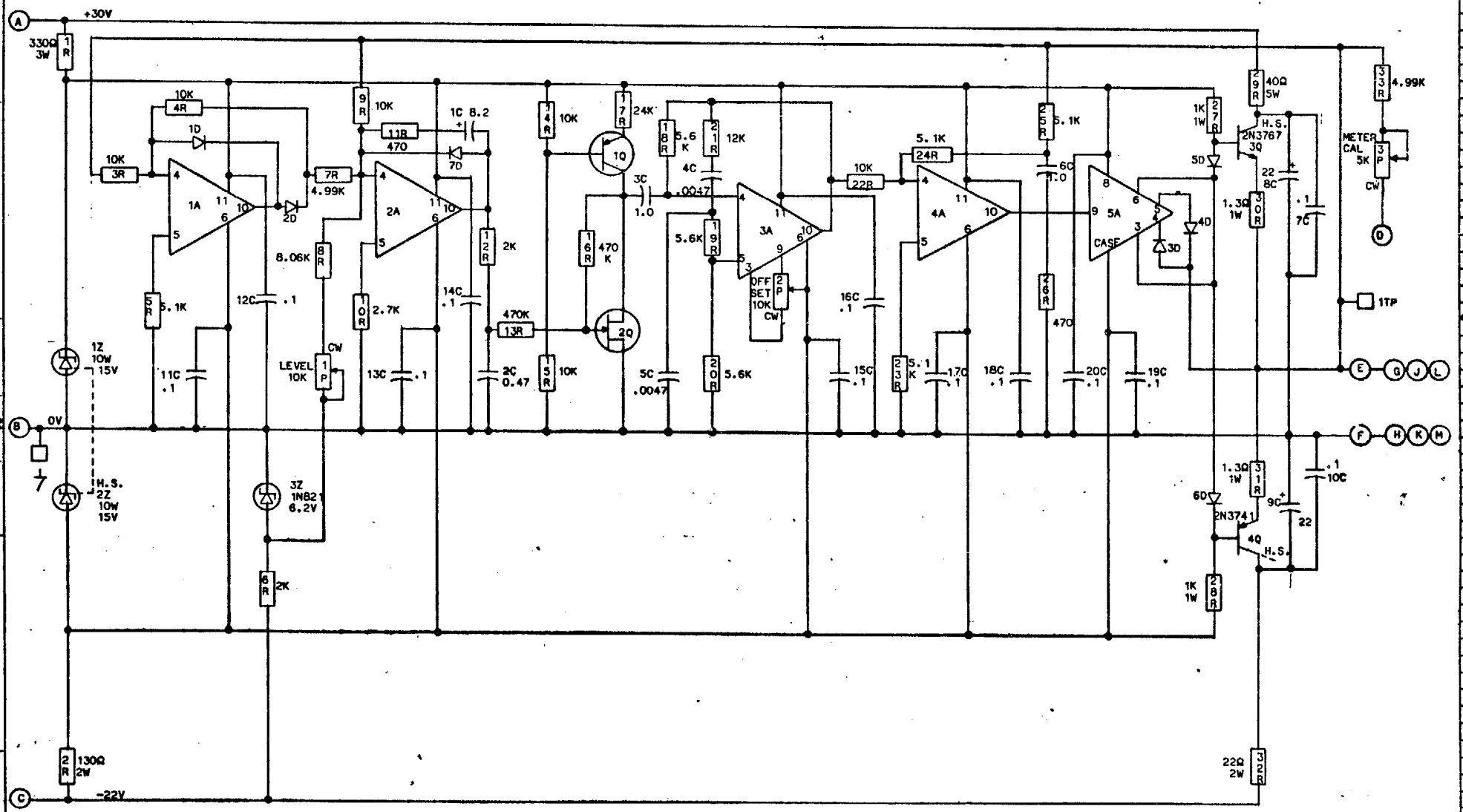
DRAWING NO. 44C304444 CODE ID NO. SHEET DO NO.
REV NO.

GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE 3KHZ OSCILLATOR

FIRST MADE FOR EHC 7-OSC



ALL CAPACITORS IN MFD UNLESS OTHERWISE NOTED.

FREQUENCY: 3000HZ±300HZ
AMPLITUDE: 6.00VRMS (17V PEAK TO PEAK)
DC OFFSET: ADJUSTABLE TO ZERO
MINIMUM LOAD IMPEDANCE: 25Ω
AMPLITUDE DRIFT: 0.1% FIFTEEN HOURS

REVISIONS

PRINTS TO 7/1 FG-W

MADE BY ISSUED 5/5/76 FEB 11 1977

APPROVED

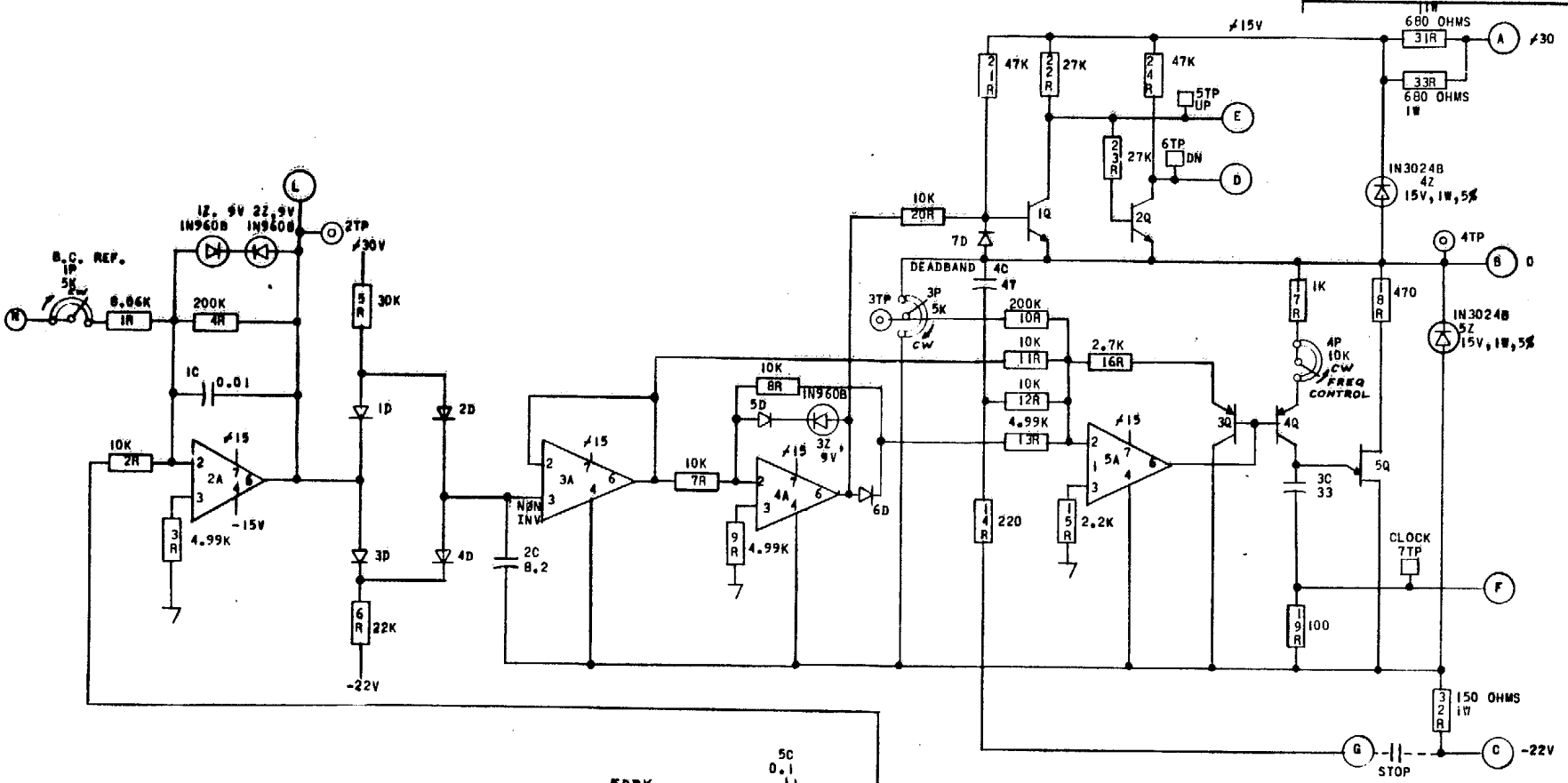
DRIVE SYSTEMS PROD. DEPT DIV OR DEPT WAYNESBORO, VA LOCATION

44C321478 CONT ON SHEET DWG NO.

1 of 1 PL

GENERAL ELECTRIC

TITLE: EHC SIGNAL CONVERTER



REV	DESCRIPTION	DATE	APPROVED
1		Aug 19, 1971	
2		Sept 20, 1971	
3		Nov 5, 1971	

COMM. & CONT. DEVICES
WAYNESBORO, VA.

44C304898 FNL

1

GENERAL ELECTRIC

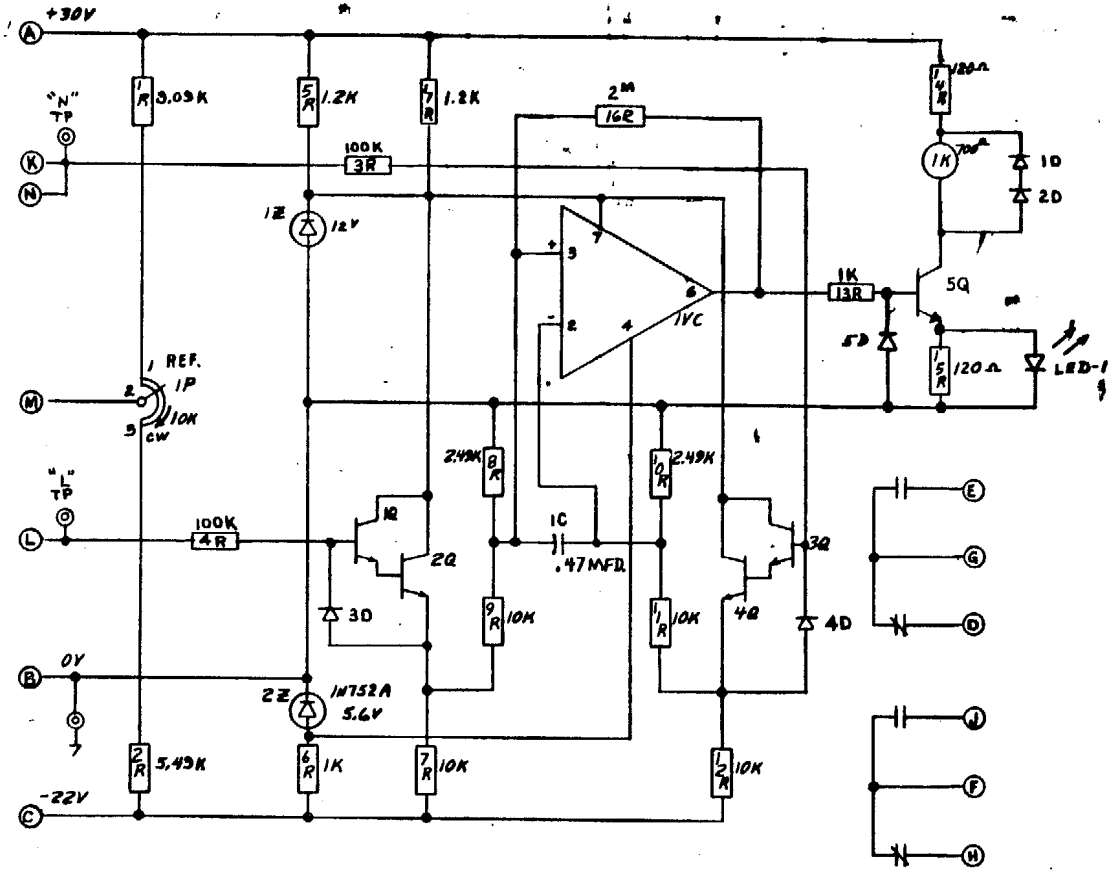
44C305259

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
VOLTAGE COMPARATOR
1VC
FIRST MATIE FOR 44B331761-G01

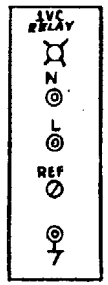
CONNECTIONS

A	+30V	N, K, L, M: INPUT SEE NOTE 2
B	0 V	
C	-22V	
D		
E		
F		
G		
H		
J		



NOTES:

1. IN NORMAL OPERATION THE RELAY IS ENERGIZED. IT CAN BE ADJUSTED TO DROPOUT ON EITHER INCREASING OR DECREASING SIGNALS BETWEEN PLUS AND MINUS 10 VOLTS.
2. DROPOUT ON — INPUT — JUMPER — REF. TP
   INCREASING SIGNAL — N — L-M — L
   DECREASING SIGNAL — L — N-M — N
   DROPOUT N>L
3. RELAY CONTACTS RATED 2A, 28VDC.
4. RELAY ENERGIZED WHEN LIGHT IS ON.



FACEPLATE

44C305259 REV. NO.

REVISIONS	PRINTS TO
1 E.O. Comm Aug 7, 1973	11
	12P DM
	FG-IN

DRAWN BY E. Roth Dec 8, 1971	CHECKED BY R. Weber 12/9/71	SPECIALTY WAYNESBORO, VA.	DIV. OR DEPT. LOCATION	44C305259 CONT ON SHEET

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE: ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A18 09-1-41975
ELEMENTARY DIAGRAM LOUISVILLE GAS & ELECTRIC
MILL CREEK #4

44C321492

PRINTED CIRCUIT BOARD LOCATIONS

RACK 1

LP STOP VALVE		HP STOP VALVE		CONTROL VALVE		SPEED SIGNAL		SPEED SIGNAL		OVERSPEED TEST		REF. LVG			LOCAL	MANUAL REFERENCE		AUTO REFERENCE						
1DI		1DI		1AØ2		3AØ2		3AØ2		1ØS1		1ØA	1ØA	1ØA	2LVG	1VC	CL & SMD	1DI	7BCI		3SC		CL SMD	1DI
X	2	X	4	X	6	X	8	X	10	X	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28

RACK 2

PRI SPEED			SEC. SPEED			SIGNAL LVG				REGULATION & COMPENSATION		SERVO VALVE LOOP				3KHZ OSC	SPEED SENSING 10% 105% 110%			200 RPM	SPEED INDICATION AND SWITCHING			LOW SPEED SW.		LSS PWR SUP	
1ØA	2FVC		1ØA	2FVC		1ØA	1ØA	1LVG		1ØA	1CR	2SA		1DI		60SC	1VC	1VC	1VC	1VC	1ØA	2FVC		LSS		LPS	
1	2	3	4	X	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28

RACK 3

				30 VOLT SUPPLY LINE A				LINE B				22 VOLT SUPPLY LINE A				LINE B				24 VOLT SUPPLY LINE A				LINE B			
1DB	1DB	1DB		1REC		1SR		1SR		1REC		1REC		1SR		1SR		1REC		1REC		1SR		1SR		1REC	
1	2	3	X	X	6	X	8	X	10	X	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28

FRONT VIEW

REVISIONS

PRINTS TO
11
12P-FS2
FG-W

ISSUED MAY 20 1976 | APPROVALS | DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA. | 44C321492 | CONT ON SHEET 1/4 | REV NO. 1

REV NO.

4 2

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

APPLIED PRACTICES	SURFACES	TOLERANCES OR MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE

ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A18 091-11975

ELEMENTARY DIAGRAM LOUISVILLE GAS & ELECTRIC

MILL CREEK #4

PRINTED CIRCUIT BOARD AND SUB. ASM. LIST

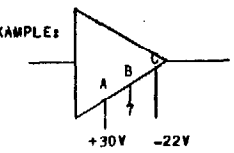
PRINTED CIRCUIT BOARD

SYMBOL	TITLE	ELEM. DIAG.	ORDER NO.	QTY.
1AØ2	ANALOG OUTPUT BOARD	44C301216	44D236379-G01	1
CL& SMD	CLOCK & STEPPING-MOTOR DRIVE	44C303567	44D331918-G01	2
1CR	COMPENSATION & REGULATION	44A301400	44D236388-G01	1
1DB	DIODE BOARD	44A300243	44D236370-G01	3
1DI	DETECTOR INDICATOR	44C300609	44D381585-G01	5
3AØ2	ANALOG OUTPUT	44C301622	44D236379-G03	2
1LVG	SIGNAL LOW VALUE GATE	44A304763	44D236372-G02	1
2FVC	FREQUENCY TO VOLTAGE CONVERTER	44C301682	44D236365-G02	3
1REC	SERIES REG. BD. #1	44A306392	44B317827-G01	6
1SR	SERIES REG. BD. #2	44B306197	44D236371-G01	6
1ØA	OPERATIONAL AMPLIFIER	44C300610	44D381589-G01	9
1ØSI	OVERSPEED INTEGRATOR	44A302692	44D236373-G02	1
2SA	SERVO AMPLIFIER	44C304445	44D236389-G02	1
3SC	SIGNAL CONVERTER	44C304898	44B331743-G01	1
LPS	LOW SPEED SWITCH POWER SUPPLY	44B306144	44B337328-G01	1
LSS	LOW SPEED SWITCH	44C306444	44B337327-G01	1
1VC	VOLTAGE COMPARATOR	44C305259	44B331761-G01	5
7BCI	BOILER CONTROL INTERFACE	44C306492	44B331746-G03	1
2LVG	REF. LOW VALUE GATE	44A300281	44D236378-G01	1
60SC	3KHZ. OSC. BOARD	44C304462	44B331717-G01	1

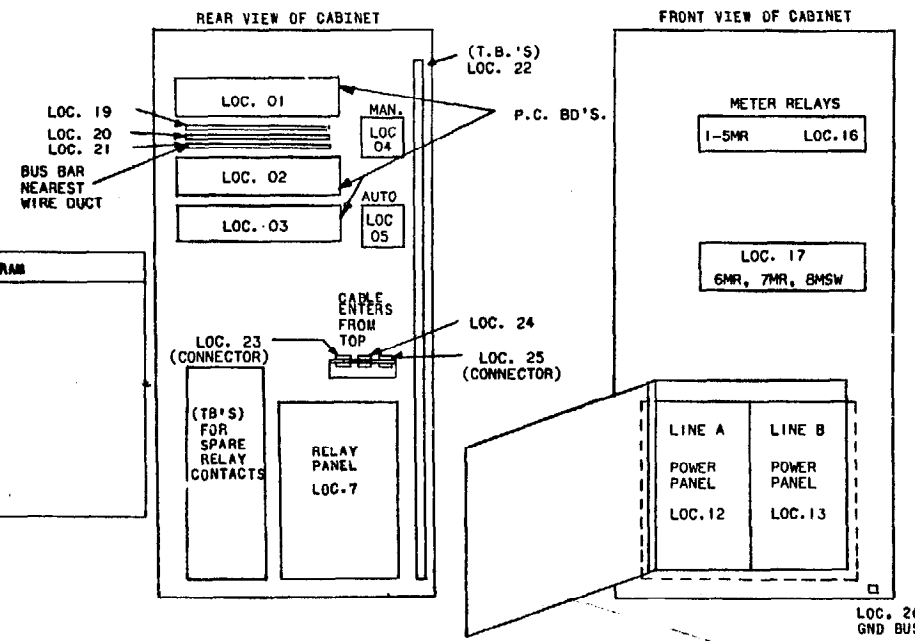
CUSTOMER NOTES:

1. ⏚ INDICATES ZERO REFERENCE POTENTIAL, NOT CASE GROUND.
2. ⏚ INDICATES STATION GROUND.
3. ALL OPERATIONAL AMPLIFIERS HAVE ADDITIONAL CONNECTIONS NOT SHOWN ELSEWHERE

EXAMPLE:



4. THE LOCATION OF THE PRINTED CIRCUIT BOARDS IS SHOWN BY THE NUMBERS APPEARING IN OR NEAR THE SQUARE WHICH REPRESENTS THE P.C. BD. EXAMPLE 0215 WOULD BE RACK #2 AND SLOT #15.
5. 1MR TO 7MR (SHEET 10) ARE METER RELAYS.
6. * INDICATES ITEMS NOT FURNISHED BY WAYNESBORO.
7. INDICATES SHIELDED CABLE.
8. INDICATES TWISTED WIRES.
9. INDICATES TERMINALS AT EHC CASE FOR CUSTOMER CONNECTION
10. INDICATES TERMINALS AT TURBINE
11. INDICATES SUB-PANEL & PC BOARD TERMINALS IN EHC CASE
12. CONTACTS FROM VOLTAGE COMPARATOR RELAY (1VC)



SUB. ASSEMBLIES — PANELS —

TITLE	ORDER NO.	CONNECTION DIAGRAM
CONTROL PANEL	3S7513KL300A13	44D308138
POWER PANEL	3S7513KF300A5	44C306526
METER PANEL	3S7513KF306A9	44C305834
METER PANEL	3S7513KF306A11	44C306262
RELAY PANEL	3S7513K307A16	44D308137
DIFF. XFMR. DRIVE	44D315178-G01	44C300882

REVISIONS

PRINTS TO

11
FG-W
12P-FS2

DRAWN BY W.J. Hyman DATE May 19 1976

APPROVALS R. Clifton MAY 20 1976

DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.

GENERAL ELECTRIC

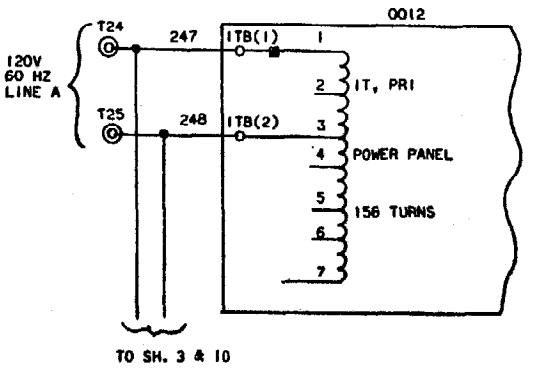
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS, DECIMALS, ANGLES

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

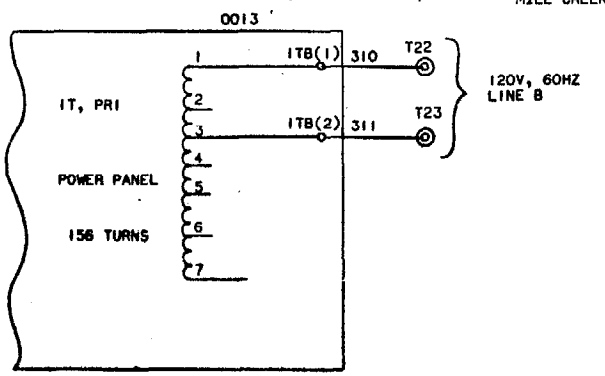
FIRST MADE FOR 3S7513TC300A18 091-41975

ELEMENTARY DIAGRAM — LOUISVILLE GAS & ELECTRIC
MILL CREEK #4

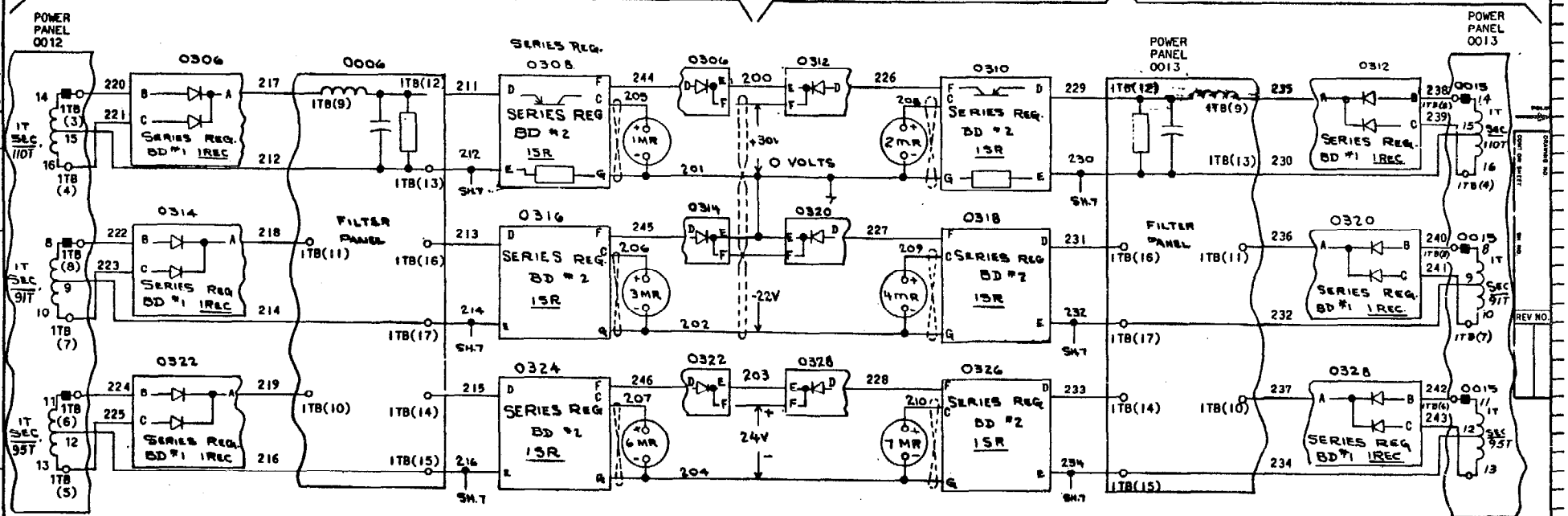
44C321492 SH 3 OF 3 SH NO 2



LINE A POWER SUPPLY



LINE B POWER SUPPLY



REVISIONS

PRINTS TO
11
12PGS
FG-W

DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.

44C321492 CONT ON SHEET 3 SH NO 2

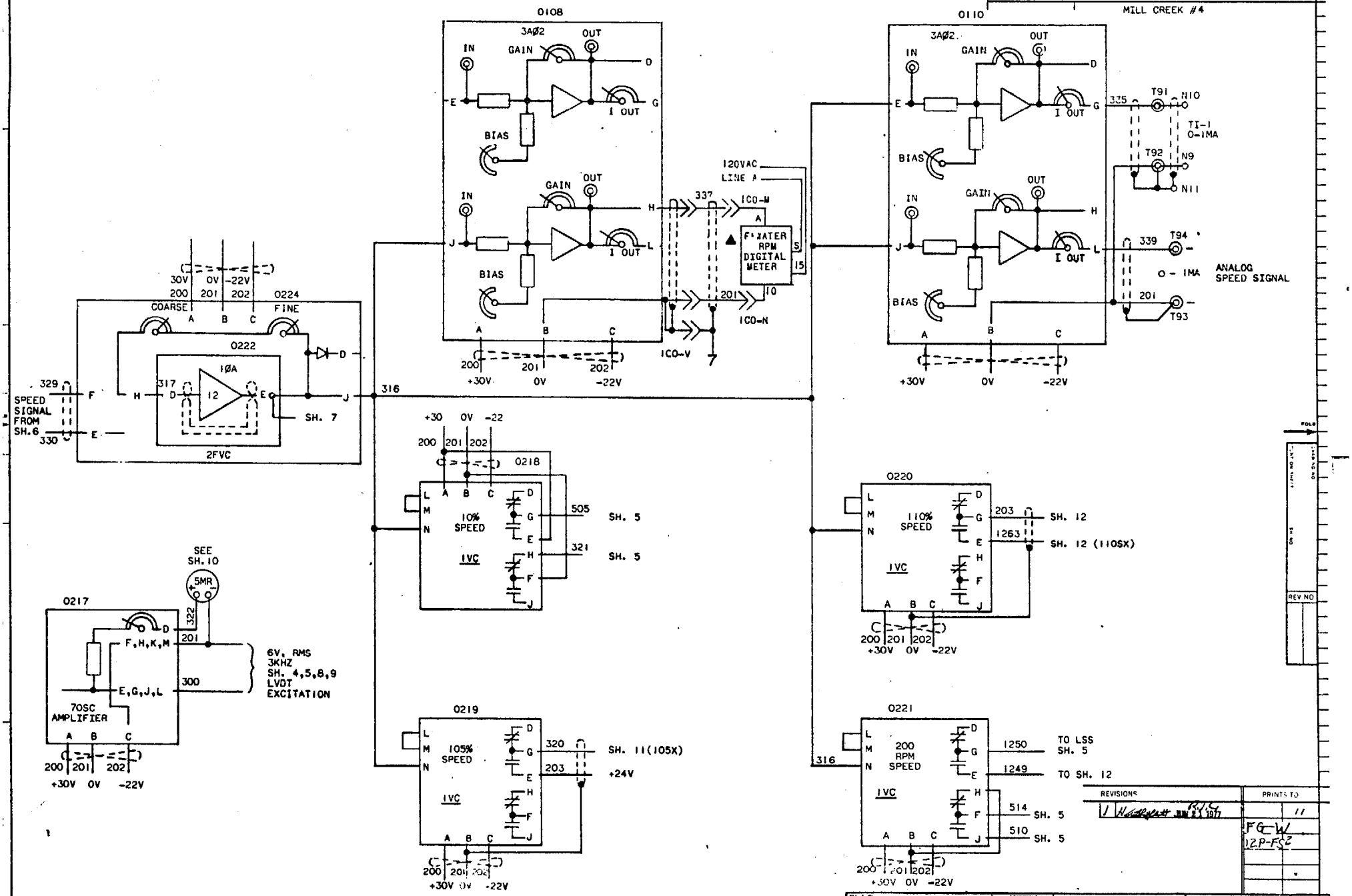
MAY 8 0 1978

GENERAL ELECTRIC 44C321492

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A18 091-41975

ELEMENTARY DIAGRAM LOUISVILLE GAS & ELECTRIC
MILL CREEK #4



▲ INDICATES ... MOUNTED ON REMOTE CONTROL ...

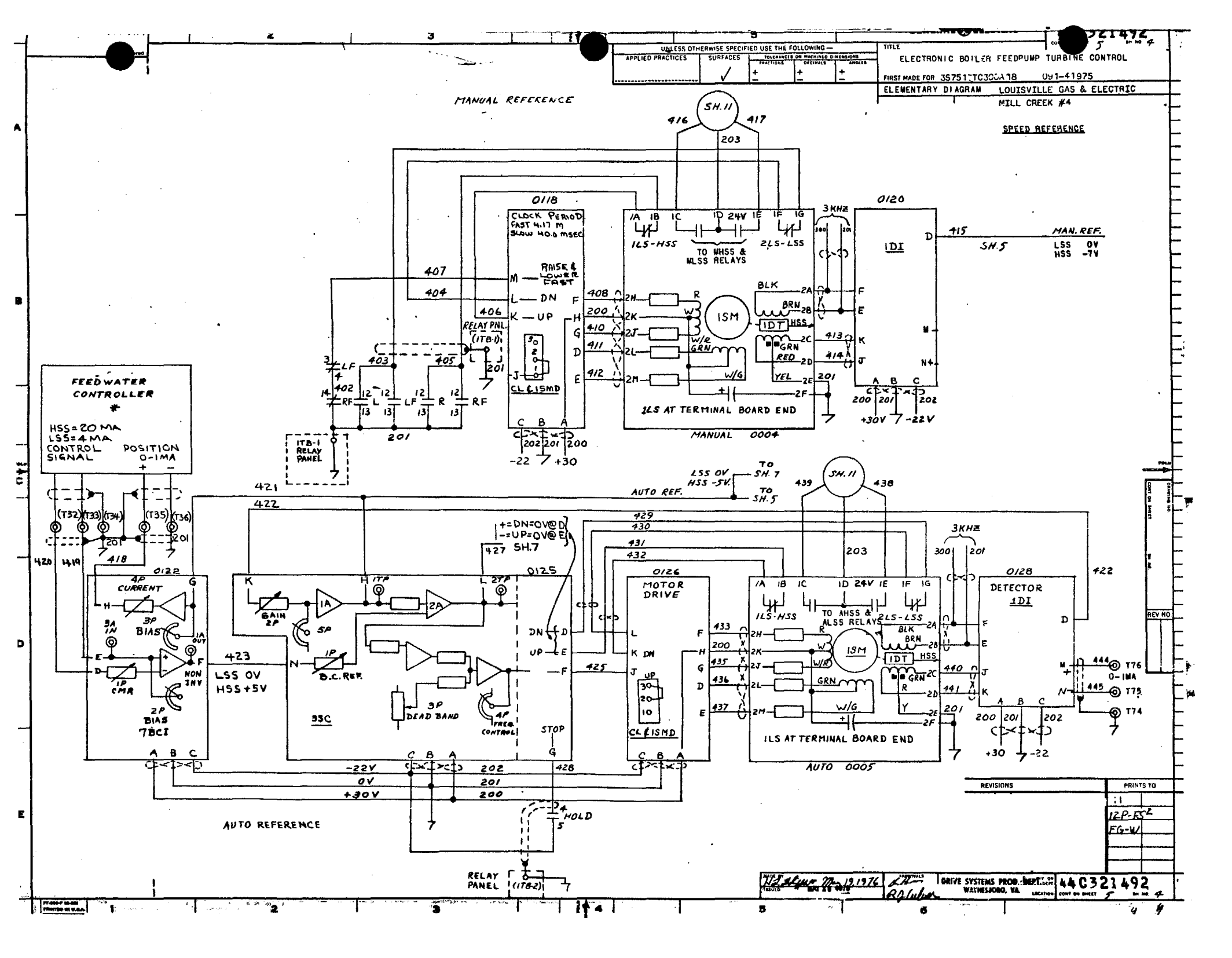
REVISIONS

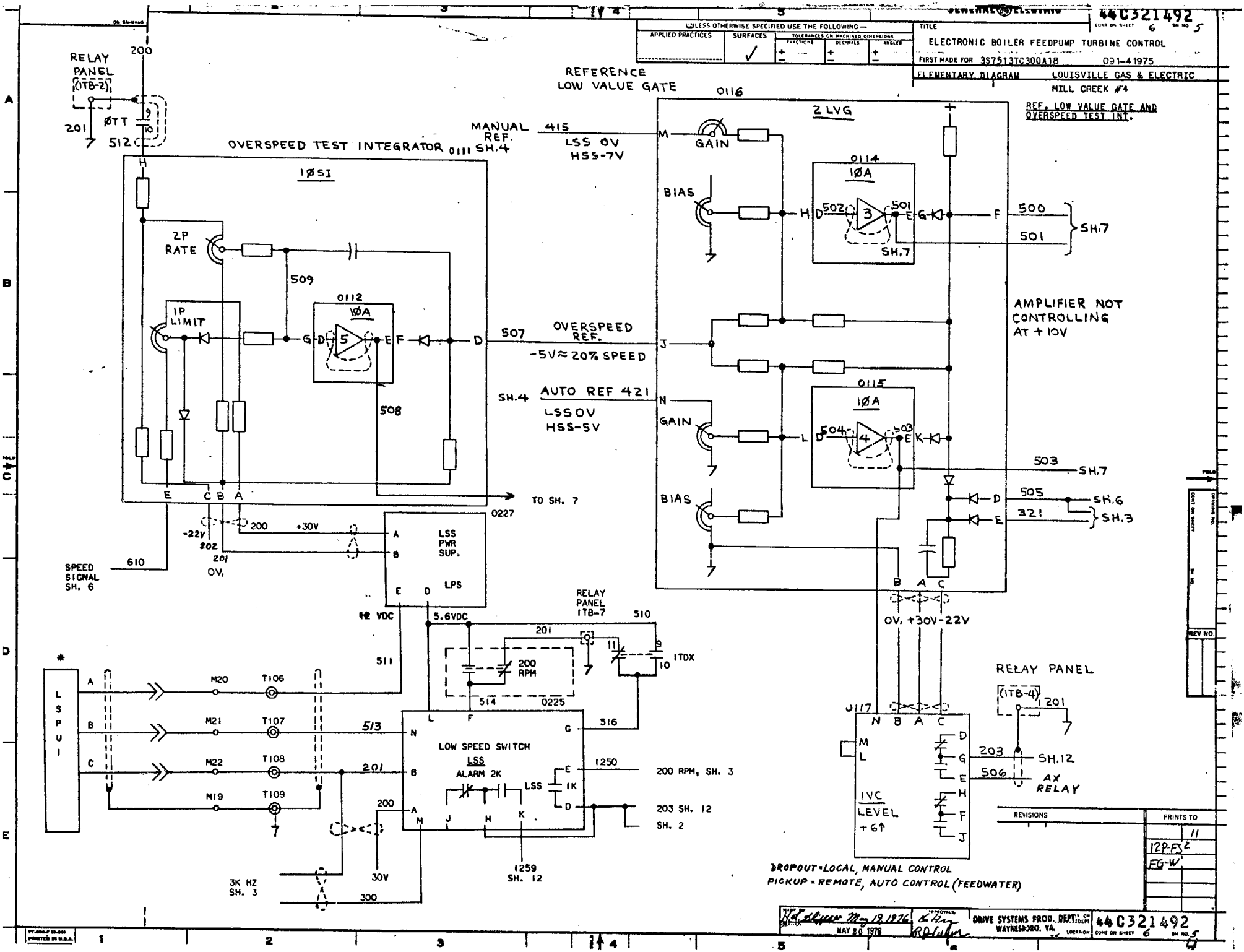
PRINTS TO
11
FC W/ 12P-FSC

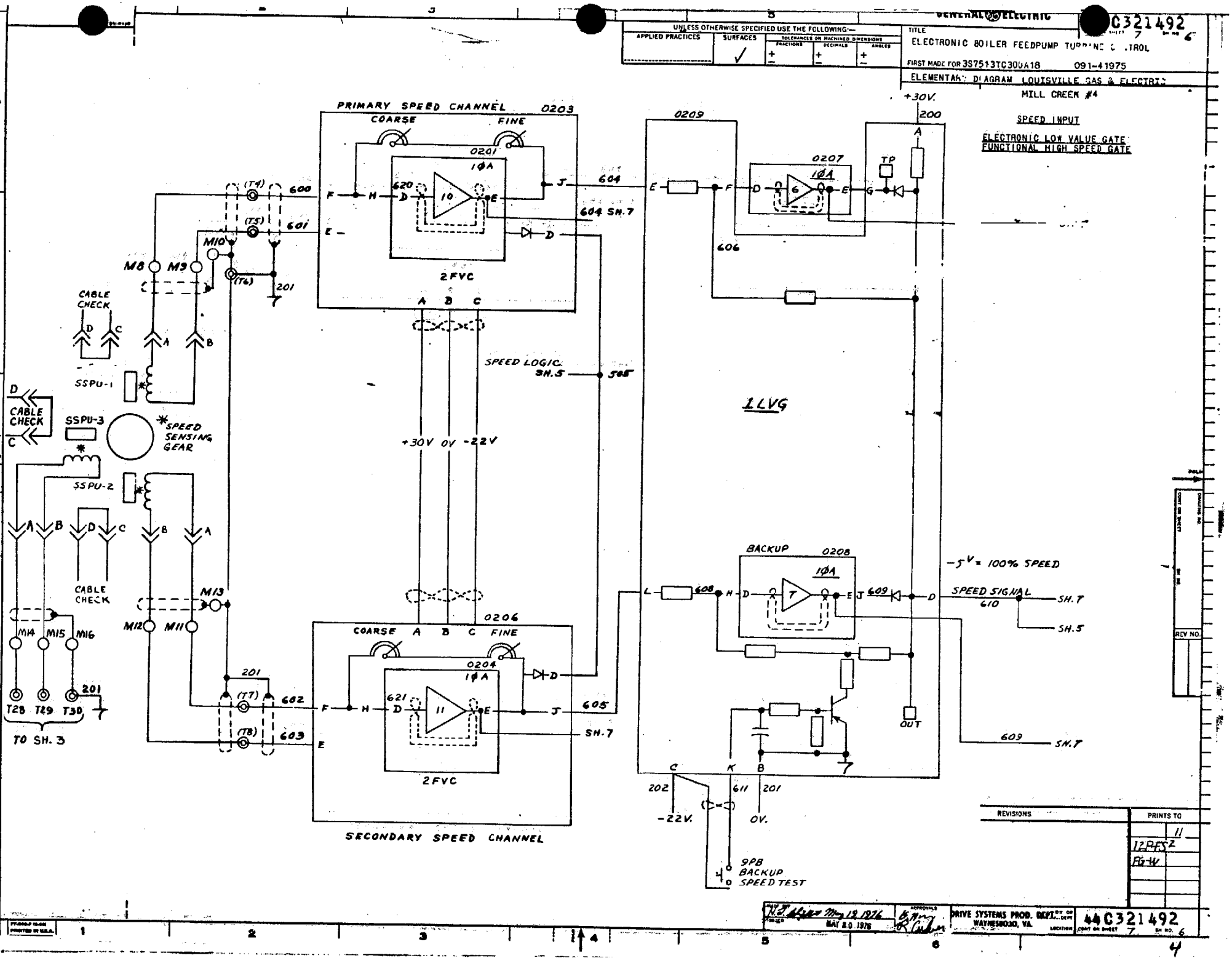
APPROVALS
MAY 20 1976
DRIVE SYSTEMS PROD. DEPT. OF SECTION
WAYNESBORO, VA. LOCATION

44C321492
CONT ON SHEET 4 SH. NO. 3

3+1PL


UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 35751-TC300A78 091-41975
ELEMENTARY DIAGRAM LOUISVILLE GAS & ELECTRIC
MILL CREEK #4
SPEED REFERENCE
MANUAL REFERENCE
FEEDWATER CONTROLLER
HSS=20 MA
LSS=4 MA
CONTROL SIGNAL
POSITION 0-1MA
ITB-1 RELAY PANEL
RELAY PNL. (ITB-1)
CLOCK PERIOD
FAST 4.17 M
SLOW 40.0 msec
RAISE & LOWER FAST
DN
UP
CL & ISMD
ILS-HSS
ILS-LSS
TO MHSS & MLSS RELAYS
ILS AT TERMINAL BOARD END
MANUAL 0004
ISM
IDT
HSS
3KHz
1DI
MAN. REF.
LSS 0V
HSS -7V
SH.5
SH.11
SH.7
AUTO REF.
LSS 0V HSS -5V
TO SH.7
TO SH.5
+=DN=0V@D
-=UP=0V@E
MOTOR DRIVE
DETECTOR 1DI
4P CURRENT
3P BIAS
NON INV
2P BIAS
TBCI
GAIN 2P
B.C. REF.
3P DEAD BAND
4P FREQ. CONTROL
HOLD
STOP
LSS 0V
HSS +5V
-22V
0V
+30V
AUTO REFERENCE
ILS AT TERMINAL BOARD END
AUTO 0005
T76 0-1MA
T75
T74
REVISIONS
PRINTS TO
12P-RC2
FG-W
DRIVE SYSTEMS PROD. DEPT.
WAYNESBORO, VA.
44C321492


44C321492
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
LOUISVILLE GAS & ELECTRIC
MILL CREEK #4
REF. LOW VALUE GATE AND OVERSPEED TEST INT.
REFERENCE LOW VALUE GATE
OVERSPEED TEST INTEGRATOR 0111 SH.4
MANUAL REF. 415
LSS OV HSS-7V
AUTO REF 421
LSS OV HSS-5V
OVERSPEED REF. -5V≈20% SPEED
AMPLIFIER NOT CONTROLLING AT +10V
RELAY PANEL (TB-2)
RELAY PANEL (TB-4)
RELAY PANEL TB-7
SPEED SIGNAL SH. 6
LSS PWR SUP.
LPS
LOW SPEED SWITCH
LSS ALARM 2K
200 RPM
1TDX
LSPUI
IVC LEVEL +6↑
AX RELAY
200 RPM, SH. 3
203 SH. 12 SH. 2
1259 SH. 12
3K HZ SH. 3
DROPOUT=LOCAL, MANUAL CONTROL
PICKUP=REMOTE, AUTO CONTROL (FEEDWATER)
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.
MAY 20 1976



GENERAL ELECTRIC

C321492

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC30UA18 091-41975

ELEMENTARY DIAGRAM LOUISVILLE GAS & ELECTRIC
MILL CREEK #4

REVISIONS

PRINTS TO

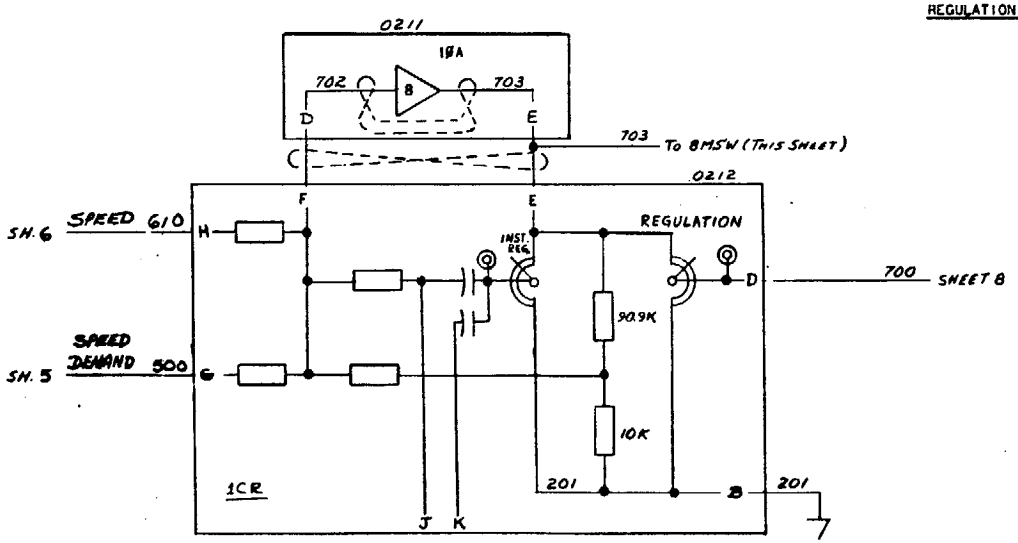
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.

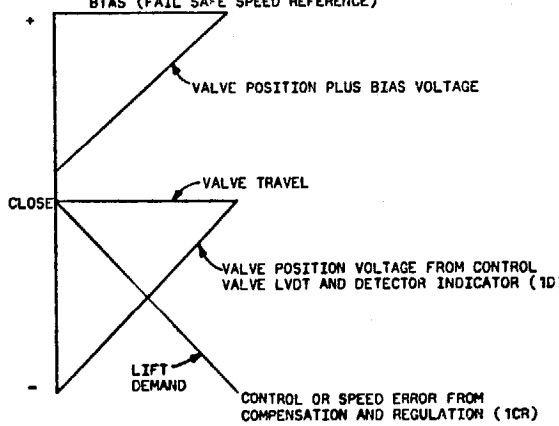
44C321492

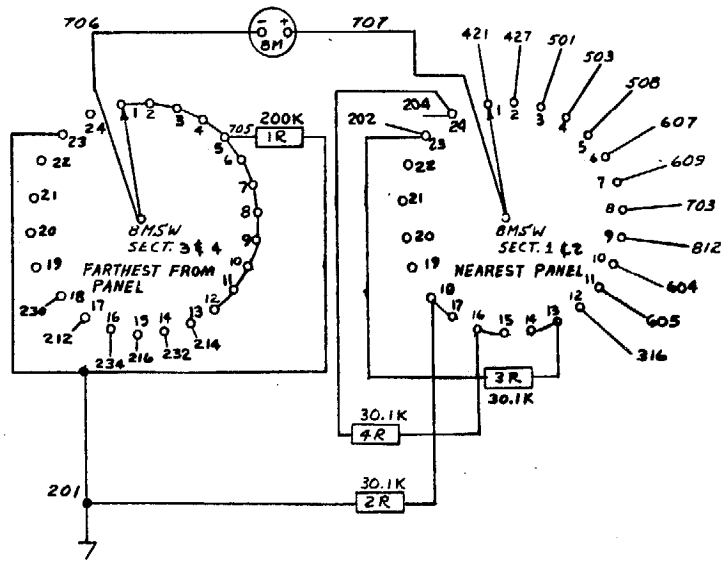
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A18 091-41975
ELEMENTARY DIAGRAM LOUISVILLE GAS & ELECTRIC
MILL CREEK #4

REGULATION AND STABILIZATION







AMPLIFIER

SWITCH POSITION	NUMBER	SYMBOL	FUNCTION	RACK LOCATION	APPROXIMATE OUTPUT VOLT RANGE	SHEET NO.
1	1	3SC	SIG. CONV.	0125	0 TO -5VDC	4
2	2	3SC	SIG. CONV.	0125	10VDC TO -10VDC	4
3	3	2LVG	MAN. REF.	0114	10VDC TO -5VDC	5
4	4	2LVG	AUTO REF.	0115	10VDC TO -5VDC	5
5	5	O'SPD	OVERSPEED	0113	10VDC TO -5VDC	5
6	6	1LVG	PRI. SPD.	0207	10VDC TO -7VDC	6
7	7	1LVG	SEC. SPD.	0208	10VDC TO -7VDC	6
8	8	1CR	COMP. & REG.	0211	10VDC TO -10VDC	7
9	9	2SA	SERVO AMP.	0214	7VDC TO -7VDC	8
10	10	2FVC	PRI. SPD.	0201	10VDC TO -7VDC	6
11	11	2FVC	SEC. SPD.	0204	0 TO -7VDC	6
12	12	2FVC	SPD. IND.	0222	0 TO -7VDC	3

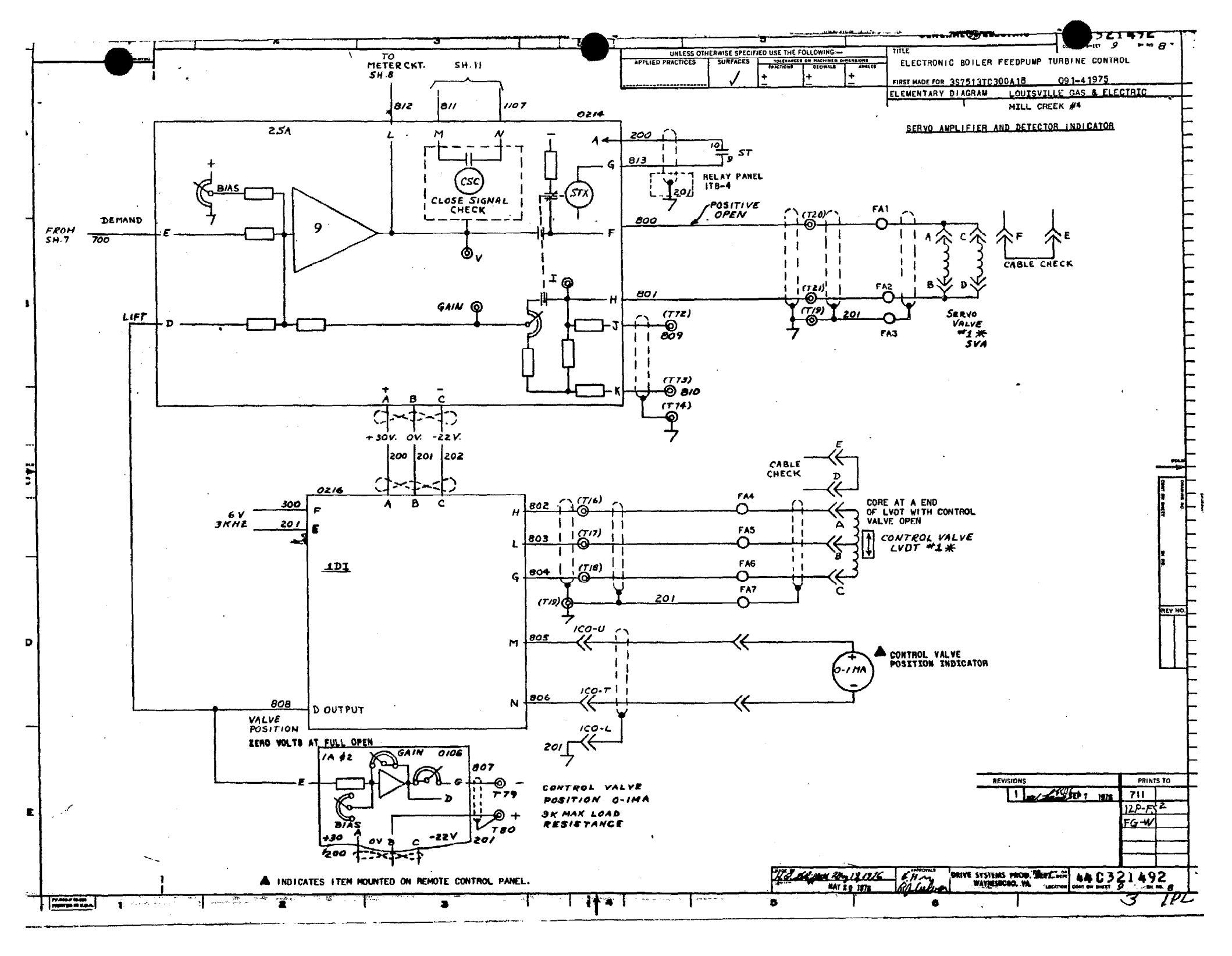
POWER SUPPLY

					APPROXIMATE OUTPUT CURRENT RANGE	
13	LINE A	1SR	-22VDC	0316	5 AMPS	2
14	LINE B	1SR	-22VDC	0318		2
15	LINE A	1SR	+24VDC	0322		2
16	LINE B	1SR	+24VDC	0328		2
17	LINE A	1SR	+30VDC	0308		2
18	LINE B	1SR	+30VDC	0310	5 AMPS	2

REVISIONS

PRINTS TO
11
12P-FS 2
FG-W

DRAWN W.J. Skinner DATE May 19 1976
APPROVALS
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA. LOCATION



UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:
APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A18 091-41975
ELEMENTARY DIAGRAM LOUISVILLE GAS & ELECTRIC
MILL CREEK #4

SERVO AMPLIFIER AND DETECTOR INDICATOR

REVISIONS
1 SEP 7 1978

PRINTS TO
711
12P-F5 2
FG-W

DRIVE SYSTEMS PROD. WAYNESBORO, VA.
44C321492

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:— APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

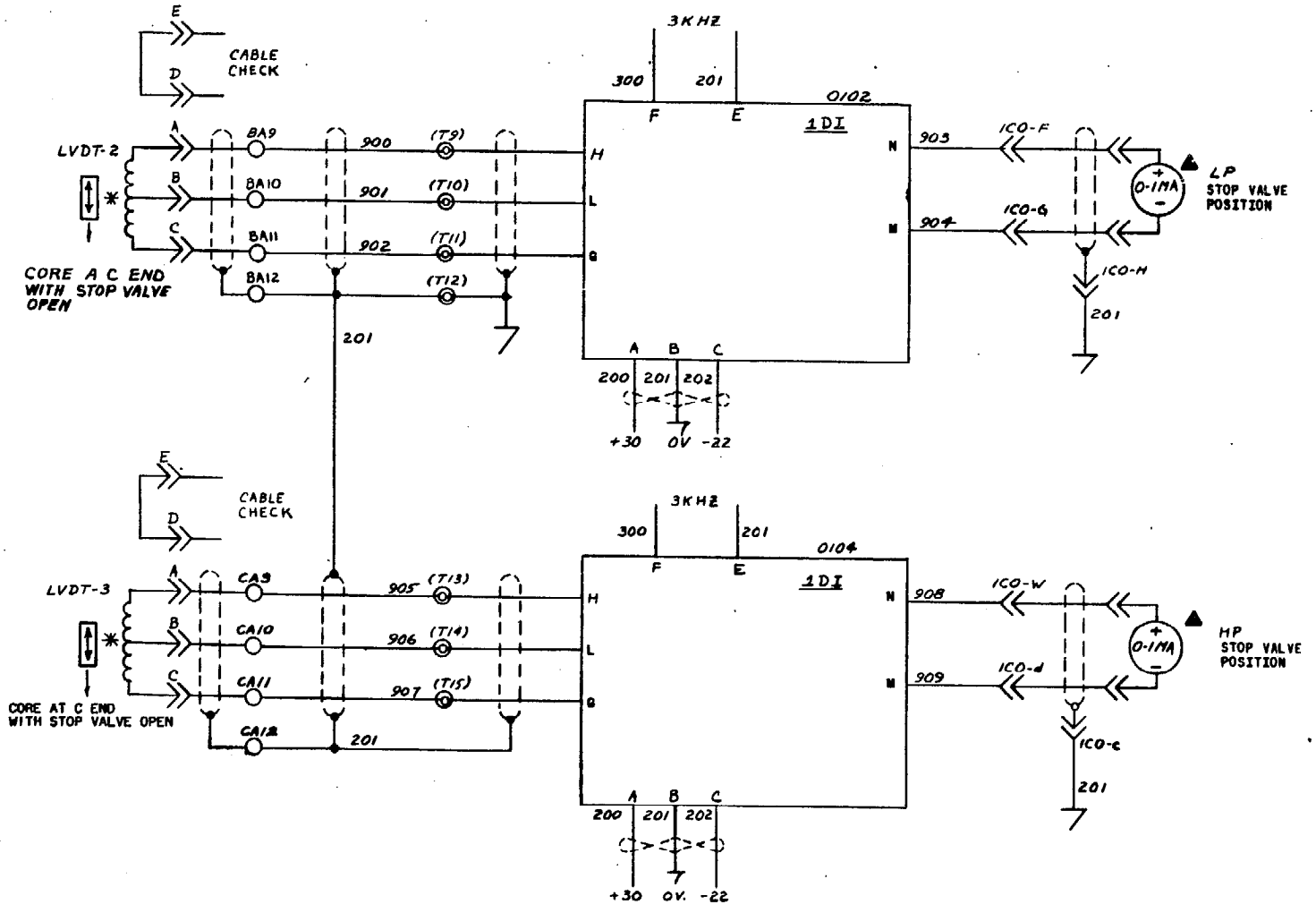
TITLE: ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A1B 091-41975

ELEMENTARY DIAGRAM LOUISVILLE GAS & ELECTRIC

MILL CREEK #8

STOP VALVE POSITION



▲ INDICATES ITEMS MOUNTED ON REMOTE CONTROL PANEL

REVISIONS

PRINTS TO: 11; 12P-FS2; FG-W

MADE BY: H.J. Aldrich May 19 1976 ISSUED MAY 20 1976 | APPROVALS | DRIVE SYSTEMS PROD. DEPT. OR DEPT. WAYNESBORO, VA. LOCATION | 44C321492 CONT ON SHEET 10 OF NO 9

GENERAL ELECTRIC

44C321492 SHEET 11 OF 10.

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE

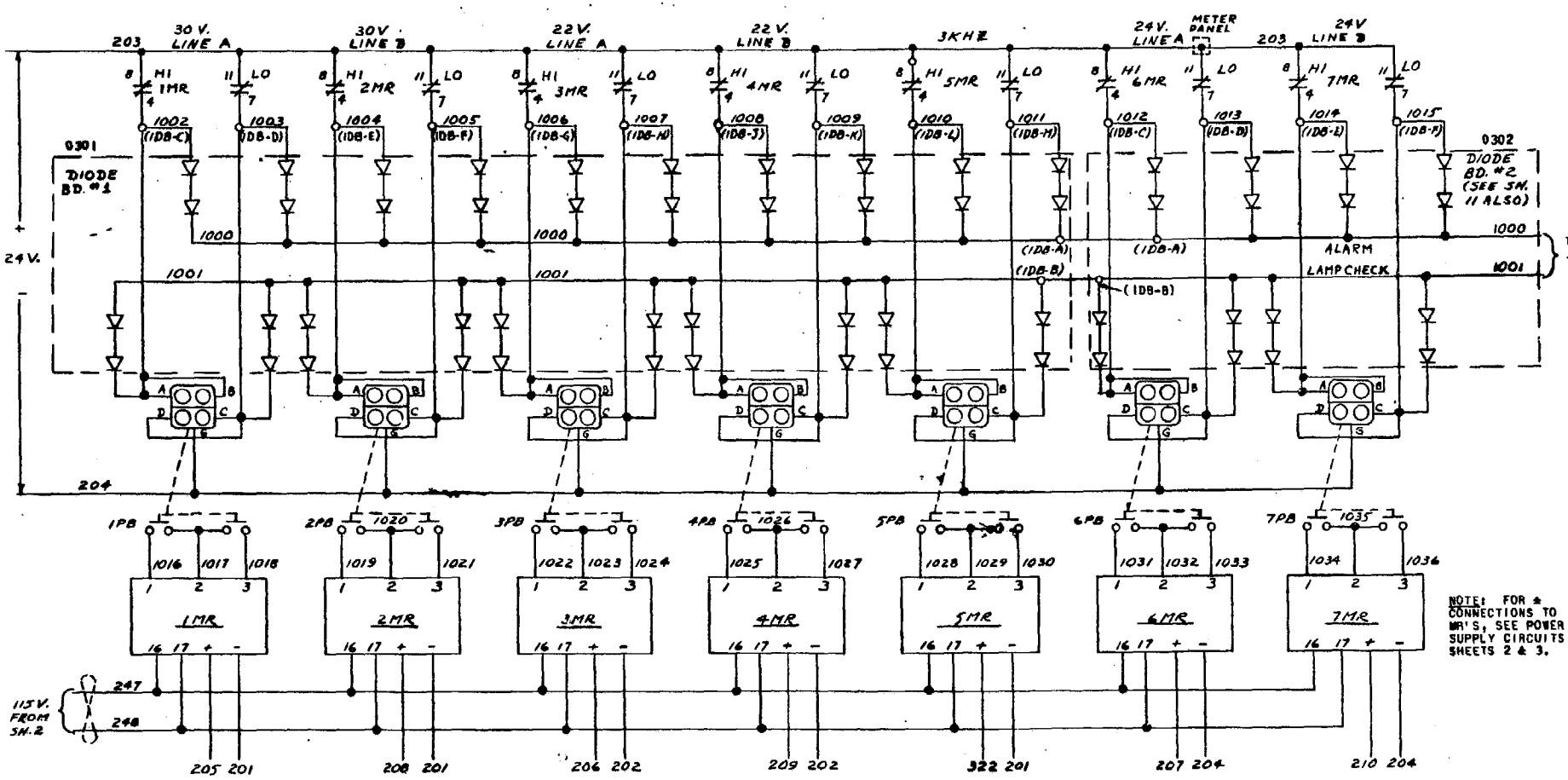
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A1B 091-41975

ELEMENTARY DIAGRAM LOUISVILLE GAS & ELECTRIC

MILL CREEK #4

MONITOR



NOTE: FOR * CONNECTIONS TO MR'S, SEE POWER SUPPLY CIRCUITS SHEETS 2 & 3.

REVISIONS

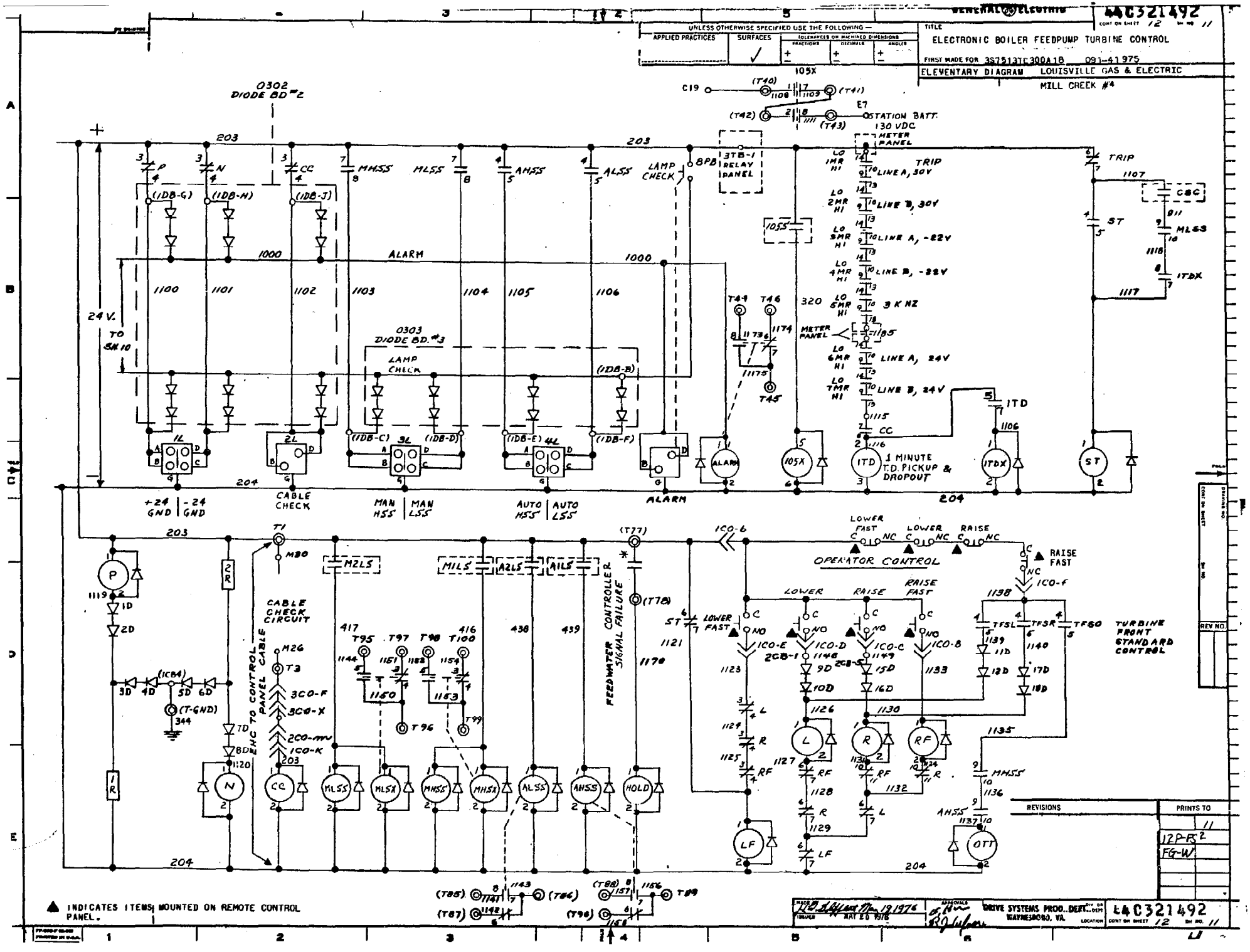
PRINTS TO 11

12PR2

FGW

DRIVE SYSTEMS PROD. MGT. WAYNESBORO, VA.

44C321492 SHEET 11 OF 10


GENERAL ELECTRIC
44C321492
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES
SURFACES
TOLERANCES ON MACHINED DIMENSIONS
FRACTIONS
DECIMALS
ANGLES
TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A18 091-41975
ELEVENTARY DIAGRAM LOUISVILLE GAS & ELECTRIC
MILL CREEK #4
105X
C19
E7
STATION BATT. 130 VDC
METER PANEL
3TB-1 RELAY PANEL
0302 DIODE BD #2
0303 DIODE BD. #3
203
204
1000
ALARM
24 V. TO SH 10
LAMP CHECK
TRIP
LINE A, 30V
LINE B, 30V
LINE A, -22V
LINE B, -22V
3 K HZ
LINE A, 24V
LINE B, 24V
1 MINUTE T.D. PICKUP & DROPOUT
ITD
ITDX
ST
CSC
MLSS
+24 GND
-24 GND
CABLE CHECK
MAN HSS
MAN LSS
AUTO HSS
AUTO LSS
CABLE CHECK CIRCUIT
TO CONTROL PANEL CABLE
FEEDWATER CONTROLLER SIGNAL FAILURE
OPERATOR CONTROL
LOWER FAST
LOWER
RAISE
RAISE FAST
TURBINE FRONT STANDARD CONTROL
HOLD
LF
OTT
▲ INDICATES ITEMS MOUNTED ON REMOTE CONTROL PANEL.
REVISIONS
PRINTS TO
12P-R
FG-W
DRIVE SYSTEMS PROD. DEPT.
WAYNESBORO, VA
44C321492

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A1B 091-41975
ELEMENTARY DIAGRAM LOUISVILLE GAS & ELECTRIC
MILL CREEK #4

OPERATOR'S CONTROL PANEL INDICATING LIGHTS

3TB RELAY PANEL
OPEN WHEN ON MANUAL CONTROL
REM
1K LSS OPEN ABOVE 2 RPM 1250
200 RPM OPEN ABOVE 200 RPM 1249
1TDX
T110 T112 1256 1258 1267 T111
1248
24 V.
506
2TB RELAY PANEL
AX
LSSX ENERGIZED BELOW TURNING GEAR SPEED
(T82)
(T83) 13 1246A 14 1247 (T84)
ENERGIZED WHEN ON AUTO CONTROL
2" LSS ALARM
T113 T115 1260 1262 1261 T114
1269
LSS ALARM X
204
110S OPEN ABOVE 110% SPEED
1263
T47 T49 1264 1266 1265 T48
110 SX
203
AX 1219
REMOTE CONTROL PANEL
ICO-R
ICO-
MANUAL CONTROL
MLSS 1222
MANUAL LSS
MHSS 1223
MANUAL HSS
ALARM 1224
ICO-P
ICO-MW
ICO-X
NORMAL RED
LAMP TEST
YELLOW FLASH ABNORM POWER
ALARM 1225
2FL
1226
21D
22D
20B-20
ICO-G
1267
2FLW

(T53) 1227 (2T85) 1X
130 V. BATTERY
1C 40MFD
(1CB-20) (T-GND) 344
2C 40MFD
1228 (T60) (2T8-6) 2X 1230

(T54) 1229
F1 RAISE
F3 LOWER
G1 G3
1240 1241
(T55) (T56)
TFSR TFSL
1230

(T27)
F5 OVERSPEED TRIP TEST
PS10 CLOSED ON TRIP
M17 M18
G5
1243 1244
(T58) (T26)
TFSO TRIP

▲ INDICATES ITEMS MOUNTED ON CONTROL PANEL

REVISIONS
PRINTS TO
12P-FS2
FGW

DRAWN W. Shipman 12-1976 MAY 8 8 1978
APPROVALS C. Lam R.J. Fabian
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.
44C321492 SHT 13 REV 12

4

GENERAL ELECTRIC

44C321492 CONT ON SHEET 14 SH NO 13

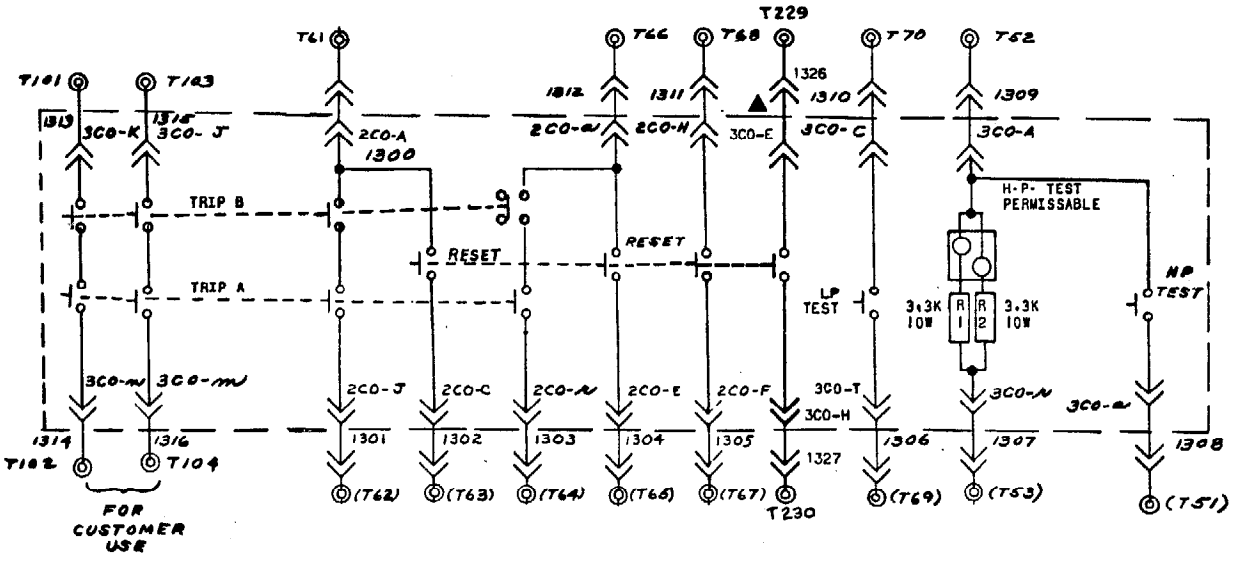
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE
ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A1B 091-41975

ELEMENTARY DIAGRAM LOUISVILLE GAS & ELECTRIC
MILL CREEK #4



▲ INDICATES ITEMS MOUNTED ON CONTROL PANEL

REVISIONS			PRINTS TO	
1	[illegible]	SEP 7 1978	711	
2	[illegible]	NOV 13 1977	12P-FS2	
			EG-W	

[illegible] MAY 20 1976	[illegible]	DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.	44C321492 CONT ON SHEET 14	SH NO 13

GENERAL ELECTRIC

#4C321492

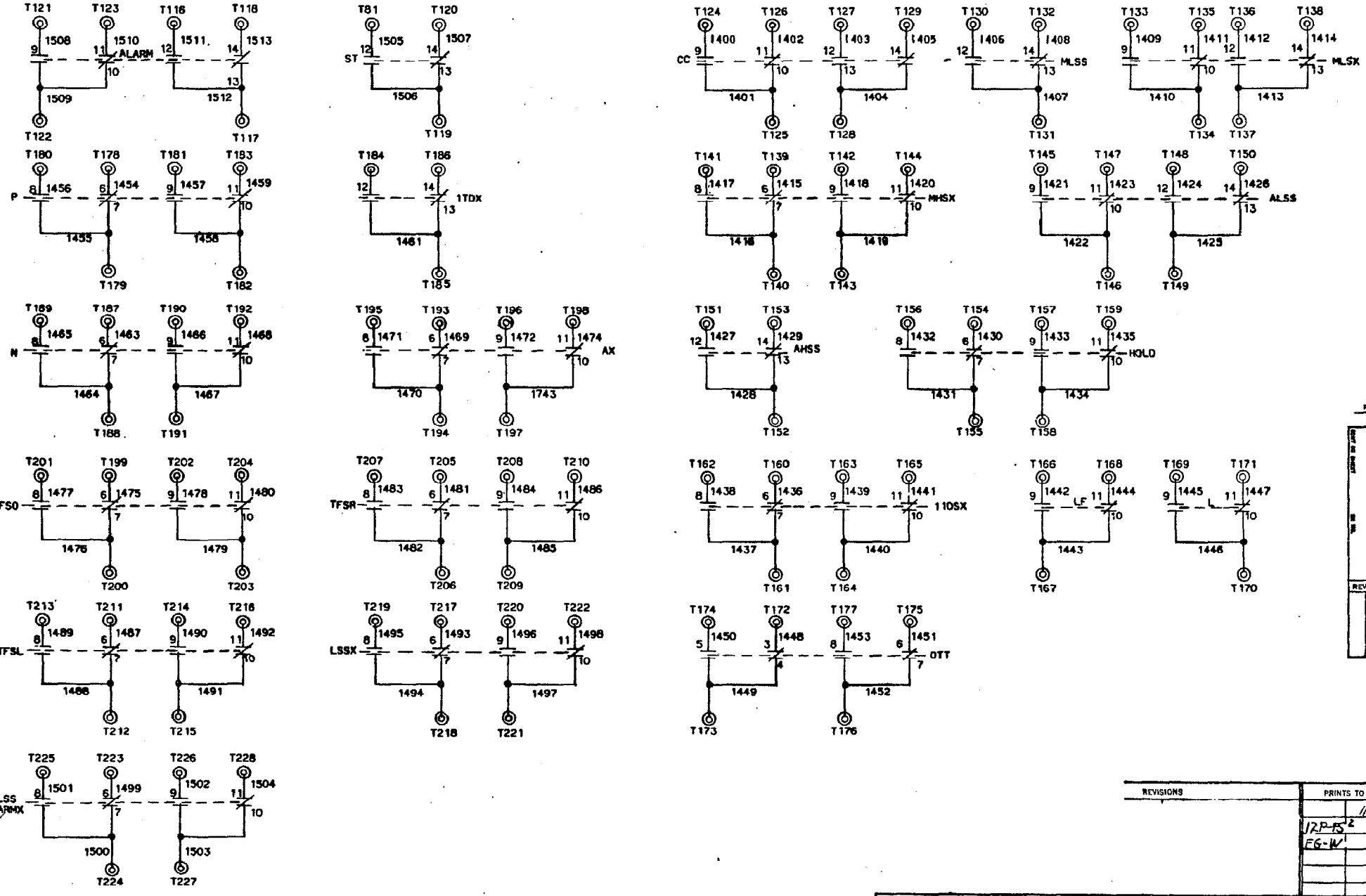
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:— APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE: ELECTRONIC BOILER FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A18 091-41975

ELEMENTARY DIAGRAM — LOUISVILLE GAS & ELECTRIC, MILL CREEK #4

SPARE RELAY CONTACTS



REVISIONS

PRINTS TO 11

12P-15 2

EG-W

DRAWN BY [signature] MAY 2 8 1976

APPROVALS [signature]

BOILER SYSTEMS PROD. DEPT. WAYNESBORO, VA.

#4C321492

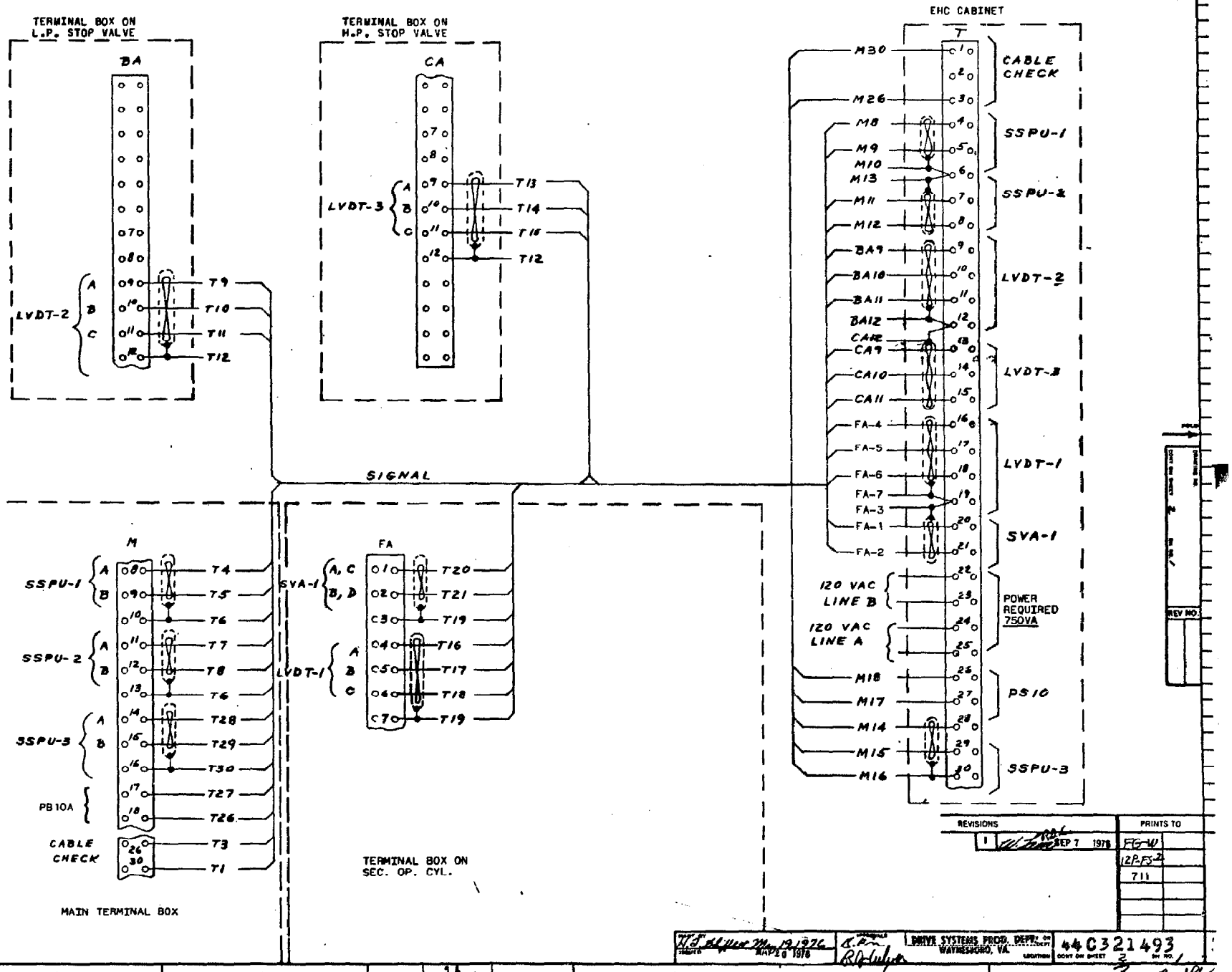
4

GENERAL ELECTRIC 44C321493 CONT ON SHEET 2 REV NO. 1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:— APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ELECTRONIC FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 3S75 I3TC300A I8 TURBINE 1153308 091-4 1975
INTERCONNECTION DIAG.
LOUISVILLE GAS & ELECT.

NOTE:
1. REFER TO 44C321492 (EHC ELEMENTARY) AND (TURBINE WIRING DWG.) FOR REFERENCE.



REVISIONS
1 RBL SEP 7 1978

PRINTS TO
FG-W
12P-FS-2
711

W.S. Miller May 19, 1976 DRAWN
APPROVED 1978
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.
44C321493 CONT ON SHEET 3 REV NO. 1

PRINTED IN U.S.A.

GENERAL ELECTRIC 44C321493 SHEET 3 OF NO. 2.

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES OR MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ELECTRONIC FEEDPUMP TURBINE CONTROL
FIRST MADE FOR 3S7513TC300A18 TURBINE: 153308 091-41975
INTERCONN. DIAG.
LOUISVILLE GAS & ELECT.

EHC CABINET

T
TRIP PB — C103, 0104 — CUSTOMER USE
LOW SPEED SWITCH — 0106 M20, 0107 M21, 0108 M22, 0109 M19
LSSX — 0110, 0111, 0112
LSS ALARMX — 0113, 0114, 0115
FOR CUSTOMER USE
RESET — 0116 D3, 0117 D7, C118
EHC REMOTE CONT. PANEL

MAIN TERMINAL BOX ON OIL TANK

D: C3 T116, 07 T117, C28 T53
G: 01 T55, 03 T56, 05 T58
C: 019 T40
M: 019 T109, A C20 T106, B 021 T107, C 022 T108 — LSPU
F: 01 T54, 03, 05
E: 07 T43, 08 T52

TERMINAL BOX ON HP STOP VALVE
CB: 07 T51, 08

EHC CABINET

T
C310
4 MA TO 20 MA INPUT FROM FEEDWATER CONTROLLER — 0320, 0330 — EHC INPUT 500 OHM MAX.
0340
FOR CUSTOMER USE — 0350, 0360 — AUTO SPEED REFERENCE POSITION 0-1 MA
C370, 0380, C390
C19 — 0400, 0410 — 105X
0420 — 105X
E7 — 0430
FOR CUST. USE — 0440, 0450, 0460 — EHC ABNORMAL CONDITION ALARM
FOR CUST. USE — 0470, 0480, 0490 — 110% SPEED
0500
ECH REMOTE CONTROL PANEL
CB7 — 0510
E30 — 0520 — HPSV TEST
D28 — 0530
F1 — 0540
G1 — 0550
G3 — 0560
0570
G5 — 0580
STATION BATTERY 125 VDC + 0590, − 0600

CONTROL PANEL
1CO
3 MULTICONDUCTOR 50 FT CABLES WITH PLUGS FURNISHED WITH EHC.
2CO
3CO

REVISIONS: 1 SEP 7 1976
PRINTS TO: FG-W 711 12P-FS2

DRAWN BY T.B. Lyon May 19 1976 ISSUED MAY 20 1976
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.
44C321493 SHEET 3 OF NO. 2

3 of 4

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ELECTRONIC FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S75 I35C300A18 TURBINE 153308 - 09 1-4 1975

INTERCONN. DIAG.

LOUISVILLE GAS & ELECT.

44C321493 CONT ON SHEET 4 SH NO. 3

EHC CABINET

T — EHC REMOTE CONTROL PANEL

C2	61	TRIP
E1	62	RESET P.B.
E2	63	
E21	64	TRIP
E22	65	RESET
C17	66	
E13	67	RESET P.B.
	68	
B87	69	LP TEST
C26	70	
	71	
	72	SERVO VALVE CURRENT 0-1 MA
	73	
	74	AUTO SPEED REFERENCE
	75	
	76	0-1 MA
	77	CLOSED ON FD WATER SIGNAL FAILURE
	78	SPEED HOLD
FOR CUSTOMER USE −	79	0-1 MA ANALOG CONTROL VALVE POSITION SIGNAL
+	80	
	81	
	82	AUTO CONTROL
	83	
	84	
FOR CUSTOMER USE	85	AUTO SIG. AT LSS
	86	
	87	
	88	AUTO SIG. AT HSS
	89	
	90	
N10	91	0-1 MA TACH. SIGNAL
N9	92	
FOR CUSTOMER USE	93	0-1MA ANALOG SPEED SIGNAL
	94	
	95	MAN. REF. AT LSS
FOR CUSTOMER USE	96	
	97	
	98	MAN. REF. AT HSS
	99	
	100	
FOR CUSTOMER USE	101	TRIP PB
	102	

CASE GND.

L.P. STOP VALVE T.B.

BB

7	T69

MAIN TERMINAL BOX ON OIL TANK

E

1	T62
2	T63
3	
4	
13	T67
21	T64
22	T65

N — 0-1MA TACH TI-1

9	T92
10	T91
11	T92

C

1	T61
17	T66
26	T70

REVISIONS

1		SEP 7 1978	

PRINTS TO: 711 FG-M 12F-F52

DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.

44C321493 CONT ON SHEET 4 SH NO. 3

DATE ISSUED MAY 20 1976

3 of 4

GENERAL ELECTRIC 44C321493 FL SH. 4

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
ELECTRONIC FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3S7513TC300A1B TURBINE, 153308 091-41975

LOUISVILLE GAS & ELECT.

SPARE RELAY CONTACTS

ALARM	121 122 123	OTT	175 176 177
CC	124 125 126	P	178 179 180
CC	127 128 129	P	181 182 183
MLSS	130 131 132	1TDX	184 185 186
MLSX	133 134 135	N	187 188 189
MLSX	136 137 138	N	190 191 192
MHSX	139 140 141	AX	193 194 195
MHSX	142 143 144	AX	196 197 198
ALSS	145 146 147	TFSO	199 200 201
ALSS	148 149 150	TFSO	202 203 204
AHSS	151 152 153	TFSR	205 206 207
HOLD	154 155 156	TFSR	208 209 210
HOLD	157 158 159	TFSL	211 212 213
110SX	160 161 162	TFSL	214 215 216
110SX	163 164 165	LSSX	217 218 219
LF	166 167 168	LSSX	220 221 222
L	169 170 171	LSS ALARM X	223 224 225
OTT	172 173 174	LSS ALARM X	226 227 228

229	RESET PB	EHC REMOTE CONTROL PANEL
230		
231		
232		
233		
234		
235		SPARES
236		
237		
238		
239		
240		
241		
242		

REV NO.

REVISIONS

PRINTS TO 11

FG-W 12-ES?

ISSUED MAY 20 1976 | APPROVALS | DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA. | 44C321493 SHT OR SHEET FL SH. NO. 4

GENERAL ELECTRIC

44C306444 SHEET 2

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING—

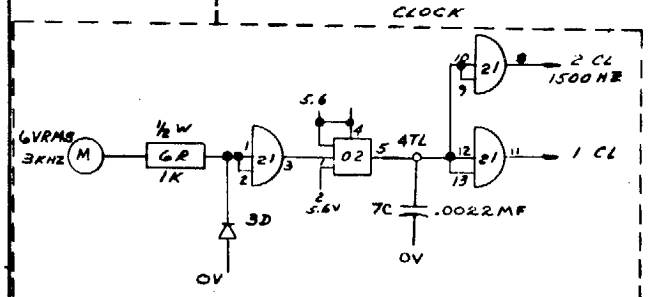
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS ±	DECIMALS ±	ANGLES ±
	√			

TITLE

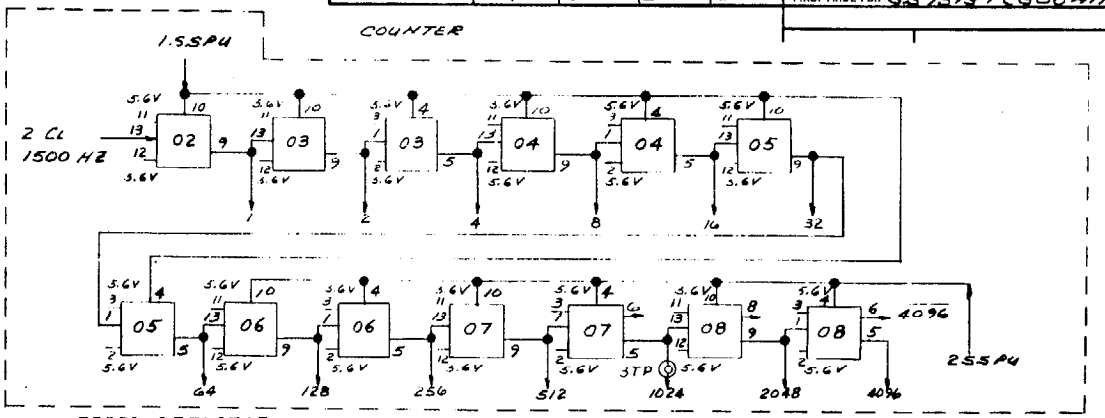
LOW SPEED SWITCH

FIRST MADE FOR 3575/8 TC3004.1

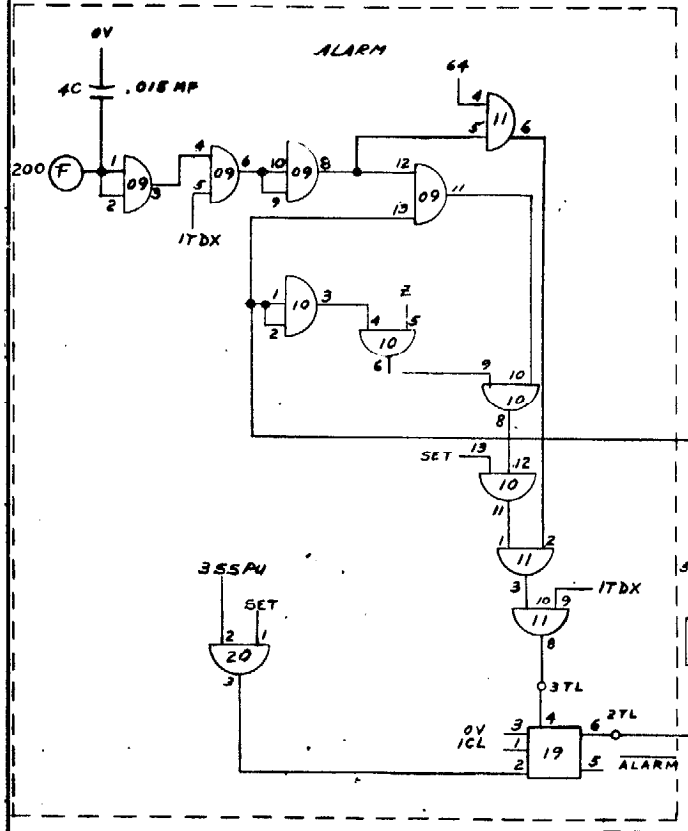
CLOCK



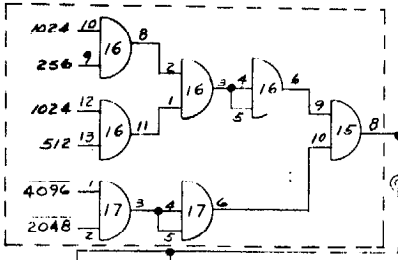
COUNTER



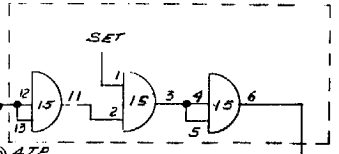
ALARM



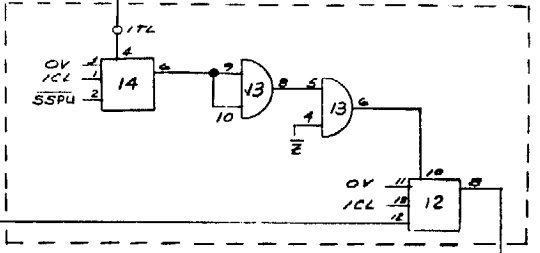
SPEED DETECTOR



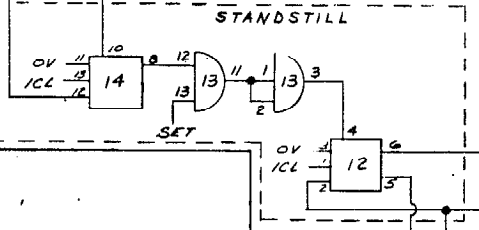
SPEED INHIBIT



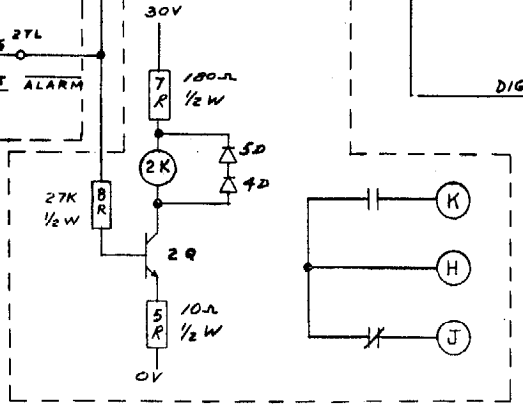
SPEED



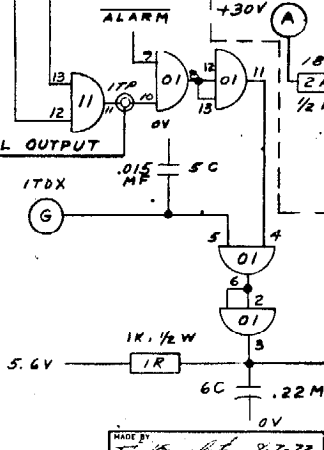
STANDSTILL



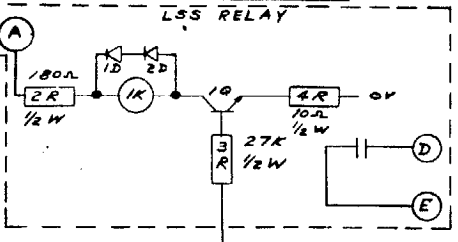
ALARM RELAY



DIGITAL OUTPUT



LSS RELAY



REVISIONS

PRINTS TO

MADE BY: [illegible] 8-7-72

ISSUED: B. Moyer Oct. 6, 1972

APPROVALS: RG Swafford

DRIVE SYSTEMS PROD. DEPT. OF WAYNESBORO, VA.

LOCATION

44C306444 SHEET 2 SH NO 1

GENERAL ELECTRIC

4C3C6444

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE

LOW SPEED SWITCH

FIRST MADE FOR 3S7513TC300A11

Z

4096

355PU

SET PULSE

+5.6V

5.6V

1/2 W

9R

75R

1C

150MF

2C

150MF

SET

0V

SHAFT SPEED PICK-UP PULSE

TOOTH WHEEL PICKUP

3C .0022MF

5.6V

2CL

200 RPM

1.55PU

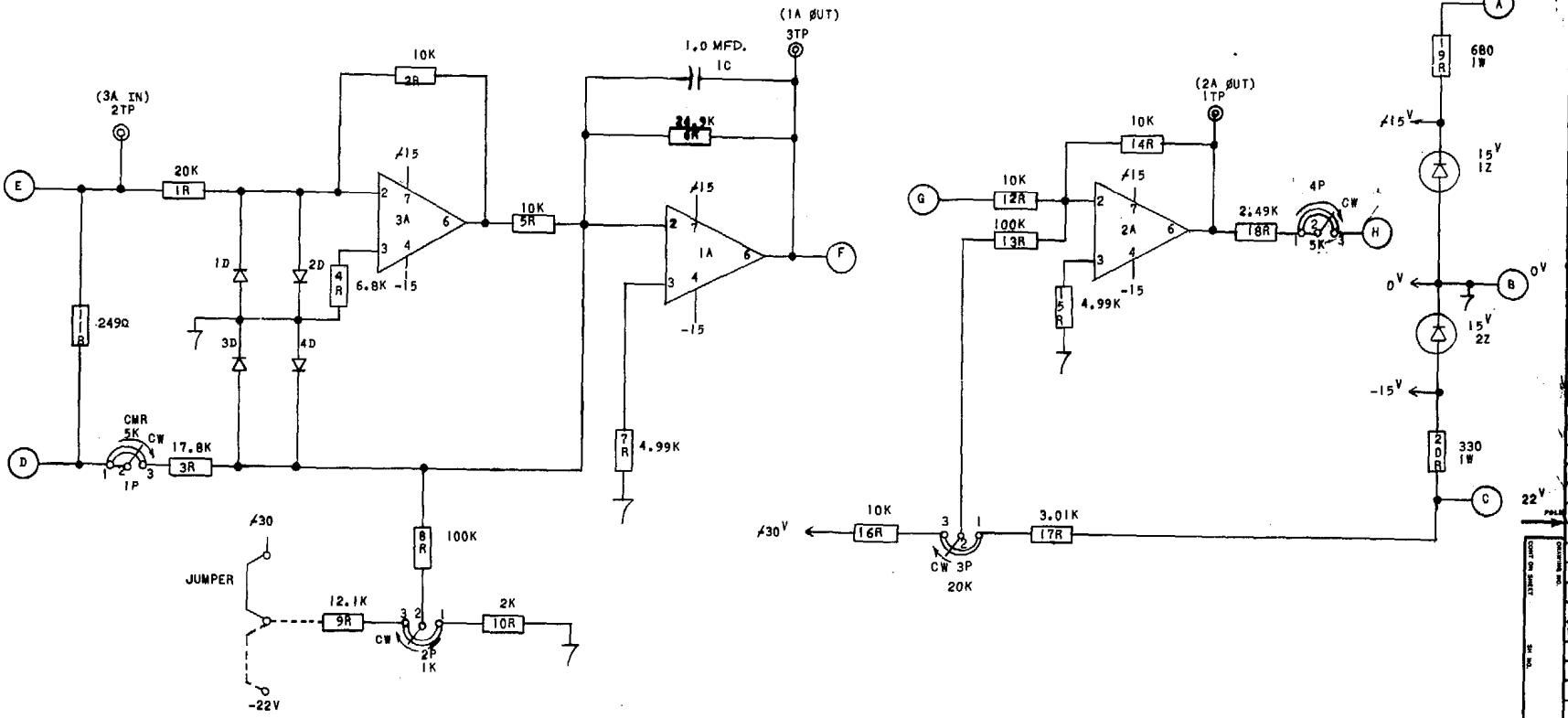
3.55PU

REVISIONS

GENERAL ELECTRIC 44C306492

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:— APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
7BDI BOILER CONTROL INTERFACE
4-20 MA INPUT
FIRST MADE FOR 44B331746-G03



NOTES:

1. 2TP INPUT

LSS	HSS
4MA	20MA

3TP OUTPUT

LSS	HSS
0V	5V

2. CONNECTIONS

A ±30 VDC
B 0V
C -22 VDC
D BOILER CONTROL COMMON
E INPUT FROM BOILER CONTROL
F OUTPUT
G EHC AUTO SPEED REFERENCE
H AUTO REF POSITION TO BOILER CONTROL

REVISIONS

PRINTS TO
VZP-DRB
FG-W
11

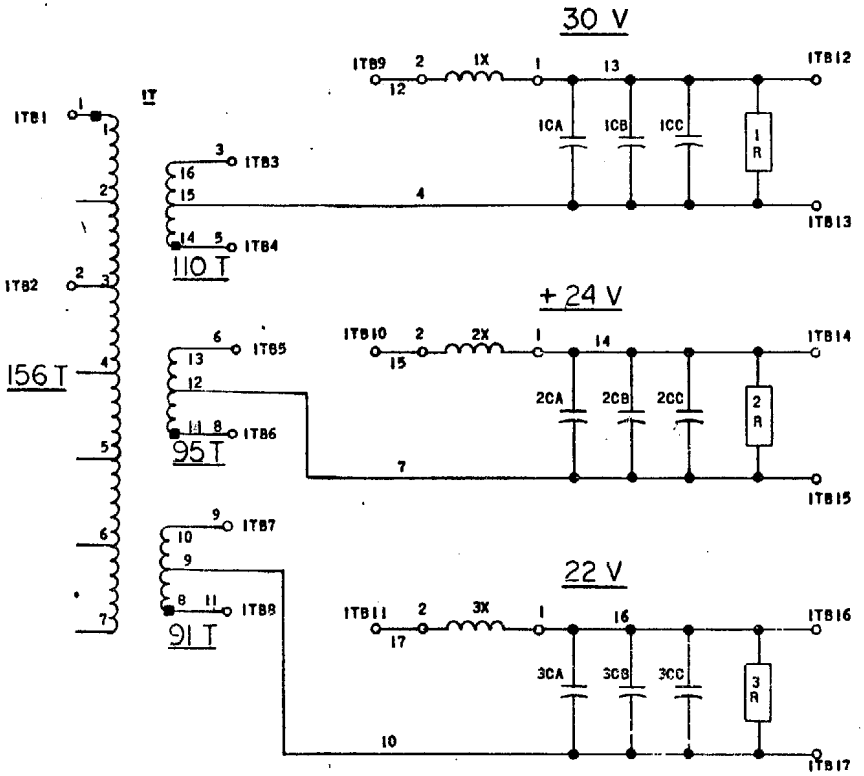
DRAWN D. Rumford 10-23-72
ISSUED D. Mayer Nov. 2, 1972
APPROVALS Q. Culver
COMBL & CONT. DEVICES WAYNESBORO, VA
44C306492

GENERAL ELECTRIC

44C306526

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES OR MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
LINE POWER SUPPLY
FIRST MADE FOR 357513KF300A5



CKT. NO.	WIRE SEQUENCE, WIRE M9729494P1918 #18 WH.
1	ITB(1),IT(1)
2	ITB(2),IT(3)
3	ITB(3),IT(16)
4	1CC(1),1CB(1),1CA(1),IT(15),1R(1),ITB(13)
5	IT(14),ITB(4)
6	IT(13),ITB(5)
7	2CC(1),2CB(1),2CA(1),2R(1),IT(12),ITB(15)
8	IT(11),ITB(6)
9	IT(10),ITB(7)
10	3CC(1),3CB(1),3CA(1),3R(1),IT(9),ITB(17)
11	IT(8),ITB(8)
13	1CC(2),1CB(2),1CA(2),1R(2),ITB(12)
14	2CC(2),2CB(2),2CA(2),2R(2),ITB(14)
16	3CC(2),3CB(2),3CA(2),3R(2),ITB(16)

1X

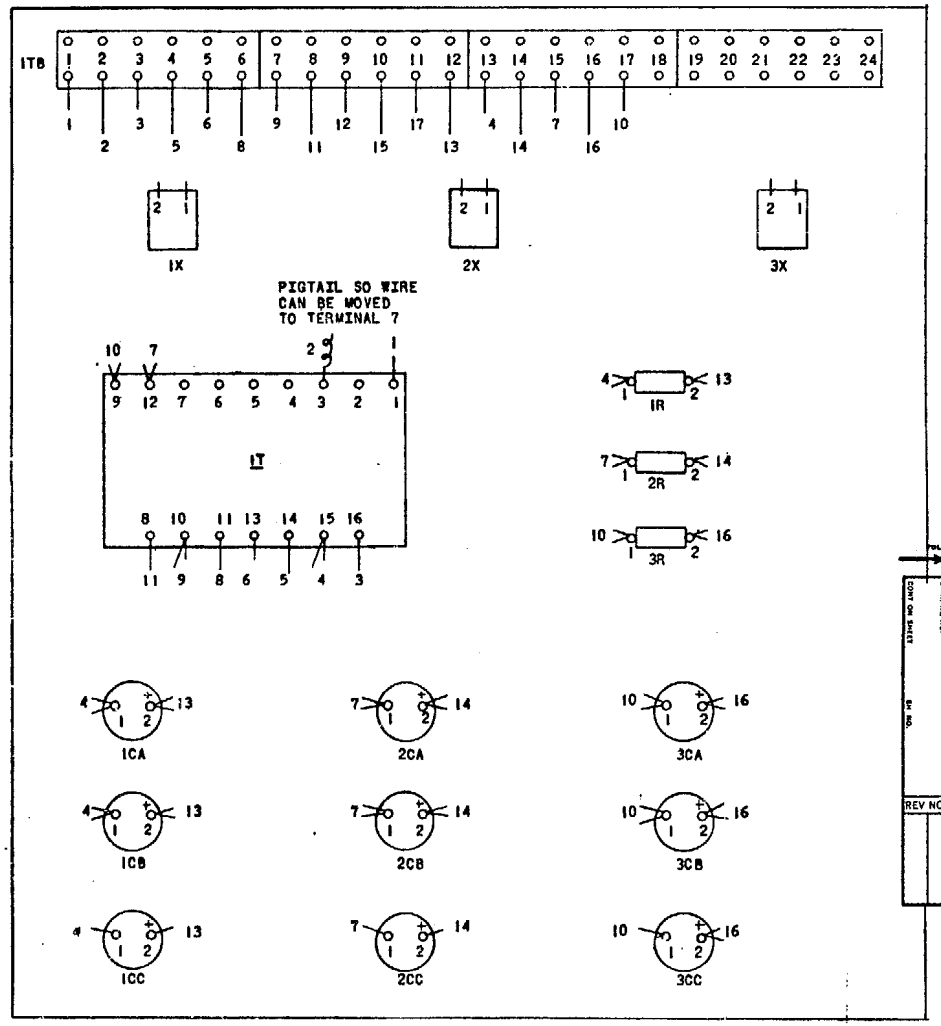
CKT.	LEAD NO.	WIRE TO
13	1	ITB(12)
12	2	ITB(9)

2X

CKT.	LEAD NO.	WIRE TO
14	1	ITB(14)
15	2	ITB(10)

3X

CKT.	LEAD NO.	WIRE TO
16	1	ITB(16)
17	2	ITB(11)



REVISIONS

			PRINTS TO
1	I.S.N.	Nov. 3, 1972	11
2	E.G. Coleman	May 10, 1973	
3	W.H. Chapman	May 30, 1973	

MADE BY: H. Templeton 10-12-72 | APPROVALS: E. Coleman | DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA. | 44C306526

Oct 20, 1972 D. Morgan

GENERAL ELECTRIC

44D300043

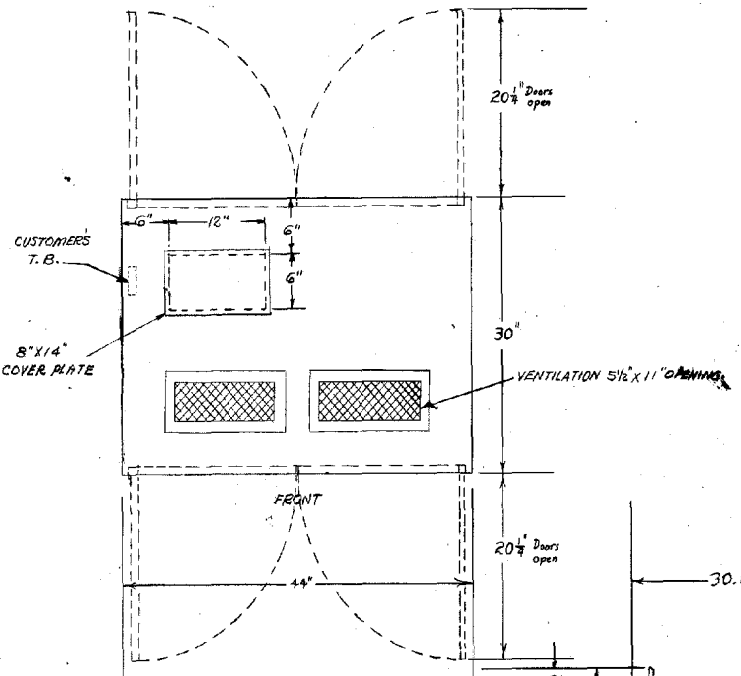
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:— APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

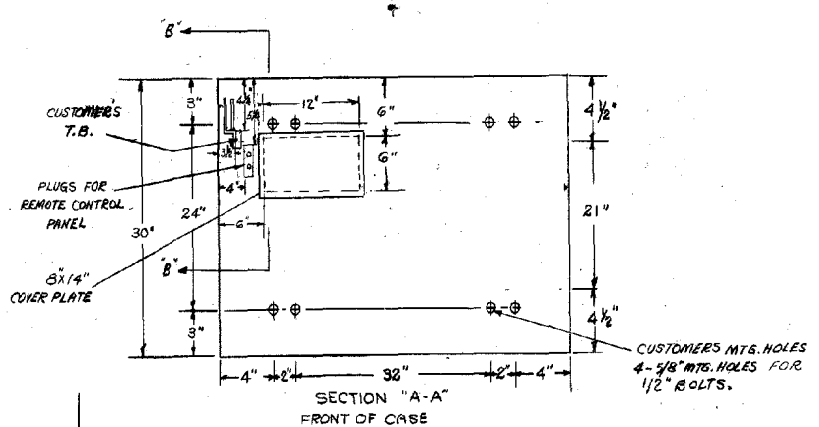
44D300043

TITLE: ELECTRONIC FEEDPUMP TURBINE CONTROL

FIRST MADE FOR 3875137C300A2

CABINET OUTLINE

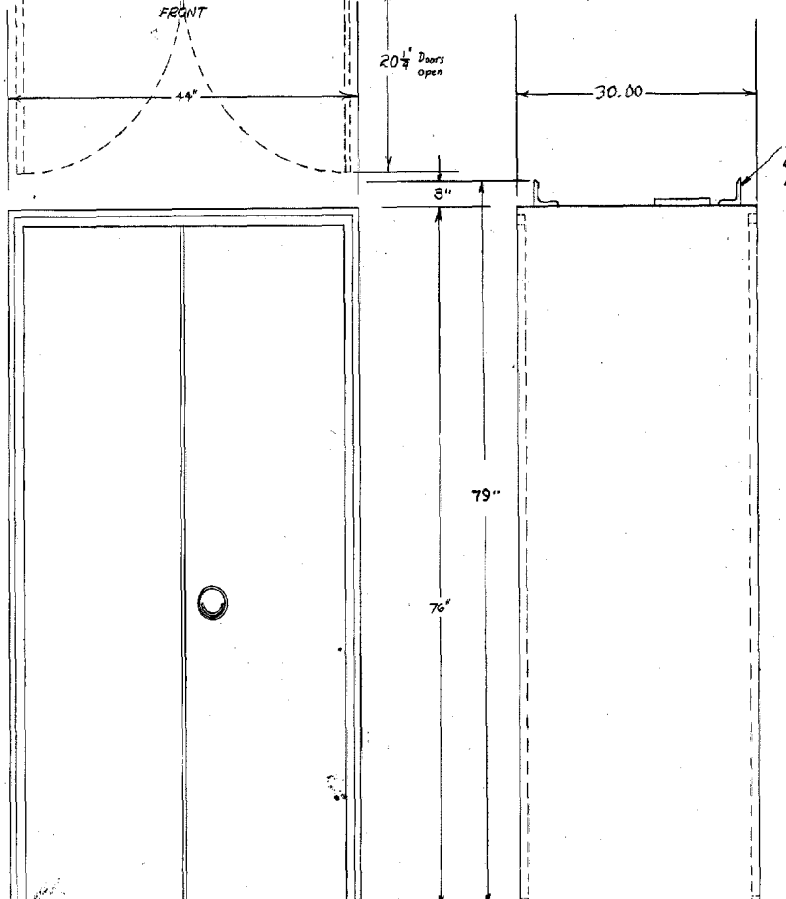




SECTION "A-A"
FRONT OF CASE

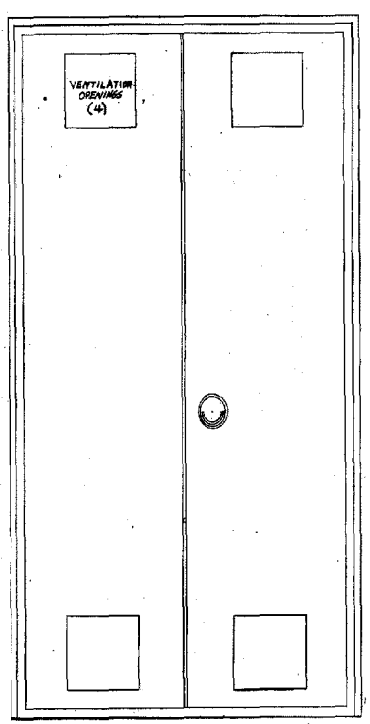
NOTES:

1. ONE CABINET REQUIRED FOR EACH BOILER FEED PUMP TURBINE.
2. ESTIMATED WT. 1500 LBS.
3. MAXIMUM AMBIENT TEMP. 40°C
4. FINISH ASA 61 GRAY
5. CABINET HAS LOOSE PIN REMOVABLE HINGED DOORS.
6. DISCONNECTION OF CAB. VET/REMOTE CONTROL CABLE ENTRANCE EITHER FROM TOP ☐ OR BOTTOM ☐ AS CUSTOMER DESIRES. (CUSTOMER TO CHECK ONE BLOCK.)
7. CUSTOMER'S PROVISIONS FOR STRAPPING CABLE RUN NEXT TO CUSTOMER'S TERMINAL BOARDS.
8. IF NON-STANDARD FINISH REQUIRED SPECIFY ON ORDERING DRAWING.
9. POWER CONSUMPTION/DISSIPATION 750W.

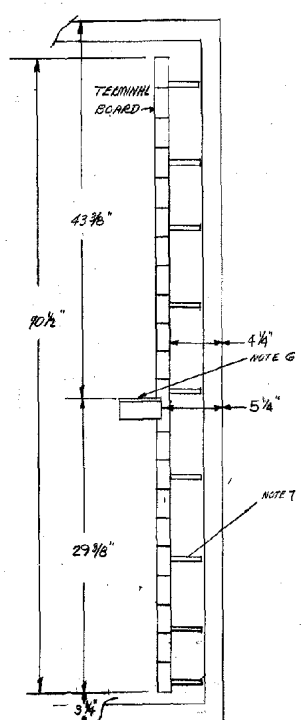


FRONT VIEW

SIDE VIEW



BACK VIEW



SECTION "B"-"B"

REVISIONS			PRINTS TO
9	REMOVED INSIDE FRONT GLOSS WHITE PAINT	1	711
8		2	722-858
7		3	756-W
		4	
		5	
		6	

ISSUED NOV 24 1967

APPROVALS R.R.M.

SPECIALTY CONTROL DEPT. WAYNESBORO, VA. LOCATION

44D300043

3+1 PL

GENERAL ELECTRIC

44D309956

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:— APPLIED PRACTICES | SURFACES √ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± ± | DECIMALS ± ± | ANGLES ±

TITLE: ENC BOILER FEEDPUMP ALIGNMENT DIAGRAM

FIRST MADE FOR 3E75131TC300A18 09-41975

LOUISVILLE GAS & ELECTRIC — MILL CREEK #4

PRIMARY CONTROL-5.00V
BACK-UP CONTROL-5.05V
} AT RATED SPEED 5300 RPM/SSP FREQ 4152 HZ

ALIGNMENT INSTR.

TURBINE RATED SPEED: 5300 RPM @ 4152 HZ SSPU

VOLTAGE COMPARATOR SETTINGS

LOC.	FUNCTION	OPERATING POINT
0218	10% SPEED	DO 415 HZ SSPU-1
0219	110% SPEED	DO 4567 HZ SSPU-1
0220	105% SPEED	DO 4359 HZ SSPU-1
0114	REMOTE/LOCAL	DO=6.5V N-TP
0221	200 RPM	DO 157 HZ SSPU-3

VALVE LIFT	STOP			"OUT" VOLTS
	CONTROL	L P	H P	1DI, 0216
C.O.T.	0"	0"	0"	(1)
EFF.O	7/8"			
FULL OPEN	13"	5.63	3.56	(2)
FULL LOAD LIFT	7/8+4.469			

GAIN = (2) − (1) VOLT / 13 INCH =

REF. STOPS 2LVG, 0116 OUTPUT TP

	MAN	AUTO
LSS	0V	L34
HSS	5.442V	5.442V

COMPENSATION & REG. - 0212

TERMINALS J-K - OPEN
INST. REG. EFF. ROT.
REG. 6% - SEE INSTRUCTIONS

OVERSPEED INTEGRATOR, 0105

RATE 12± 2% / MINUTE
LIMIT 120% SPEED — SEE INSTRUCTIONS

SERVO AMP - 0214

BIAS — SEE INSTRUCTIONS
GAIN

VOLTS LIFT=GAIN (SEE ABOVE)

VOLT 1 1.22

METER RELAY SETTING: ± 5% OF OPERATING POINT

SPEED

TO 1081

NORMAL AMP 9 OR VTP +1.44V, ITP + 0.35V.

NEG. OPENS — CLOSE SIGNAL CHECK — START CONTROL — POS. OPENS

GAIN=10

1CR-0212 — REG — INST REG

2SA-0214 — DEMAND — BIAS — LIFT — GAIN — STX

HP OIL SUPPLY — RAM — CONTROL VALVE — SERVO VALVE — TURBINE — GEAR

HP STOP VALVE — STEAM — LP STOP VALVE — STEAM

WITHDRAW CORE FOR VALVE OPEN

LSS ALARM & RELAY — LOW SPEED SWITCH — LSSX RELAY — SSPU-1 — SSPU-2 — SSPU-3

2FVC 0205 COARSE FINE — 2FVC 0206 COARSE FINE

AMPLIFIER NOT CONTROLLING AT +10V.

1LVG-0209 — PRIMARY — BACK UP — OUTPUT — BACK UP SPEED TEST

5800 RPM OR 3963SSPU HZ = 7.317V

200 RPM

1VC 0221 — OUTPUT — REF — LSSX RELAY

SPEED DEMAND — 1A 0102 — CONTROL VALVE POSITION

0V FULL OPEN

1DI-0216 — NULL — ZERO METER — CAL

CONTROL VALVE POSITION REMOTE PANEL — 0-100

LVDT-1 — LVDT-2 — LVDT-3

1DI 0104 — NULL — ZERO METER — CAL — OUT

1DI-0102 — NULL — ZERO METER — CAL — OUT

REMOTE CONTROL PANEL — HP STOP VALVE POSITION — LP STOP VALVE POSITION

AUTO LSS = L34
MAN. LSS = 0V. HSS = +5.442 V

MOUNTED ABOVE AUTO SPEED REFERENCE

CLOCK & 1SMD-0118 — CLOCK PERIOD SLOW 4.0M SEC. FAST 4.17M SEC. — DN UP — MOTOR

1DI-0120 — ZERO METER — CAL — OUT

2LS LSS — 1LS HSS — LF RF L R

AMPLIFIER NOT CONTROLLING AT + 10V

2LVG-0116 — MAN. GAIN (HSS) — MAN. BIAS (LSS) — SG GAIN (HSS) — SC BIAS (LSS) — MAN — AUTO — OUTPUT

110% SPEED — 1VC-0220 — OUTPUT — DO: CCW — REF — 110 SX RELAY

10% SPEED — 1VC-0218 — OUTPUT — DO: CCW — REF — +30VDC

105% SPEED — 1VC-0219 — OUTPUT — DO: CCW — REF — 105SX RELAY

3A02 0110 — 0-1MA — 3A02 0108 — DIGITAL DISPLAY — REMOTE CONTROL PANEL

DROPOUT+LOCAL — 1VC-0117 — REF — DO: CW — CONTROL INDICATION AX RELAY

7BCI-0122 — 0-1MA AUTO REF POSITION — 4P CURRENT — 3P BIAS — 2P BIAS — 0V LSS +5V HSS — 3A IN — 1A OUT — IP CMR — NON INV

FEEDWATER CONTROLLER — LSS= +4 MA HSS=+20MA

MOUNTED BELOW MANUAL SPEED REFERENCE

2P GAIN — ITP — 2TP — B.C. REF. 1P — DEAD BAND — 4P FREQ CONTROL — 3SC-0125

CLOCK & SMD 0126 — MOTOR — LSS 2LS — HSS 1LS

1DI-0128 — OUT — ZERO METER — CAL

HOLD — -22V

AMP 5 AT +10V NORMAL OPERATION NEGATIVE FOR OVERSPEED TRIP TEST.

1081-0112 — RATE — LIMIT — OTT — SPEED SIGNAL INPUT

POWER SUPPLIES

LPS 0227 — LOW SPEED SWITCH POWER SUPPLY — 5VDC 12VDC

7OSC-0217 — METER — LEVEL — 6V RMS — 3KHZ

5MR — 3KHZ 6V RMS

1SR-0308 — INPUT — LEVEL — METER — +30V BUS BAR — 30 LINE A — 30 LINE B

1SR-0310 — INPUT — LEVEL — METER

0 VOLT BUS BAR

1SR-0316 — LEVEL — METER — -22 LINE A — -22V BUS BAR — -22 LINE B

1SR-0318 — LEVEL — METER

1SR-0324 — LEVEL — METER — 24 LINE A — 24V — 24 LINE B

0328-F — 0326-G

1SR-0326 — LEVEL — METER

LINE B 120V, 60HZ — T22 T23 — POWER PANEL

120V LINE A 60 HZ — T24 T25 — POWER PANEL — METER RELAYS

NOTE

TEST METER IN SWITCH POSITIONS 1 TO 12 READS OUTPUT OF AMPLIFIERS 1 TO 12, APPROX. 10 VOLTS FULL SCALE.

SWITCH POSITIONS 13 TO 18 READ OUTPUT CURRENT OF SERIES REGULATORS. APPROX. .5 AMPERES FULL SCALE.

REVISIONS		PRINTS TO
1	SEP 7 1976	711
2	JAN 21 1977	12R452
		FG-W

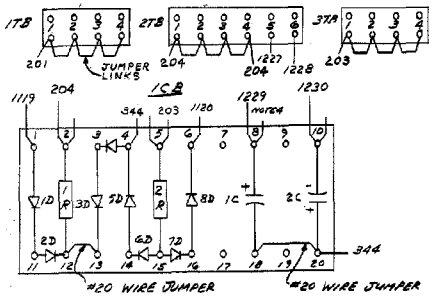
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.

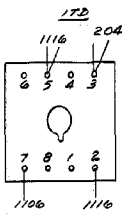
44D309956

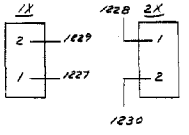
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:— APPLIED PRACTICES | SURFACES √ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ± | REV NO. | 44D308137 CONT ON SHEET SH. NO.

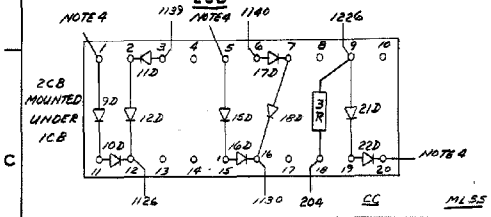
TITLE: RELAY PANEL

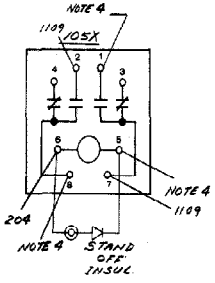
FIRST MADE FOR 3S7513KF307A16

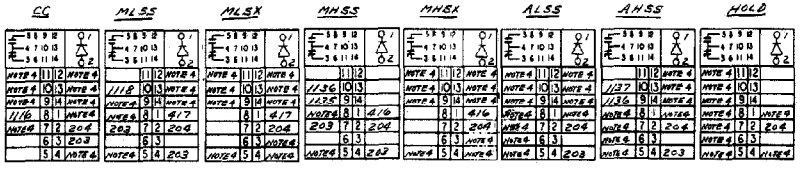


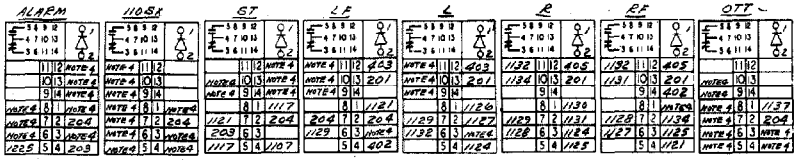


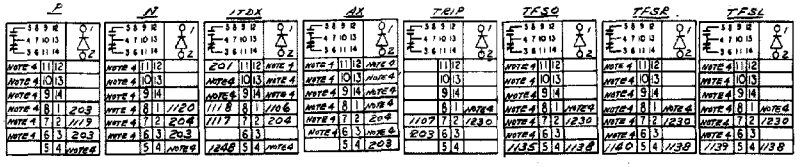


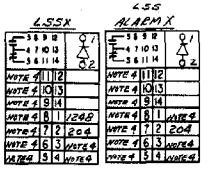


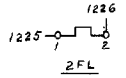












WIRE TABLE

CIRCUIT	WIRE SEQUENCE
201	RF,R,L,LF,1TDX,1TB
203	OC,MLSS,MLSS,MHSS,MHSS,ALSS,AHSS,ST, ALARM,P,P,N,AX,TRIP, 1CB,3TB
204	105X,CC,MLSS,MLSX,MHSS,MHSX,ALSS,AHSS, HOLD,OTT,LF,LF,ST,110SX,ALARM,N,1TDX, AX,LSS ALARMX,LSSX,1CB,1TD,2TB-1
204	2CB-1B, 2TB
344	1CB-4,1CB-20
402	LF,RF
403	L,LF
405	R,RF
416	MHSS,MHSX
417	MLSS,MLSX
1106	1TD,1TDX
1107	TRIP,ST
1116	1TD,1TD, CC
1117	ST,ST,1TDX
1118	1TDX,MLSS
1119	1CB,P
1120	N,1CB
1121	RF,LF,ST
1124	L,R
1125	R,RF
1126	L,2CB
1127	L,RF
1128	RF,R
1129	R,L,LF
1130	R,2CB
1131	R,RF
1132	RF,R,L
1134	RF,R
1135	TFSO,MHSS
1136	MHSS,AHSS
1137	AHSS,OTT
1138	TFSO,TFSR,TFSL
1139	TFSL,2CB
1140	TFSR,2CB
1225	ALARM,2FL
1226	2FL,2CB
1230	TRIP,TFSO,TFSR,TFSL,1CB
1248	LSSX,1TDX
1109	105X, 105X

FACTORY NOTES:

1. USE #20 WIRE, M9729494P1618
2. DO NOT CORD-TIE WIRES. INTERCONNECTING WIRES TO BE ADDED AFTER PANEL IS MOUNTED IN CASE ASM.
3. CONNECT DIODES TO EACH RELAY SOCKET AS SHOWN IN RELAY SYMBOL.
4. WIRE CONNECTION SHOWN ON MASTER WIRE TABLE.

REACTOR WIRE TABLE

CIRCUIT	CONNECTIONS
1227	1X, 2TB
1228	2X, 2TB
1229	1X, 1CB
1230	2X, 1CB

REVISIONS: 1 B Hughes SEP 4 1975

PRINTS TO: 11, 227-2167, FG-W

DRAWN BY: [illegible] 9/16/75 APPROVED: Parker

DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA. 44D308137 SH. NO.

APR 15 1975

GENERAL ELECTRIC

44D309986

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING — APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS, DECIMALS, ANGLES

TITLE: CONTROL PANEL

FIRST MADE FOR 3475/3475300A16

RPM

BACK VIEW

TPB-B

TPB-A

NEAREST PANEL

RPB

HPSV 1309

HP

CV

LP

HPPB

LPPB

HSS MANUAL LSS

RF R L LF PWR

CONNECTORS - DIODE BOARD (BACK VIEW)

PLACE 1CØ, 2CØ, 3CØ, LABELS ON FRONT & BACK OF BRACKETS

1CØ

2CØ

3CØ

DB

CABLE CHECK

1CØ, 2CØ, 3CØ

INDICATES DIODE BOARD TERMINALS

USE 44A310526-001 SHIELDED CABLE & #20 WHITE MS7294S4P16IB

	CIRCUIT	WIRE SIZE	CONNECTIONS
SHIELDED CABLE	201	WHITE	1CØ,RPM
	337	BLACK	1CØ,RPM
	201	SHIELD	1CØ-V
SHIELDED CABLE	805	WHITE	1CØ,CONT
	806	BLACK	1CØ,CONT
	201	SHIELD	1CØ-L
SHIELDED CABLE	902	WHITE	1CØ,LP
	0-	BLACK	1CØ,LP
	201	SHIELD	1CØ-H
SHIELDED CABLE	-07	WHITE	1CØ,HP
	90	BLACK	1CØ,HP
	201	SHIELD	1CØ-J
	15	#20	LF,L
	16		L,R
	17		R,RF
	203		1CØ,PWR,LF,LF,L,R,RF,1CØ
	203		1CØ-K,2CØ-m
	203		3CØ-F,3CØ-X
	204		HSS,MAN,LSS,PWR,1CØ
	1123		LF,1CØ
	1133		RF,1CØ
	1148		L,1CØ
	1149		R,1CØ
	1200		PWR,DB
	1219		MAN,MAN,DB,1CØ
	1222		LSS,LSS,DB,1CØ
	1223		HSS,HSS,DB,1CØ
	1224		PWR,PWR, ,1CØ
	1267		PWR,PWR,DB,1CØ
	1138		1CØ,RF
	247		2CØ,RPM
	248	#20	2CØ,RPM

#16 WIRE, MS7294S4P21IB

CIRCUIT	WIRE SIZE	CONNECTIONS
1300	#16	TPB-B,RPB,2CØ
1301		TPB-A,2CØ
1302		TPB-B,2CØ
1303		TPB,2CØ
1304		RPB,2CØ
1305		RPB,2CØ
1306		LPPB,3CØ
1307		R1,R2,3CØ
1308		HPPB,3CØ
1309		HPSV,HPPB,3CØ
1310		LPPB,3CØ
1311		TPB,2CØ
1312		TPB-B,RPB,2CO
1313		TPB-B,3CO
1314		TPB-A,3CO
1315		TPB-B,3CO
1316		TPB-A,3CO
1320		TPB-B,TPB-A
1321		TPB-B,TPB-A
1322		TPB-B,TPB-A
1323		TPB-B,TPB-A
1324		HPSV,R1
1325		HPSV,R2
1326		RPB,3CO-E
1327	#16	RPB,3CO-H

REVISIONS | PRINTS

DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.

44D309986

JUN 13 1977

2+1PC

GENERAL ELECTRIC CO.
WAYNESBORO, VA.

REGULATOR SUMMARY

JUN 16 1977

44X321562

COMPILED BY: R. J. Culver R Culver DATE: 6/14/77 1 SHEETS; SHEET 1

CUSTOMER: Louisville Gas and Electric, Mill Creek #4

BFTB 153308 Requisition 091-41975

ITEM	MATERIAL		QUAN.
1	3S7513TC300A8 Electronic Boiler Feedpump Turbine Control		1
	Outline Drawing	44D300043	
	Wire Table	44A308899	
	Interconnection Drawing	44C321493	
	Elementary Drawing	44C321492	
	Alignment Drawing	44D309956	
	Alignment Instructions	44A376102	
2	3S7513KL300A6		
	Outline	44B309476	
	Connection Diagram	44D309986	
3	Cabinet-Control Panel Power Cable	44B306148	2
4	Cabinet-Control Panel Control Cable	44B300423	1

REVISIONS

DISTRIBUTION

711	
FG-W	
~~12~~P-FS 2	

3+ IPL

GENERAL ELECTRIC COMPANY • DRIVE SYSTEMS DEPARTMENT
CONTROL DEVICES OPERATION • WAYNESBORO, VIRGINIA 22980

GENERAL ELECTRIC

9/77 (50)